

Development of Cardiovascular systems (CVS)



Development of Cardiovascular systems (CVS),,,

➤ *Introduction to CVS*

➤ *Development of Heart*

➤ *Development of blood vessels*

➤ *Development of lymphatic systems*

Introduction to CVS

- ❖ The 3 germ layers, **Embryonic ectoderm**, **Embryonic endoderm** and **Embryonic mesoderm** gives rise to specific tissues and organs of the body.

❑ **Embryonic ectoderm gives rise to:**

- ✓ Epidermis,
- ✓ Central and peripheral nervous systems,
- ✓ The eye, and inner ear,
- ✓ Neural crest cells,
- ✓ To many connective tissues of the head

Introduction to CVS,,

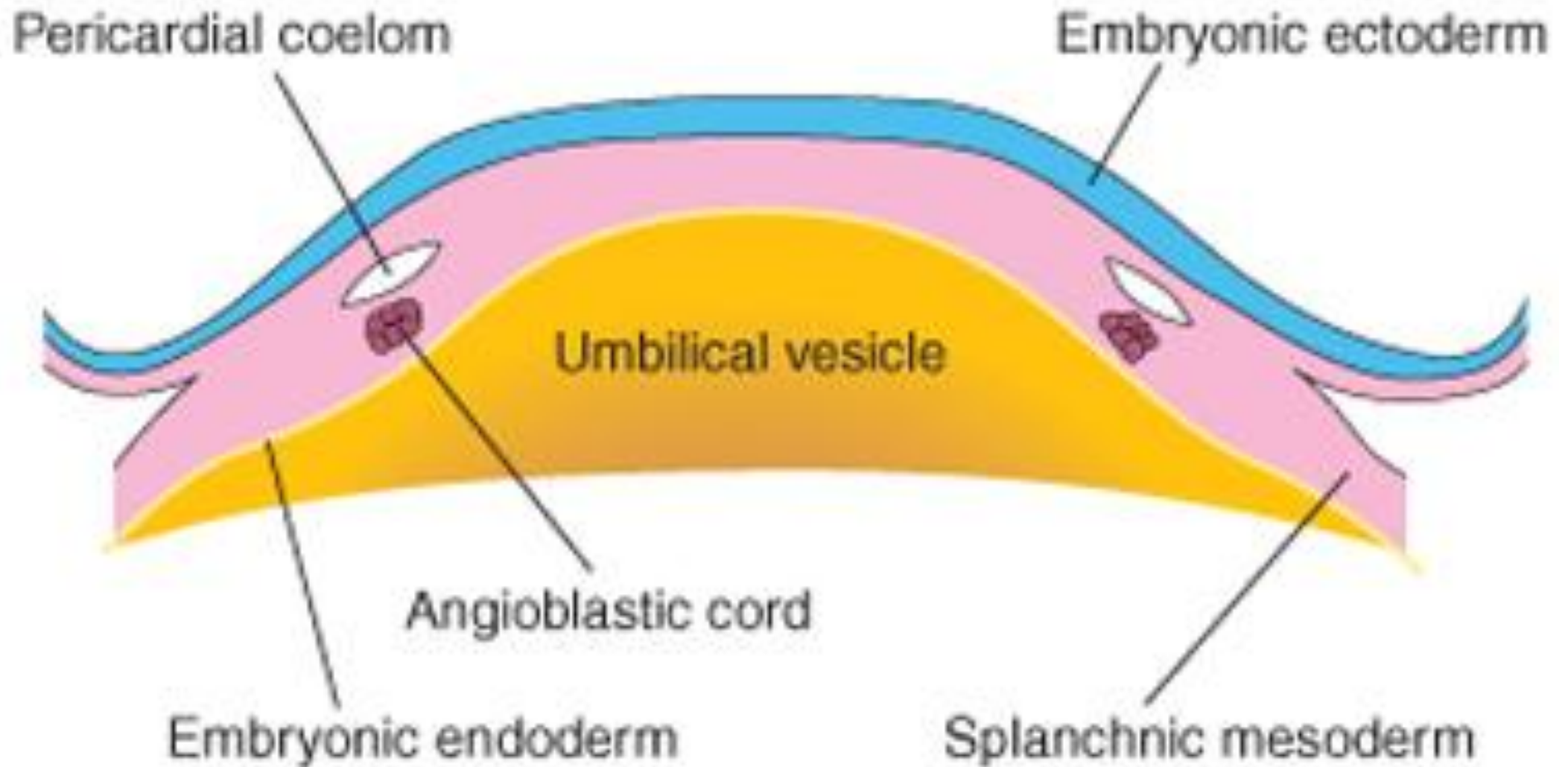
❑ Embryonic endoderm gives rise to:

- ✓ The epithelial linings of the respiratory and alimentary (digestive) tracts
- ✓ The glandular cells of the liver and pancreas

❑ Embryonic mesoderm gives rise to:

- ✓ Skeletal muscles,
- ✓ Blood cells and the lining of blood vessels,
- ✓ All visceral smooth muscular coats,
- ✓ The serosal linings of all body cavities

Introduction to CVS,,



Development of Cardiovascular System

- *The cardiovascular system is the **first major system** to **function in the embryo**.*
- The primordial heart and vascular system appear in the middle of the **3rd week**.
- *The reason why cardiovascular system develop **first** is **the rapidly growing embryo can no longer** satisfy its nutritional and oxygen requirements by diffusion alone.*
- **Consequently**, there is a need for an efficient method of acquiring oxygen and nutrients from the maternal blood and disposing of carbon dioxide and waste products.

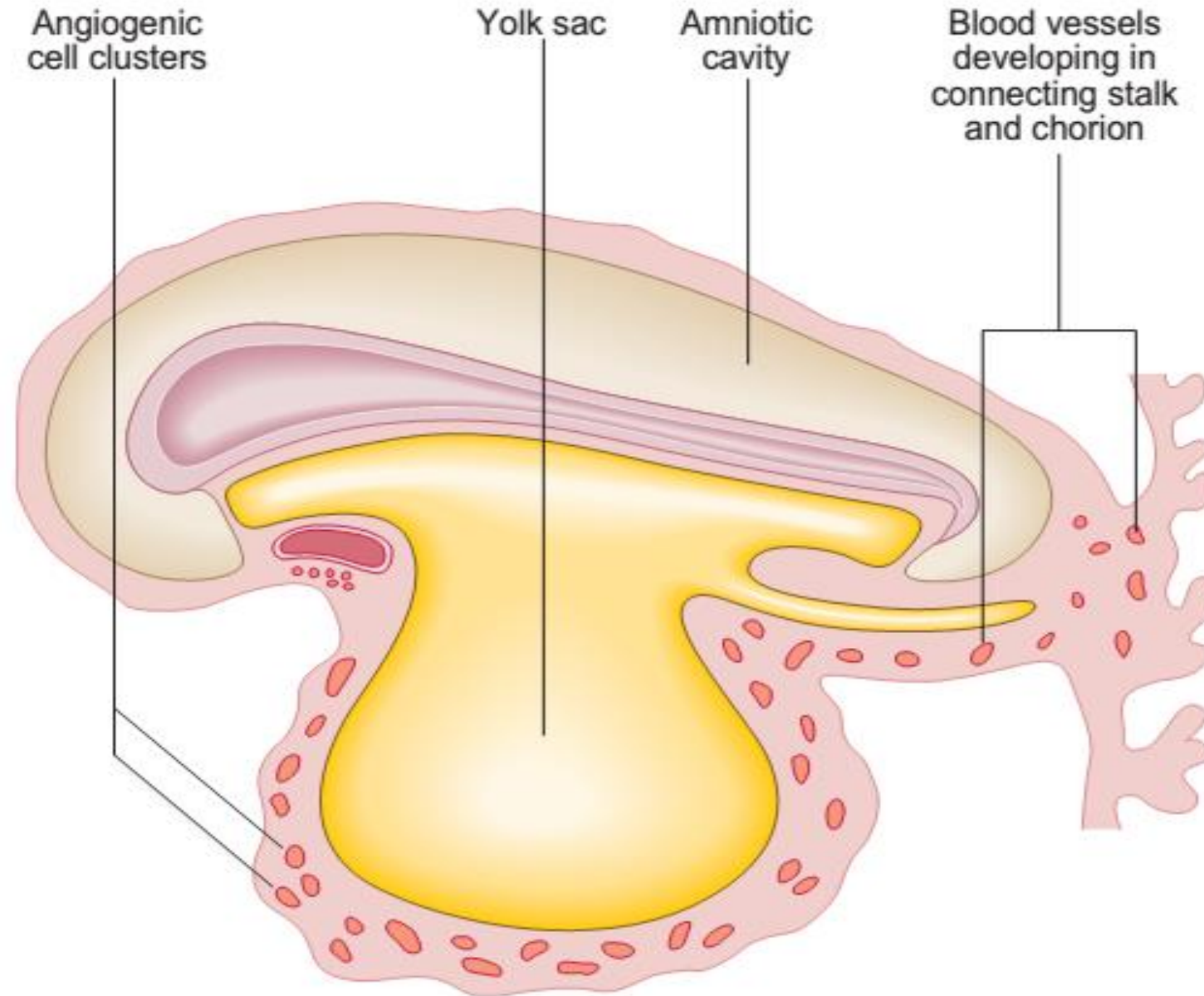
Development of Cardiovascular System ,,

- The cardiovascular system is derived mainly from:
 - *Splanchnic mesoderm*, which forms the primordium of the heart.
 - *Paraxial and lateral mesoderm* near the otic placodes from which the internal ears develop.
 - *Neural crest cells* from the region between the otic vesicles and the caudal limits of the third pair of somites.

Development of Cardiovascular System ,,

- ❑ The first sign of CVS **development** appear as **angiogenic cell** clusters in the extra-embryonic mesoderm lining the yolk sac at the beginning of 3rd week.
- ❑ Embryonic blood vessels begin to develop approximately **2 days** later.
- ❑ **Formation of embryonic/extraembryonic vascular system involves two processes:**
 - **Vasculogenesis** - formation of new vascular channels by assembly of individual cell precursors called **angioblasts**
 - **Angiogenesis** - formation of new vessels by budding and branching from preexisting vessels.

Development of Cardiovascular System ,,,

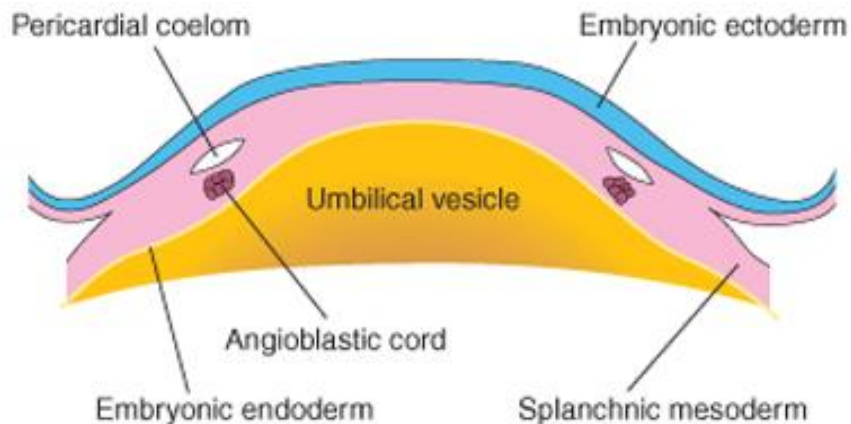


Heart development

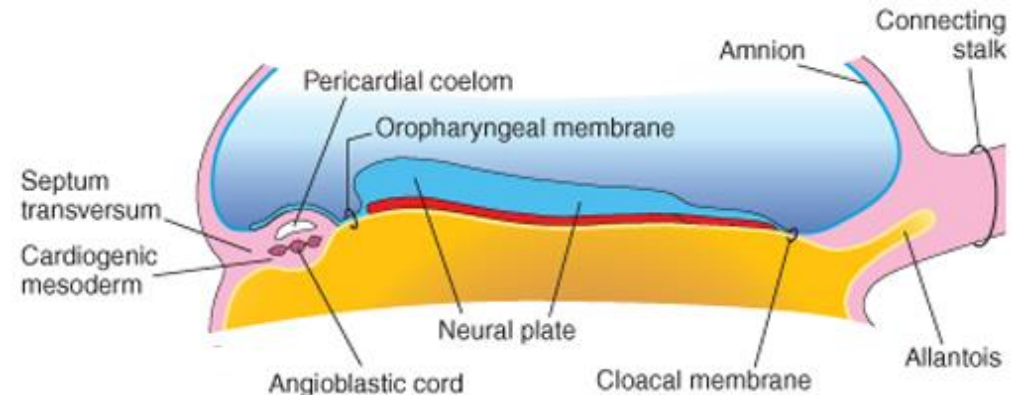
- The development of heart can be discussed by dividing into:
 1. Development of primitive heart tube
 2. Formation of Cardiac Wall
 3. Dilatation of primitive heart tube
 4. Partitioning of Primordial Heart and development of its chambers
 - Development of Atrioventricular septum
 - Partitioning of Primordial Atrium
 - Development of Rt & Lt atrium
 - Partitioning of Primordial Ventricle
 - Partitioning of outflow part of heart
 4. Development of cardiac valve
 5. Development of conduction systems of heart

Development of primitive heart tube

- The primordial heart and vascular system appear in the middle of 3rd week.
- The earliest sign of heart development appear as **paired endothelial strands- angioblastic cords**- from aggregation of the splanchnic mesoderm in **cardiogenic area**, ventral to pericardial coelom, and dorsal to yolk sac.
- The 2 angioblastic cords that canalize to form 2 endocardial heart tubes.

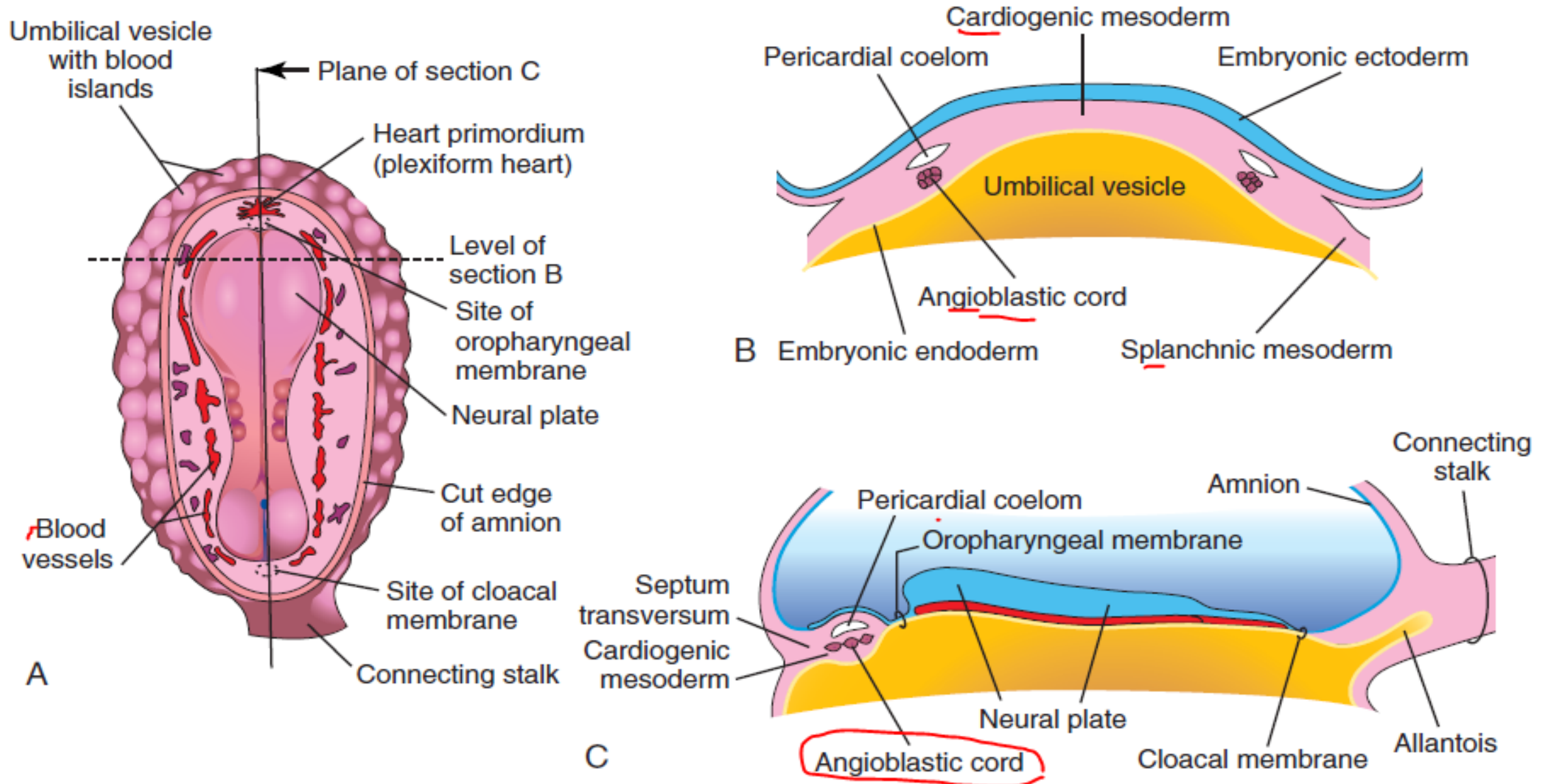


Transverse section



Longitudinal section

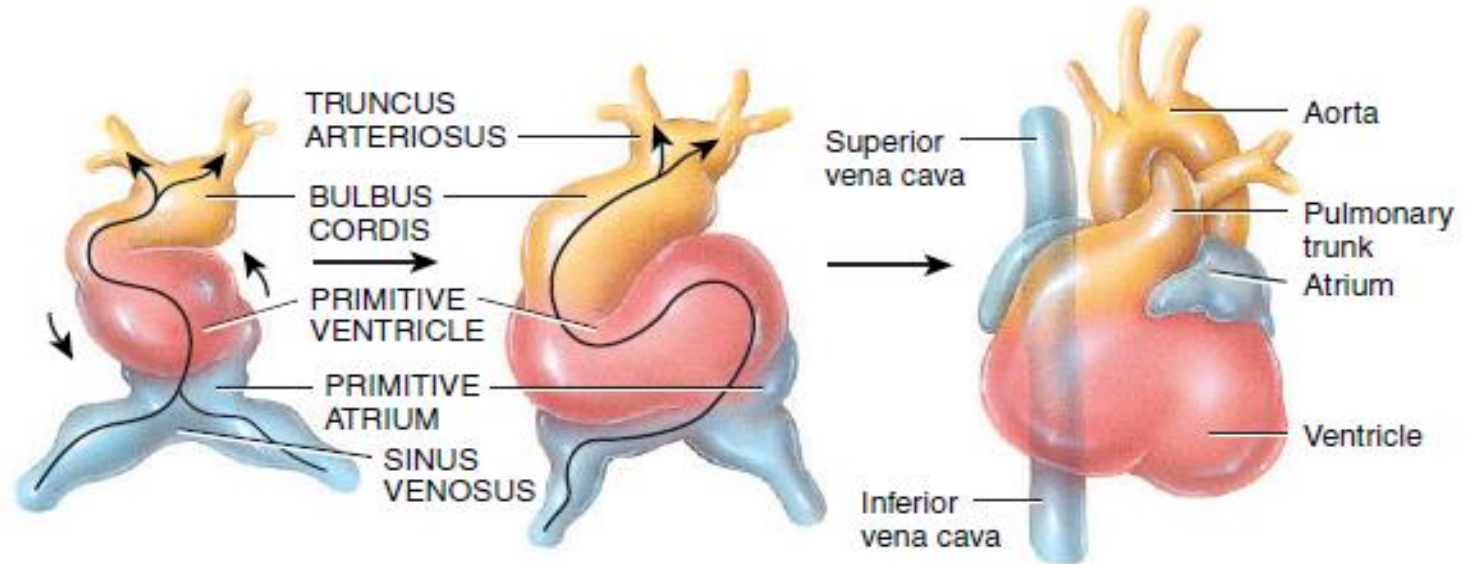
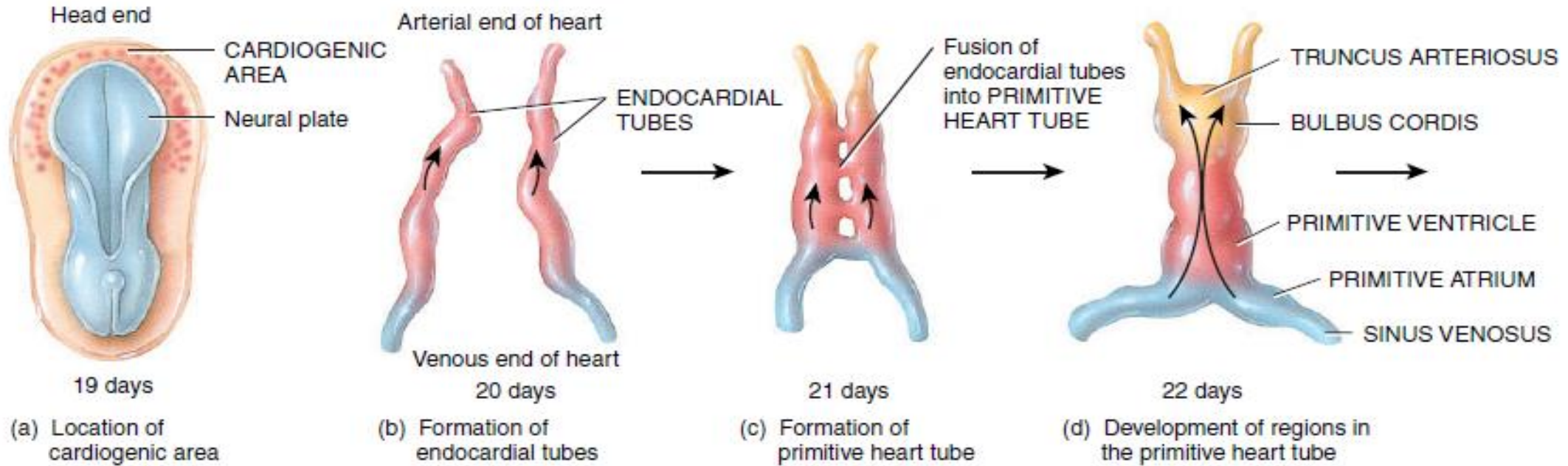
Development of primitive heart tube,,



Development of primitive heart tube,,,

- ❑ The cardiovascular system is one of the first systems to form in an embryo, and the heart is the first functional organ.
- This order of development is essential because of the need of the rapidly growing embryo to obtain oxygen and nutrients and get rid of wastes.
- The *heart* begins its development from **mesoderm** on day 18 or 19 after fertilization.
- **In the head end** of the embryo, the heart develops from a group of mesodermal cells called the **cardiogenic area**.
- In response to signals from the underlying endoderm, the mesoderm in the cardiogenic area forms a pair of elongated strands called **cardiogenic cords**.
- Shortly after, these cords develop a hollow center and then become known as **endocardial tubes**.

Development of primitive heart tube,,

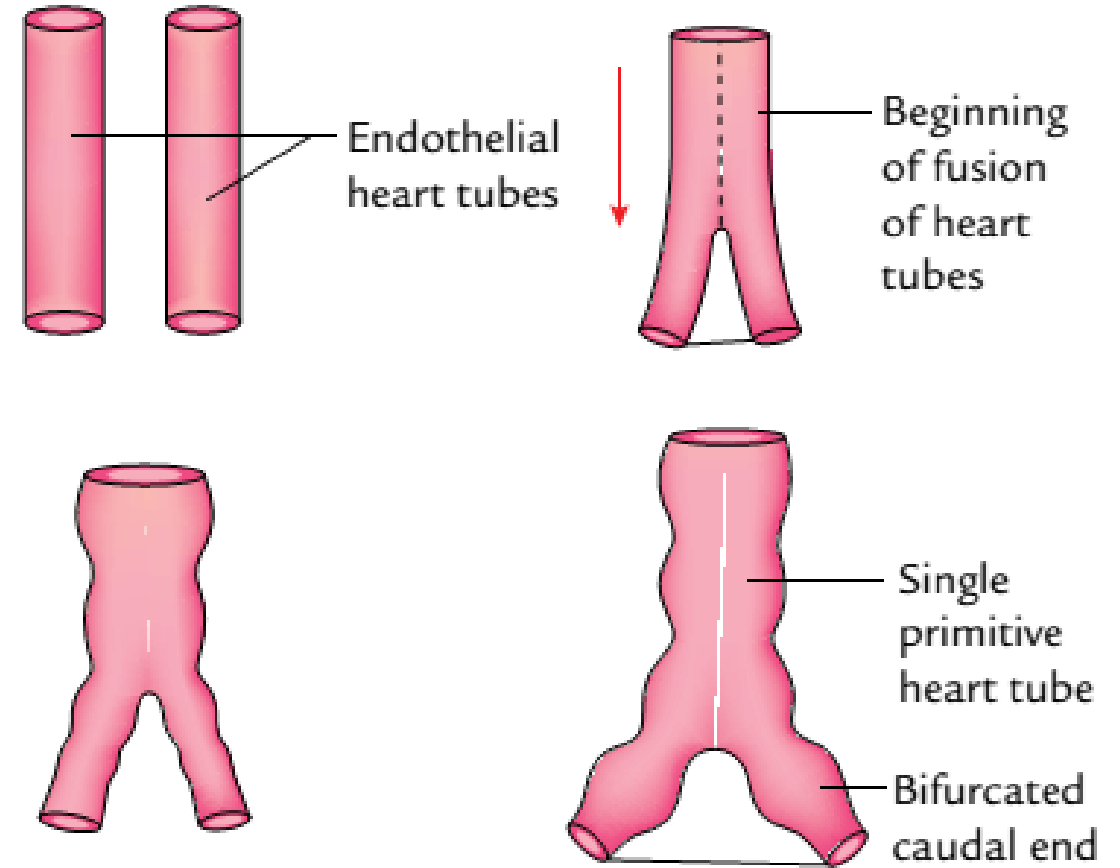


Development of primitive heart tube,,

- This two separate heart tube approach each other & form a single heart tube.

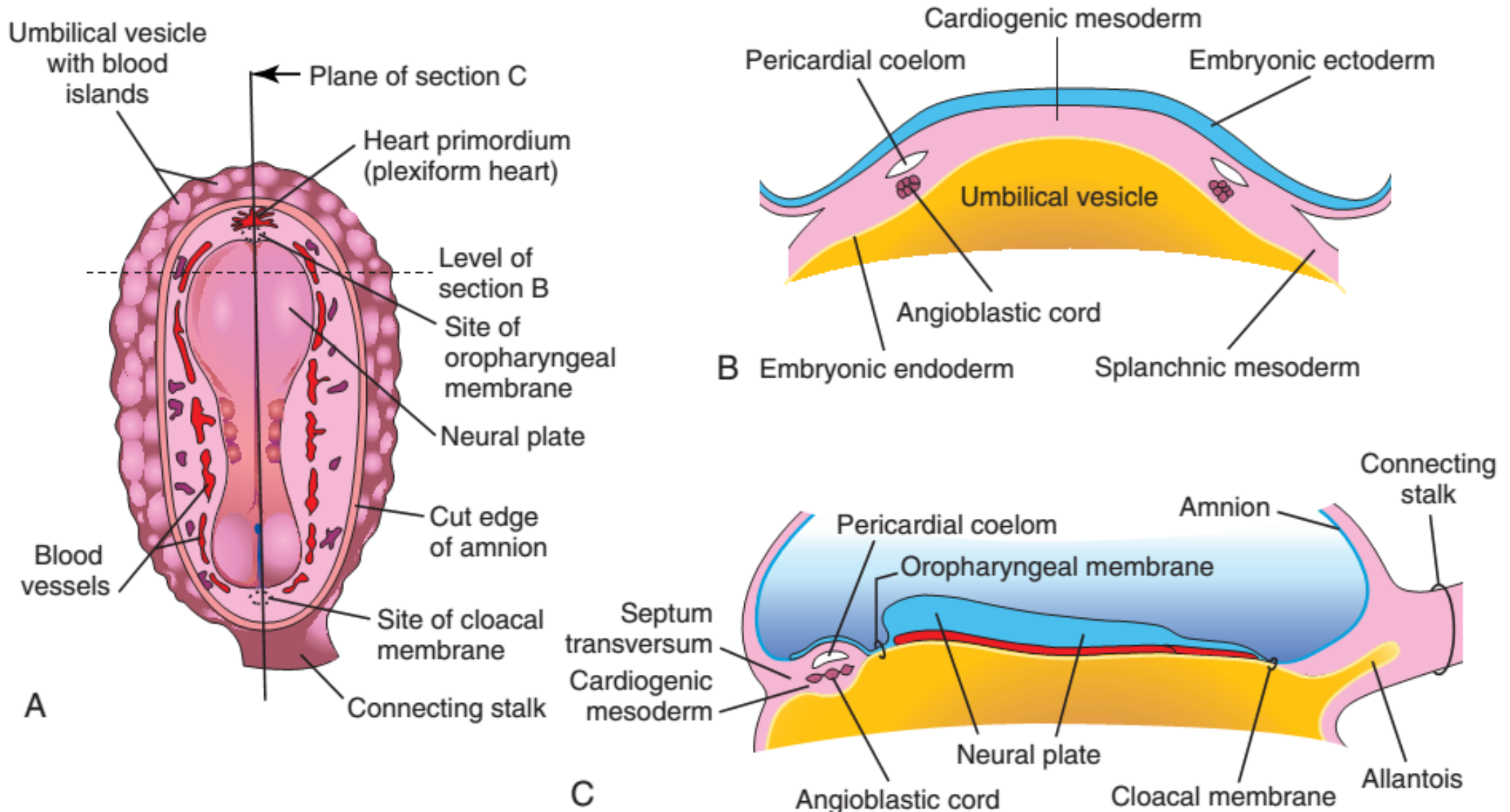
➤ *Fusion begins cranially and extend caudally.*

- The position of heart & pericardial cavity also changes b/c of embryonic folding.

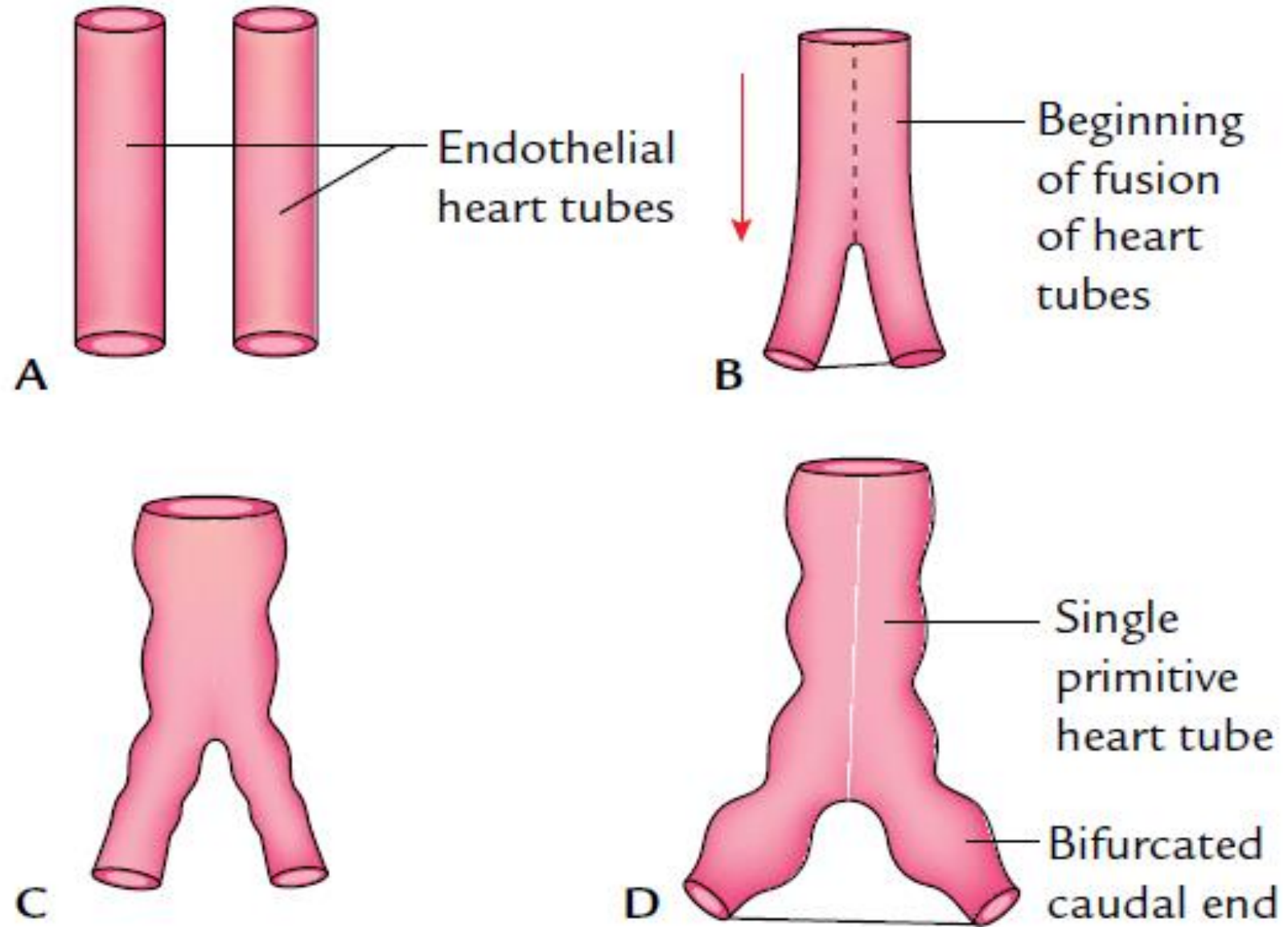


- **The heart begins to beat at 22 to 23 days.**
- Blood flow begins during the 4th wk and can be visualized by Doppler ultrasonography.

Development of primitive heart tube,,,



Development of primitive heart tube,,,

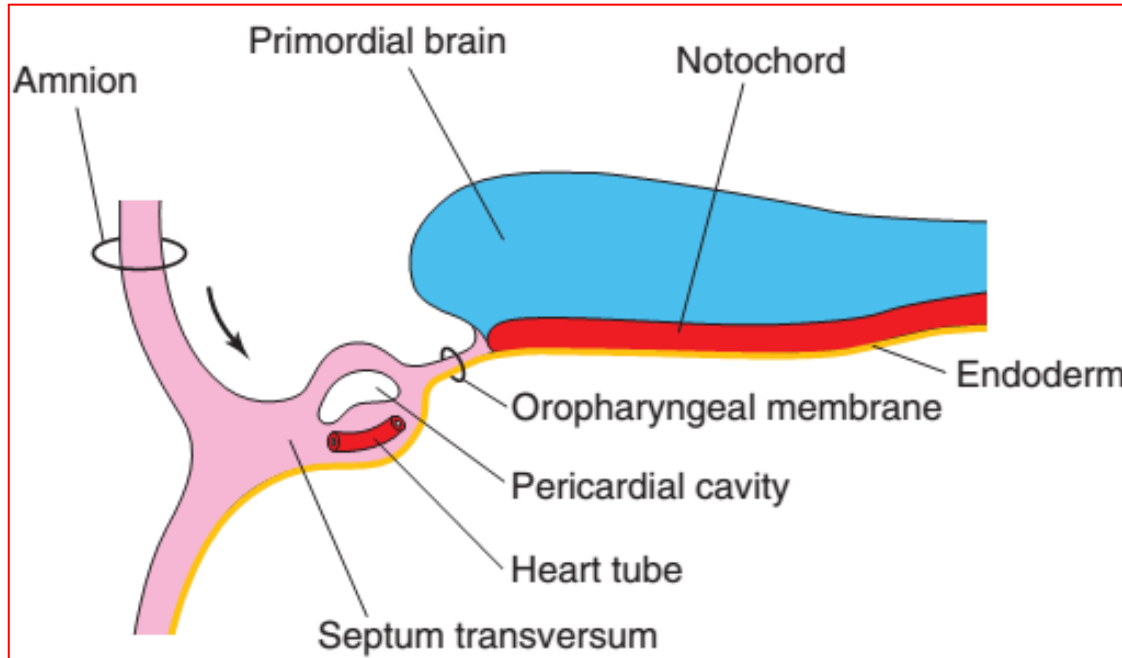


Fusion of the endothelial heart tubes

Development of primitive heart tube,,,

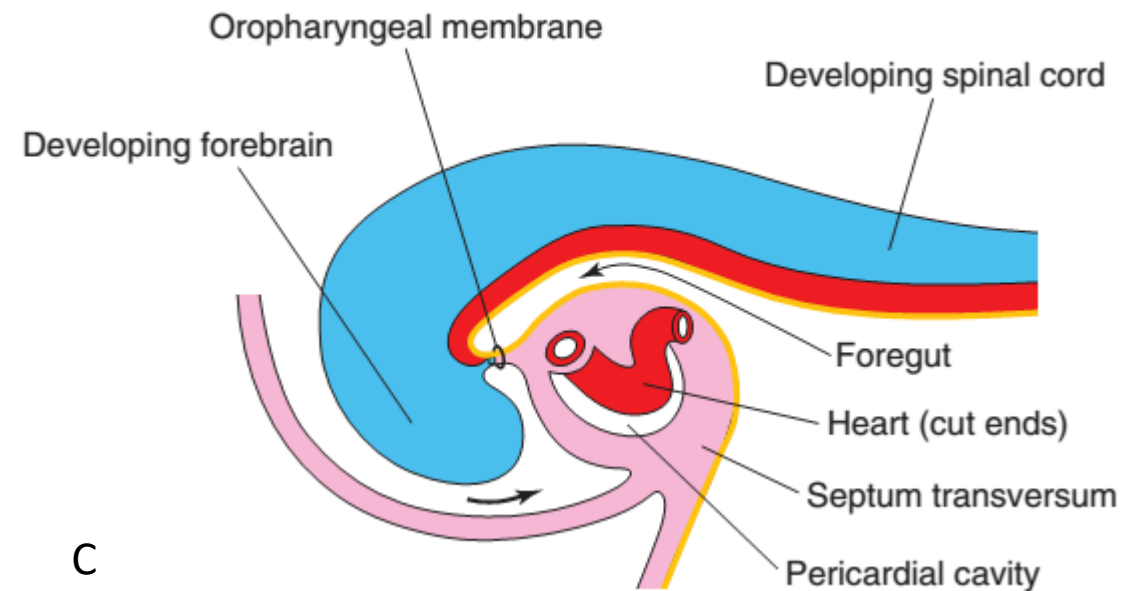
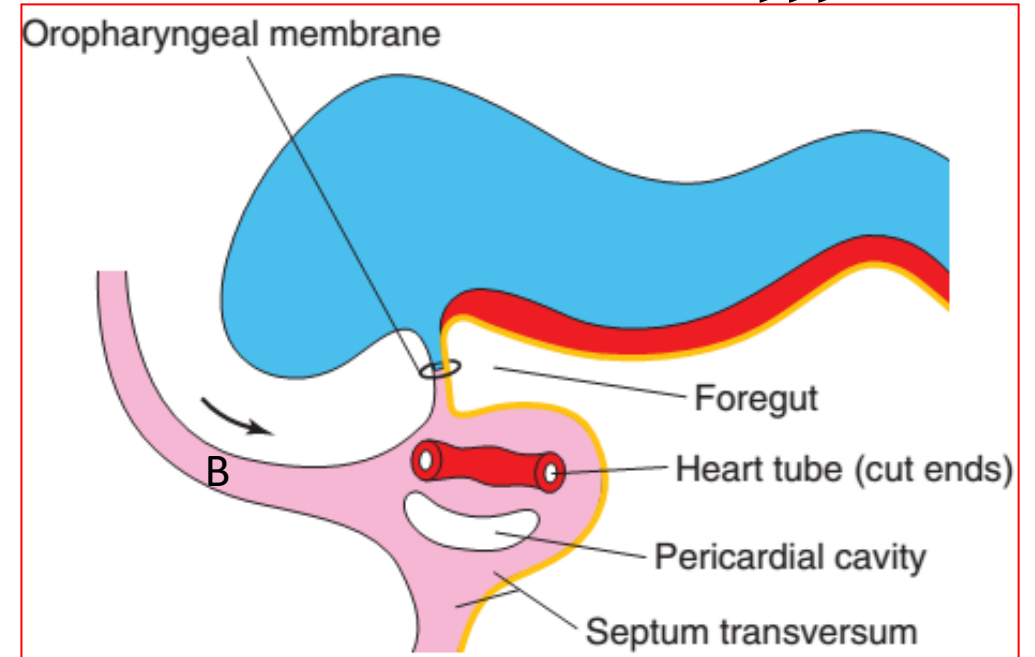
- Initially, the central portion of **the cardiogenic area is anterior to the buccopharyngeal membrane and the neural plate.**
- With closure of the neural tube and formation of the brain vesicles, however, the central nervous system grows cranially so rapidly that it extends over the central cardiogenic region and the future pericardial cavity.
- As a result of growth of the brain and cephalic folding of the embryo, the **buccopharyngeal membrane** is pulled forward, while the heart and pericardial cavity move first to the cervical region and finally to the thorax.
- **As the embryo grows & bends cephalo- caudally**, it also folds laterally. As a result, the caudal regions of the paired cardiac tube merge except at their caudal most ends.

Development of primitive heart tube,,



A

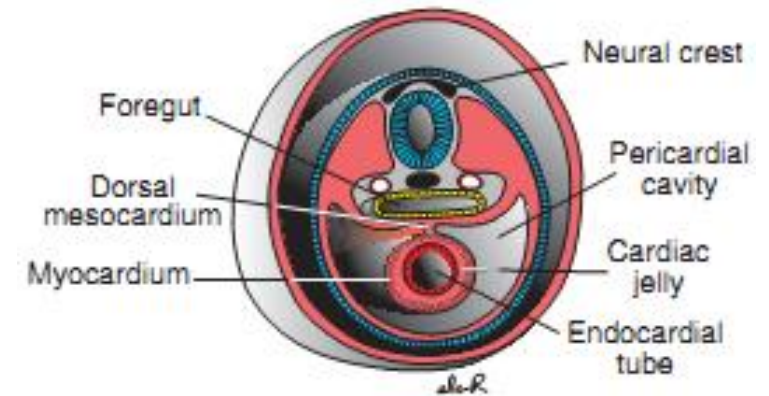
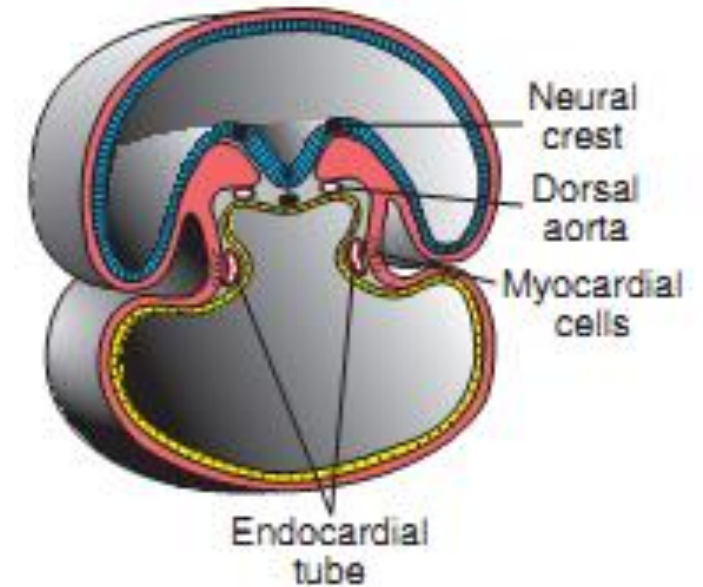
- ❑ A and B, & C As the head fold develops, the tubular heart and pericardial cavity move ventral to the foregut and caudal to the oropharyngeal membrane.



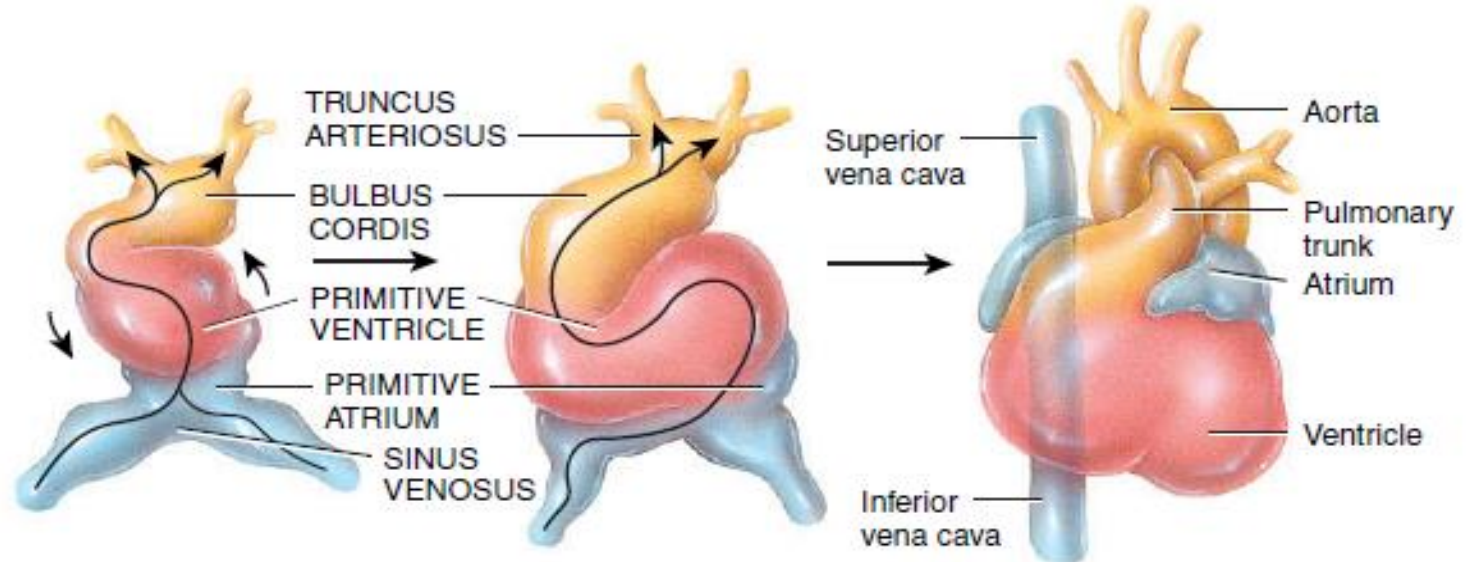
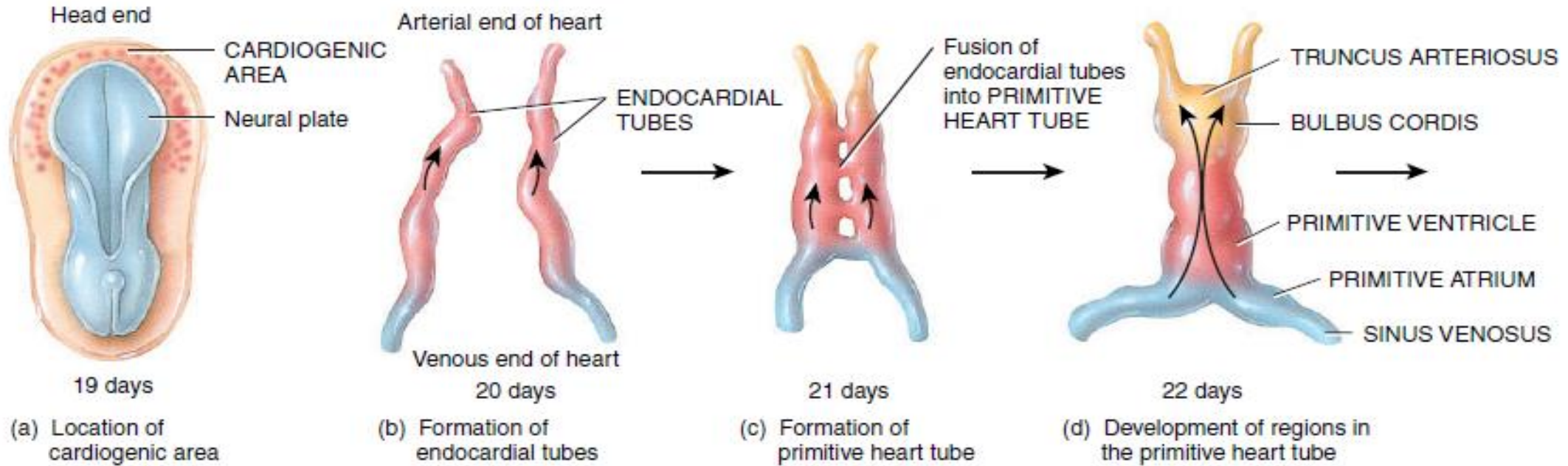
Development of primitive heart tube,,,

❑ After lateral folding of embryo,

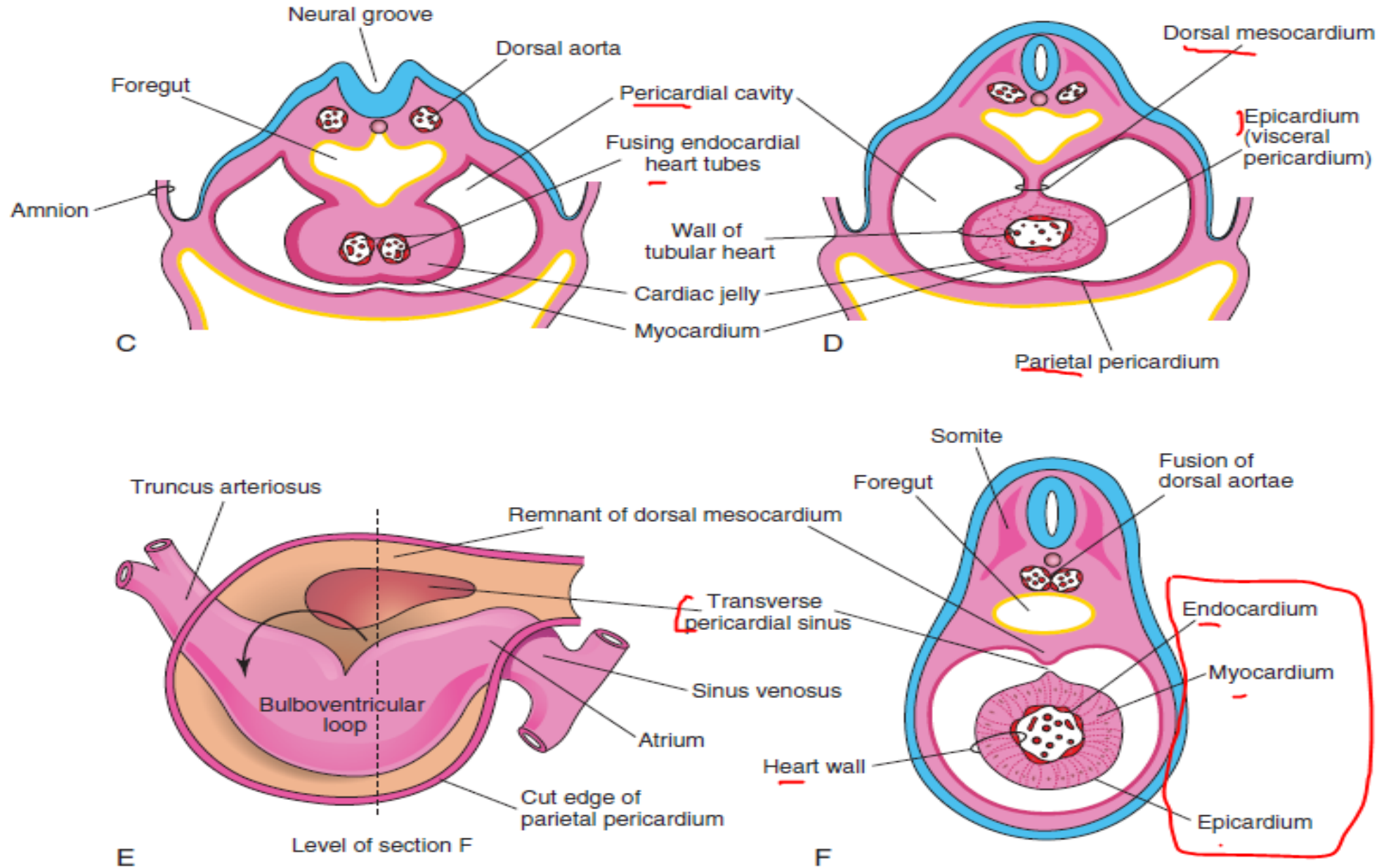
- 2 endocardial tubes fuse cephalocaudal direction to form a single heart tube by **21,22** days.
- This heart tube lies inside the pericardial cavity, its dorsal wall is connected to **foregut** by **dorsal mesocardium** (day 22).
- The central part of dorsal mesocardium degenerates, forming transverse passage **dorsal to heart ,called transverse sinus of pericardium** by day 28.



Development of primitive heart tube,,

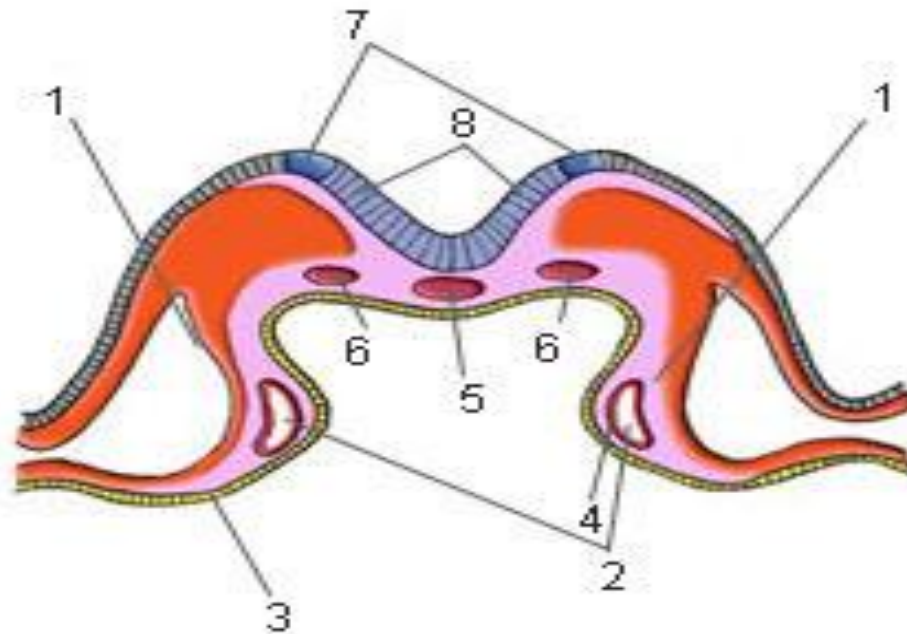


Development of primitive heart tube,,,



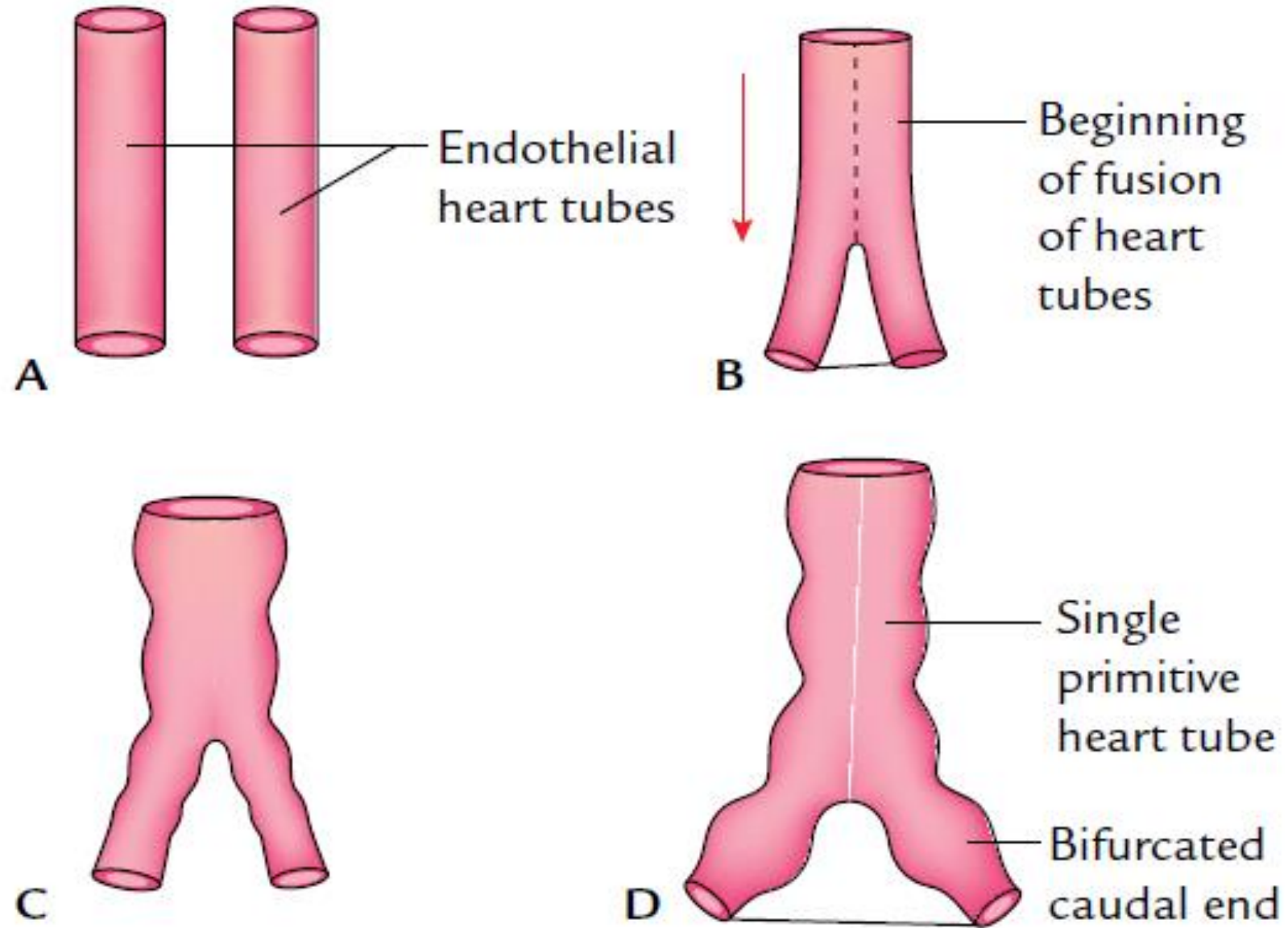
Development of primitive heart tube,,

- These heart tubes approach each other and fuse to form a single heart tube, temporarily attached to the dorsal side of the pericardial cavity by the **dorsal mesocardium**.
- The splanchnic mesoderm around the heart tube (future **endocardium**) thickens and forms the **myoepicardial mantle** (future **myocardium** and **epicardium**).



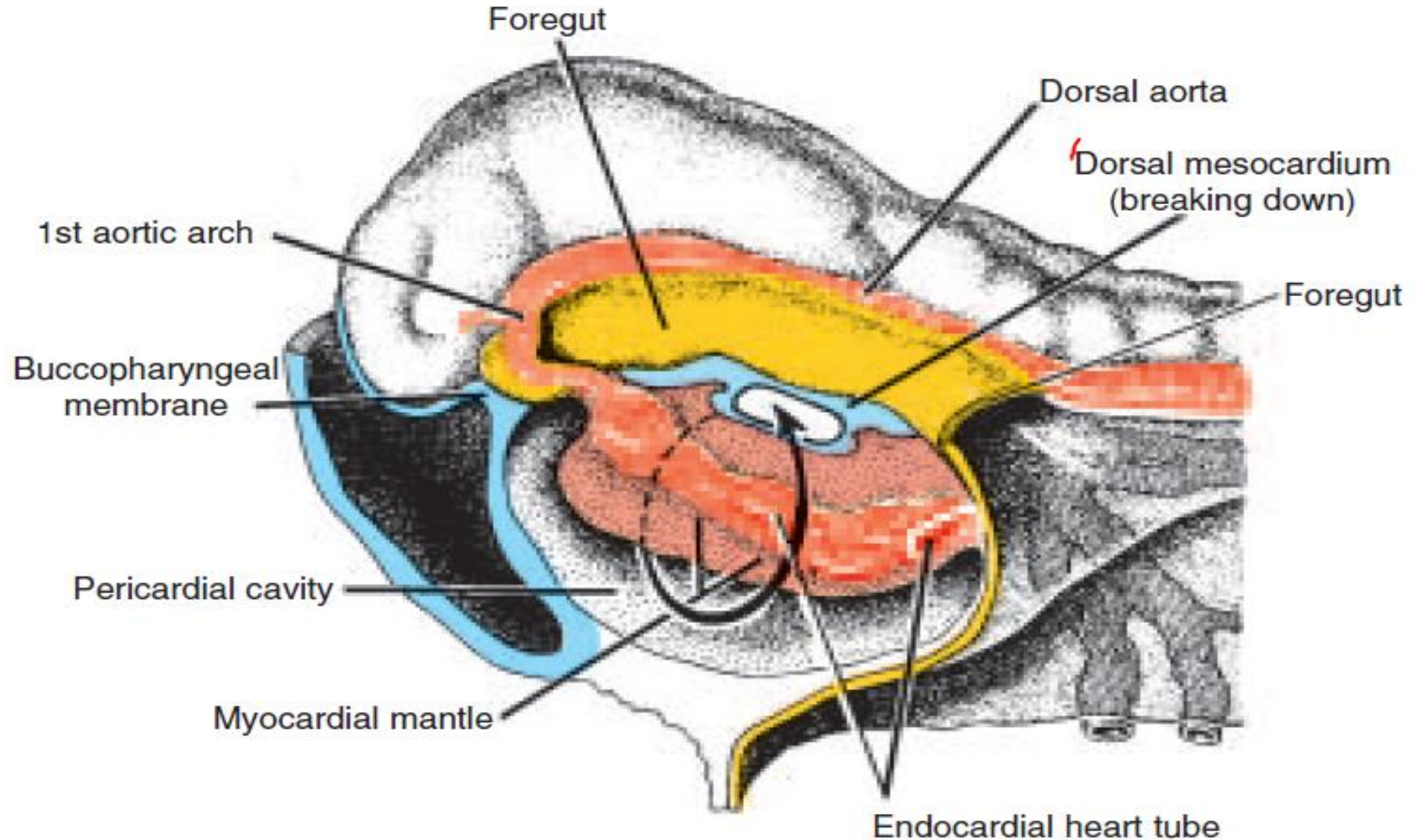
1. *Epimyocardial cells*
2. *Heart tube*
3. *Endoderm*
4. *Endocardial cells*
5. *Notochord*
6. *Dorsal aorta*
7. *Neural crest*
8. *Neural fold*

Development of primitive heart tube,,,

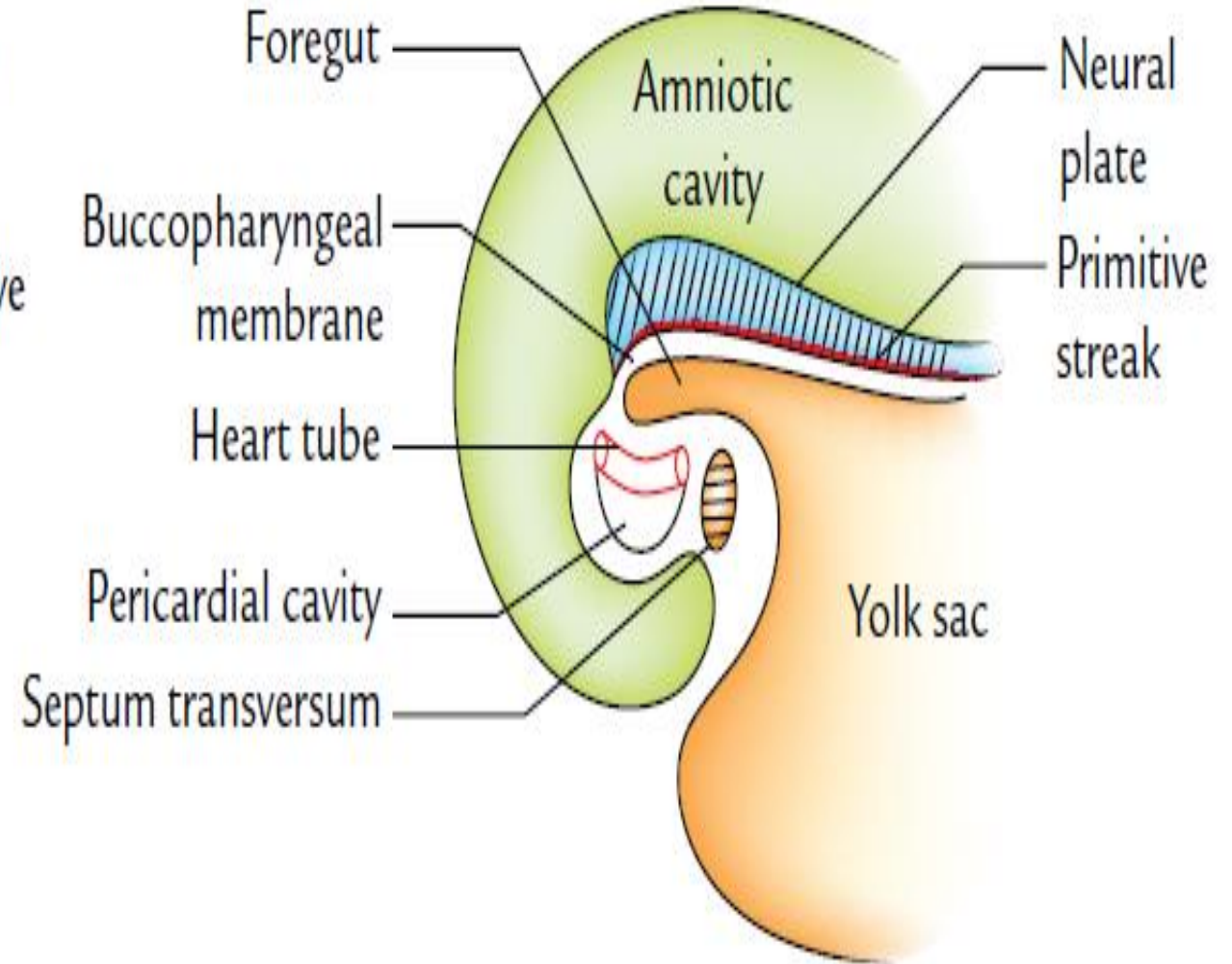
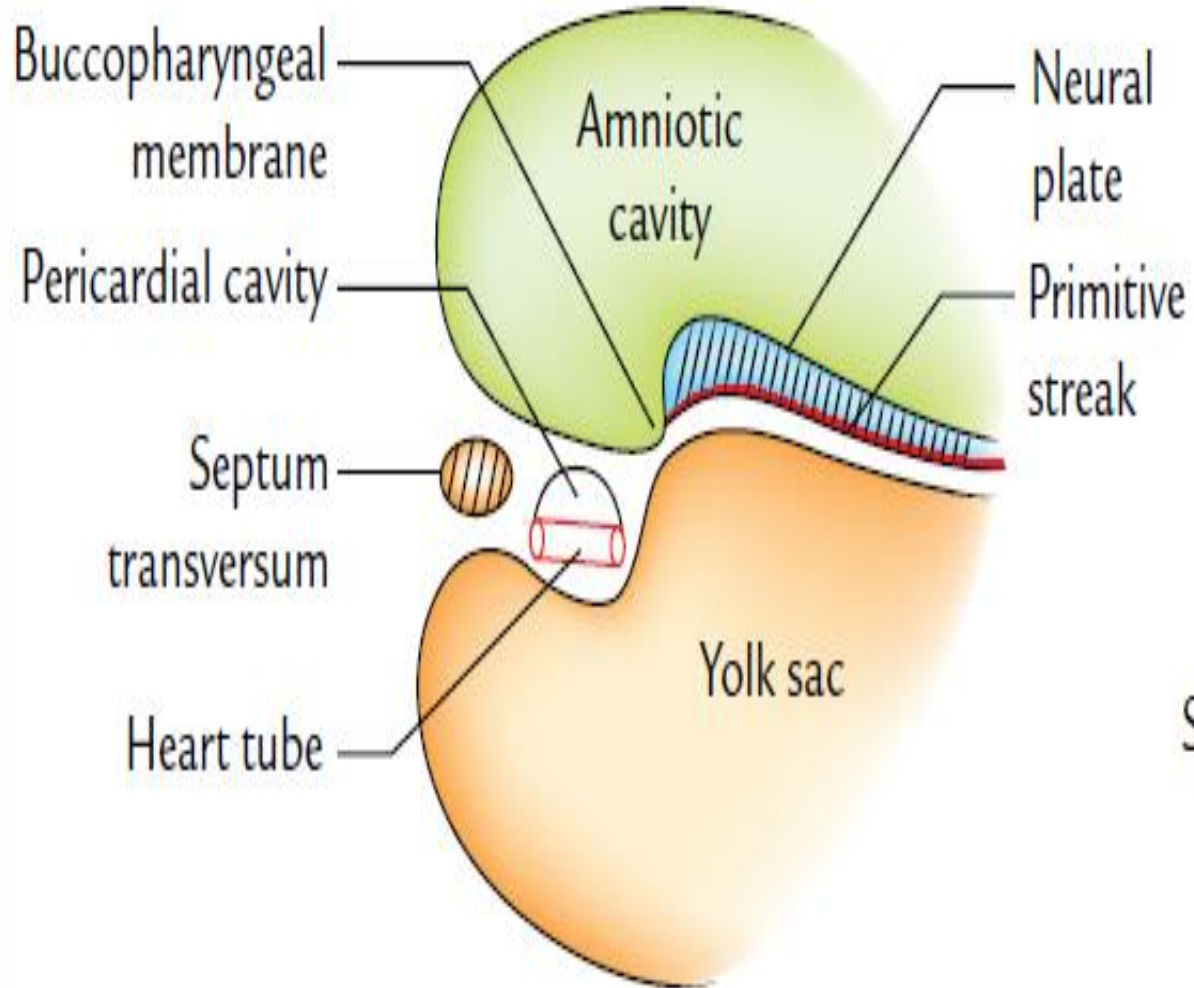


Fusion of the endothelial heart tubes

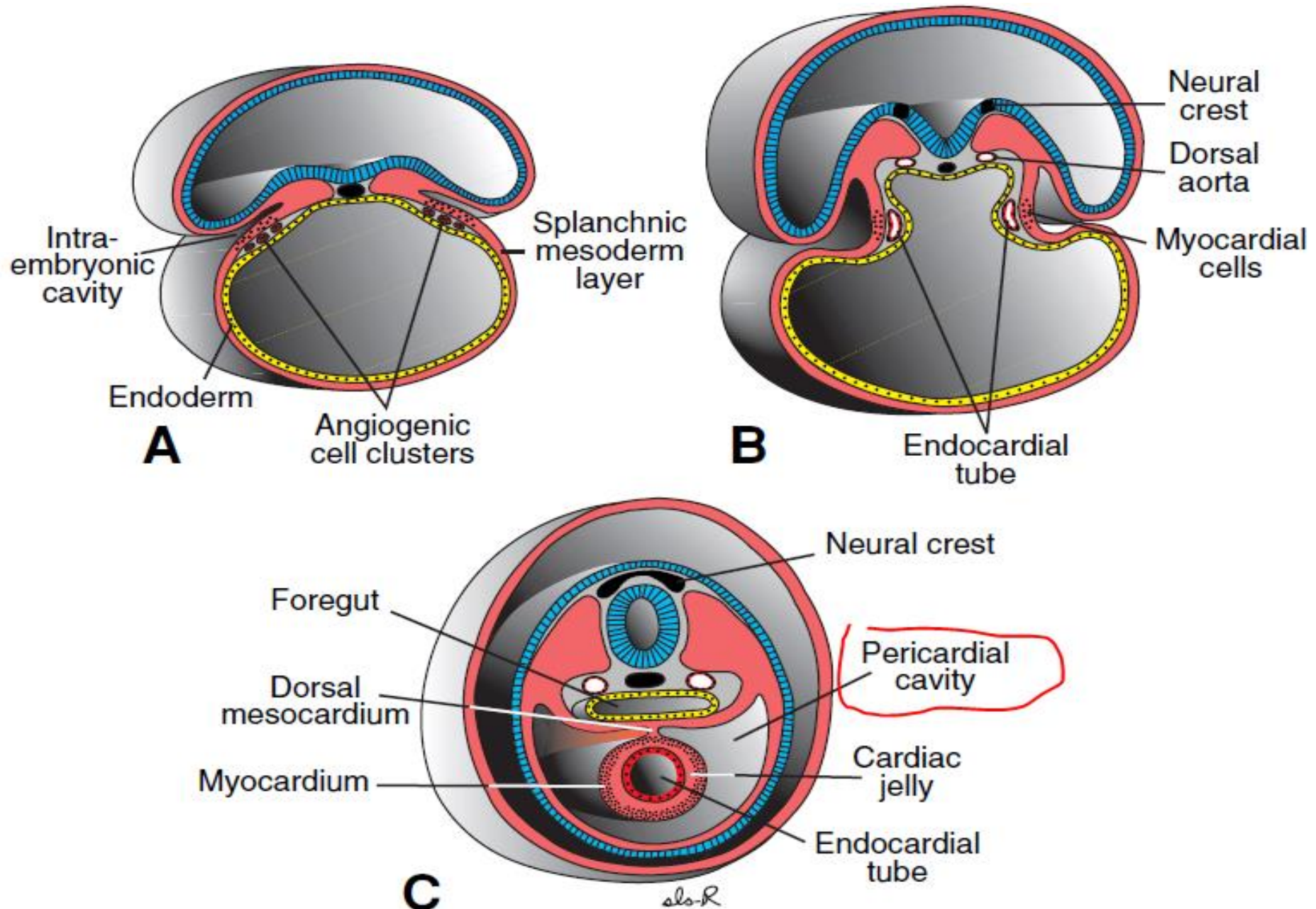
Development of primitive heart tube,,



Development of primitive heart tube,,

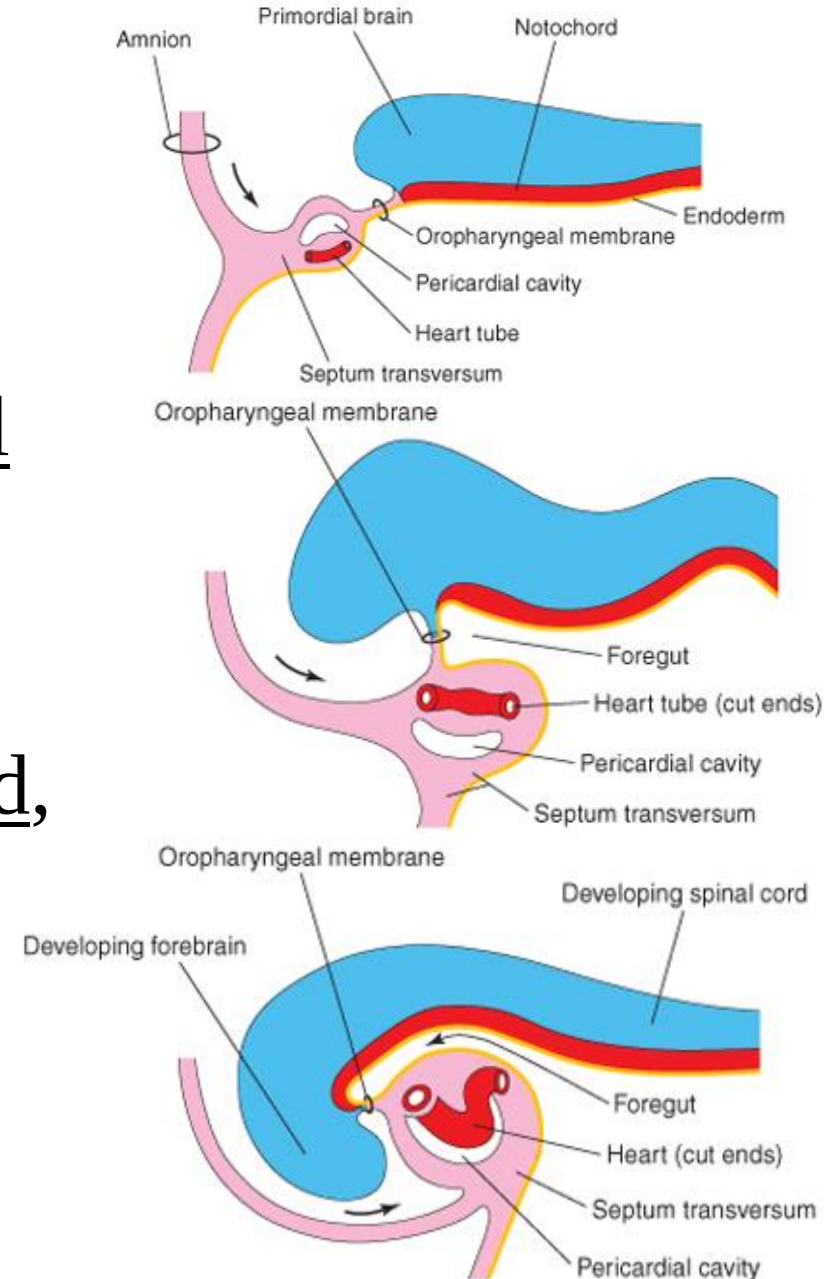


Development of primitive heart tube,,



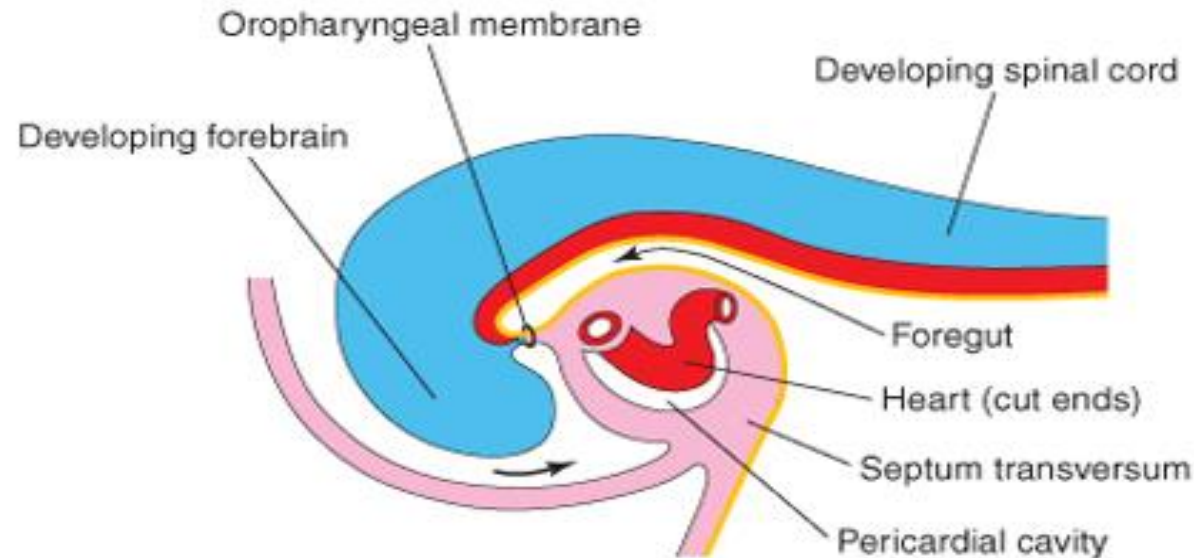
Development of primitive heart tube,,

- As the head fold develops (during 4th week), heart tube & pericardial cavity lie ventral to foregut and caudal to **oropharyngeal** membrane.
- The position of heart tube is reversed, it lies dorsal to pericardium.

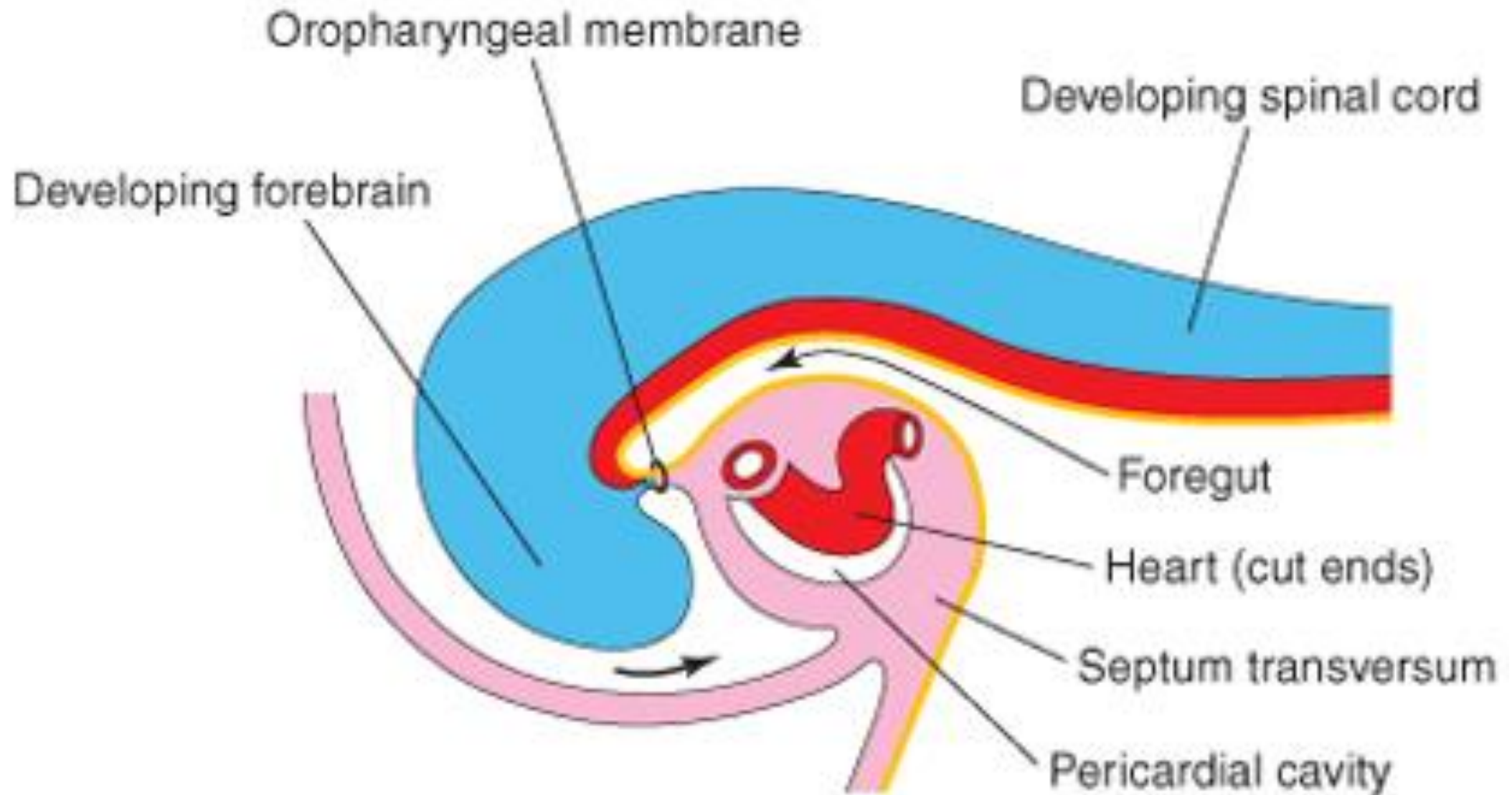


Development of primitive heart tube,,,

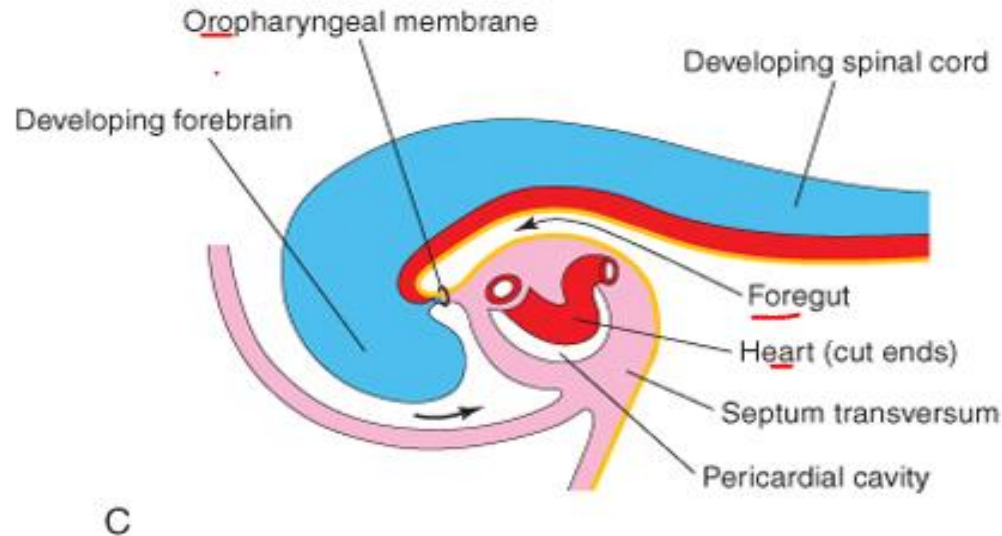
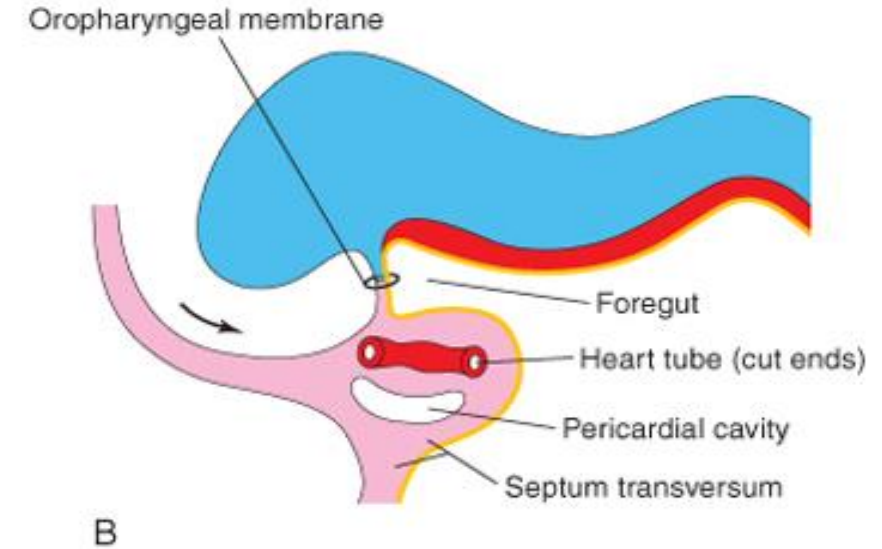
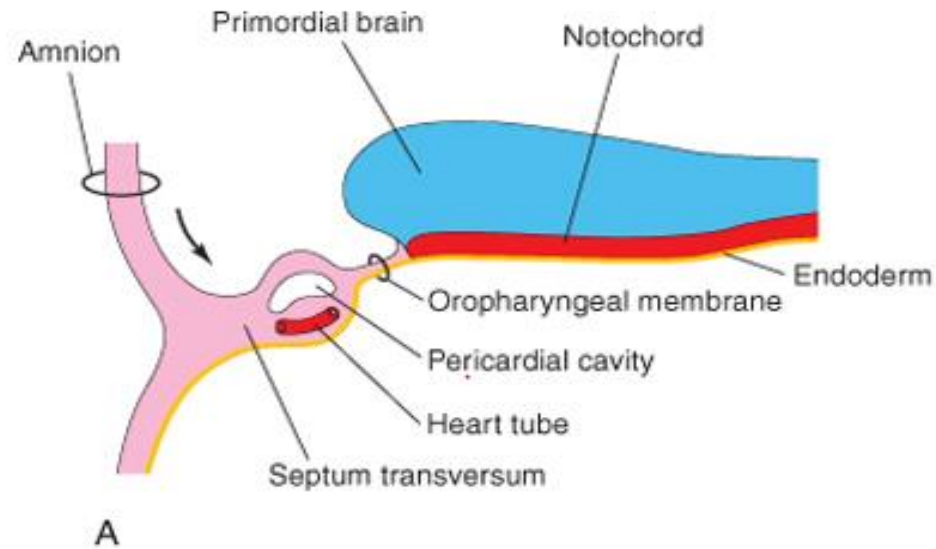
- Complete head folding and reversion of **heart tube**, **pericardium** & **septum transverse** (future central tendon of diaphragm).
- Note also the heart tube lies inside the pericardial cavity.
- The growth of the central nervous system pulls the cardiogenic area ventrally and caudally to the buccopharyngeal (oropharyngeal) membrane.



Development of primitive heart tube,,

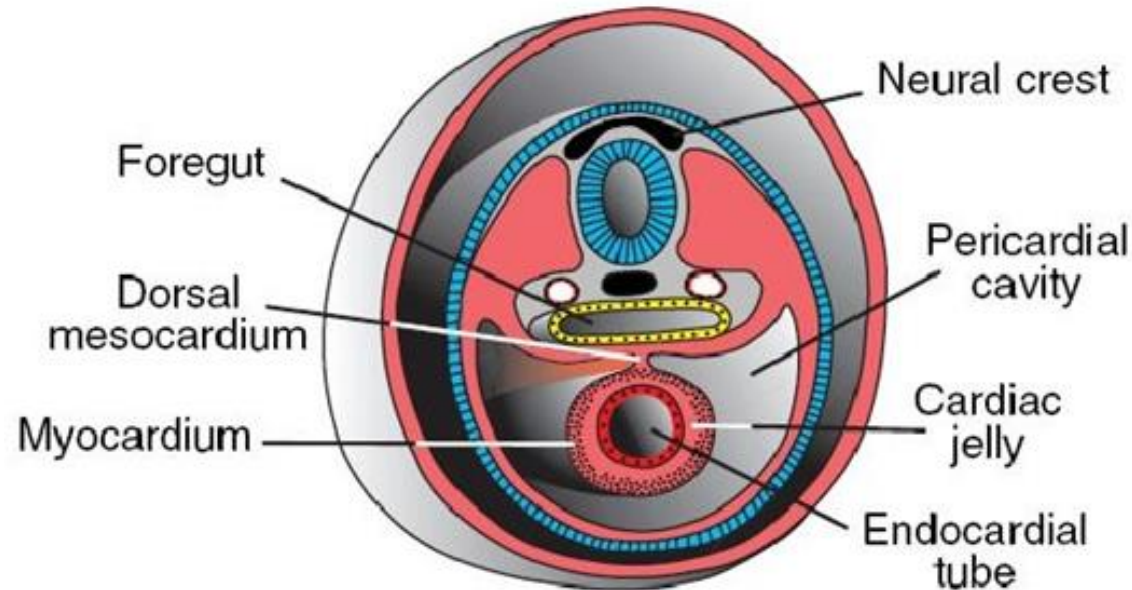


Development of primitive heart tube,,

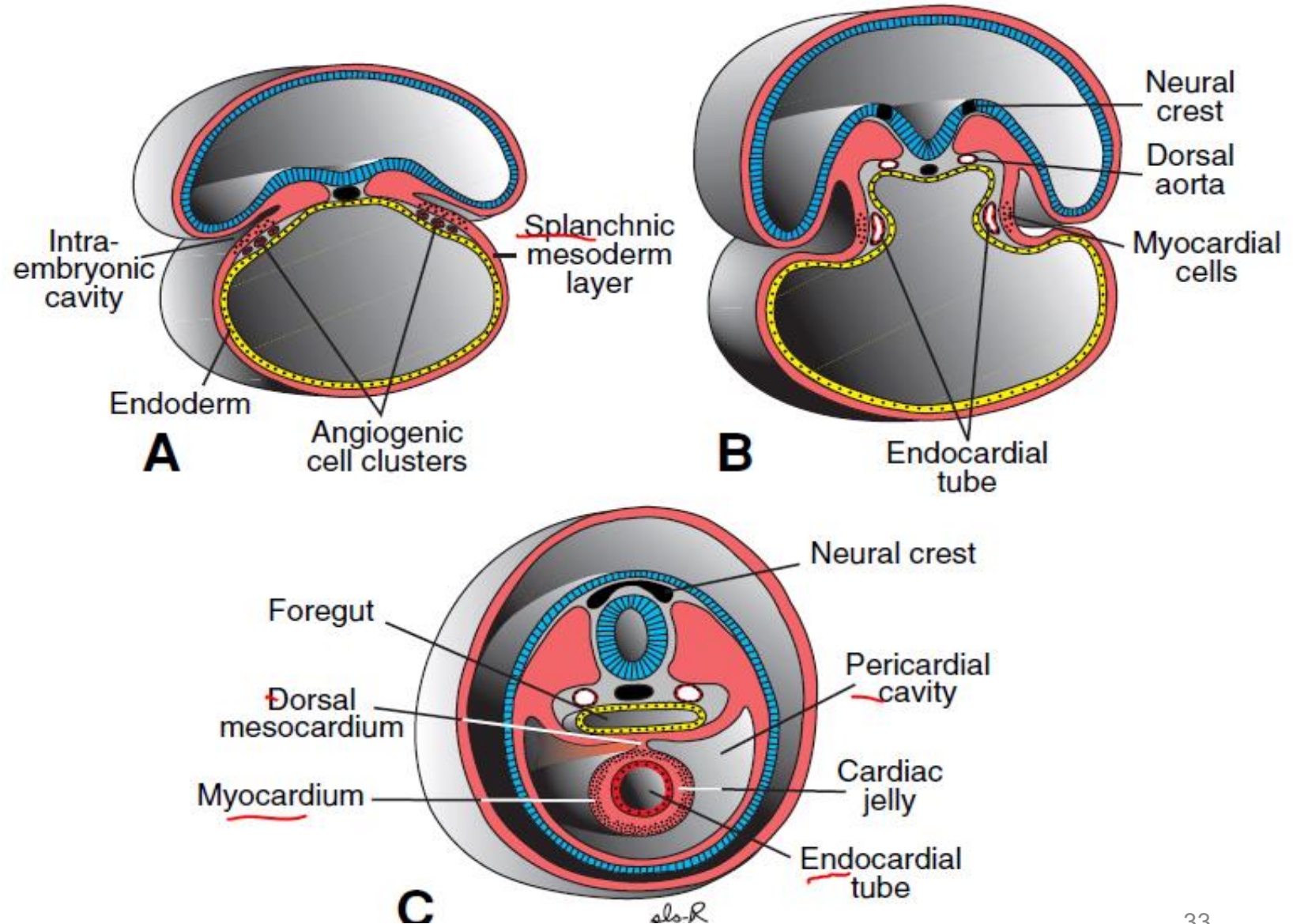
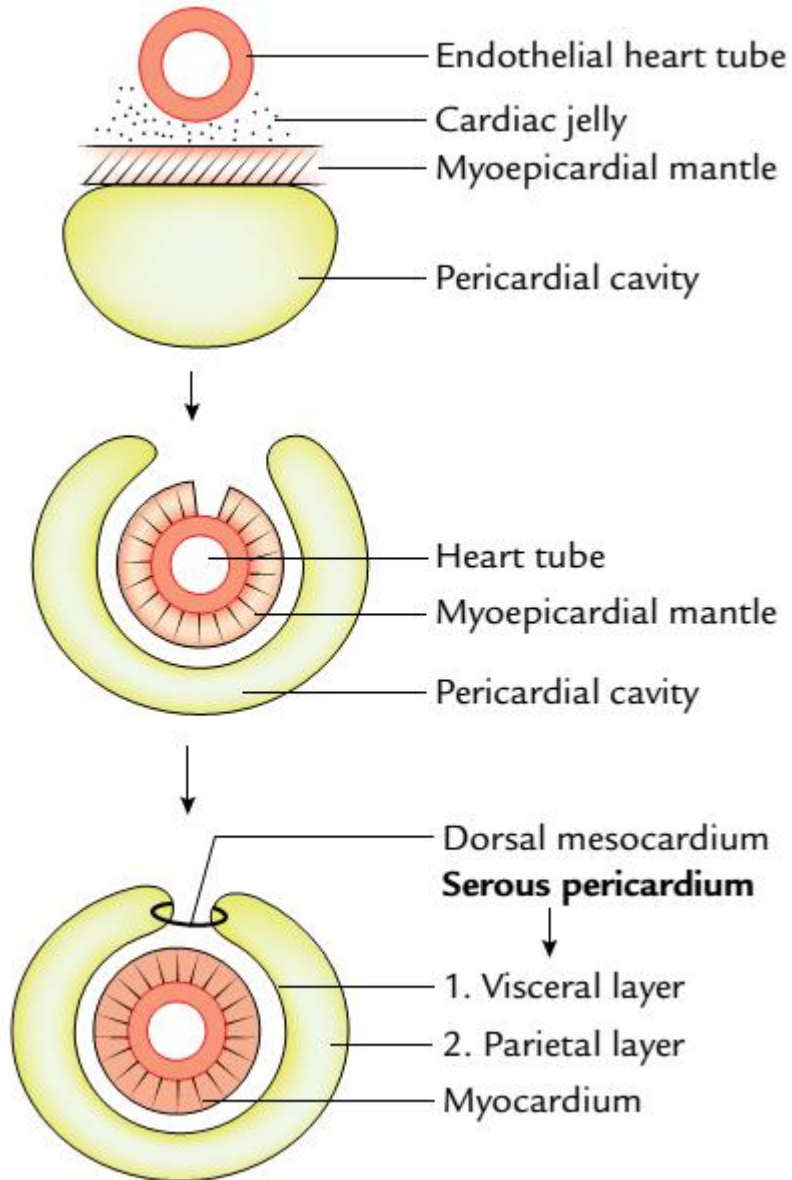


The layers of primitive heart wall

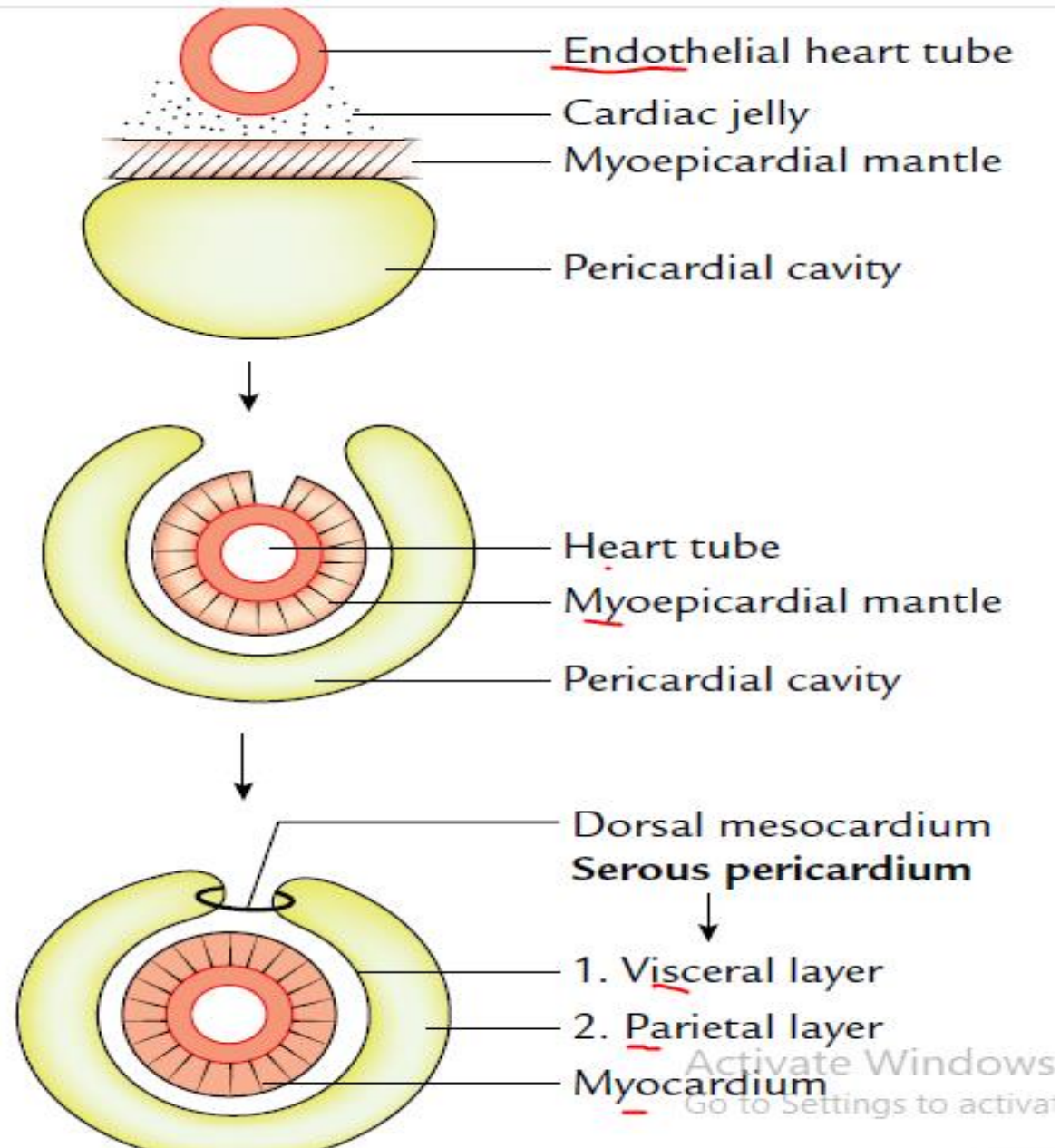
- Thin endothelial tube becomes... **internal endothelial lining** of the heart or **endocardium**.
- **Splanchnic mesoderm** surrounding the pericardial coelom becomes..... **primordial myocardium** (muscular wall of heart).
- Thin endothelial tube is separated from thick muscular tube (**myocardium**) by gelatinous C.T. (cardiac jelly).... Forming **AV septum & valves**.
- **Visceral pericardium** is derived from **mesothelial cells** and forms the **epicardium**.

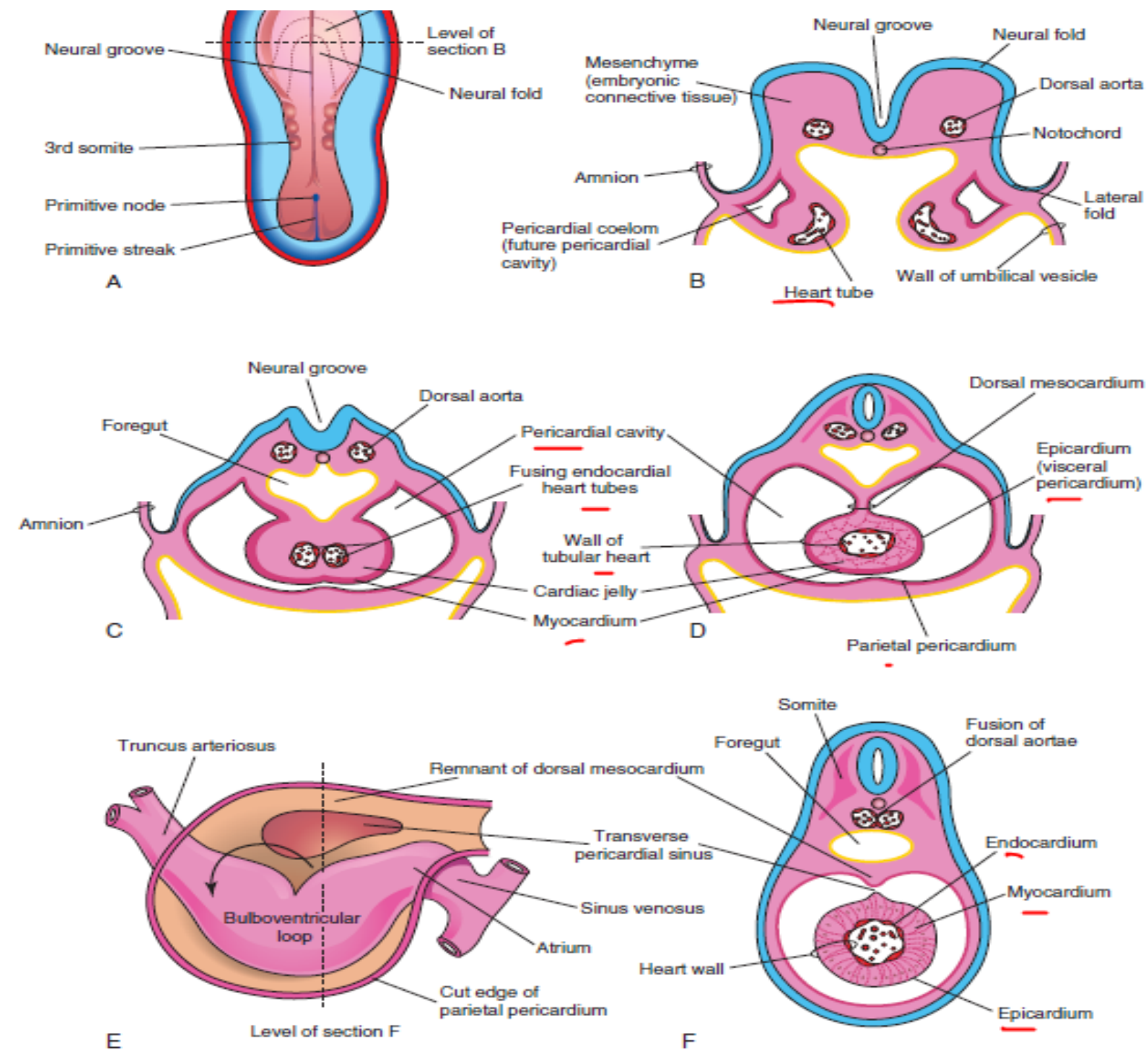


The layers of primitive heart wall,,,



The layers of primitive heart wall,,,





Dilatation of primitive heart tube

❑ The primitive heart tube elongates and develops alternate **dilatations** and **constrictions**:

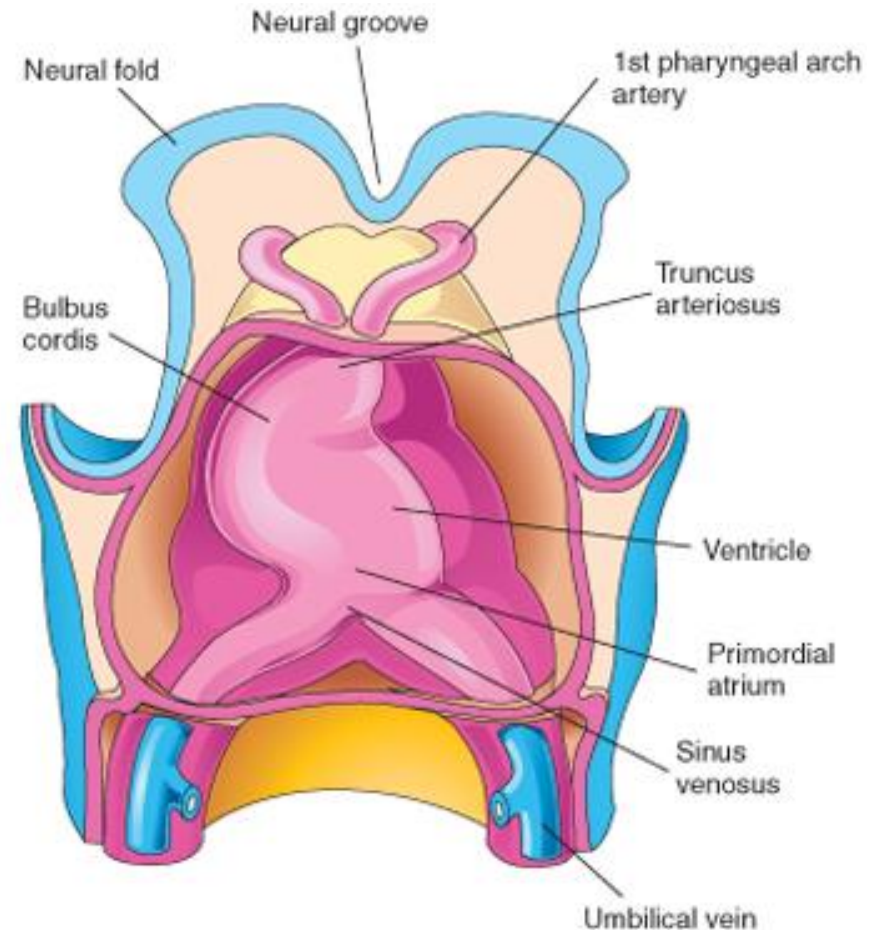
1-truncus arteriosus

2-bulbus cordis

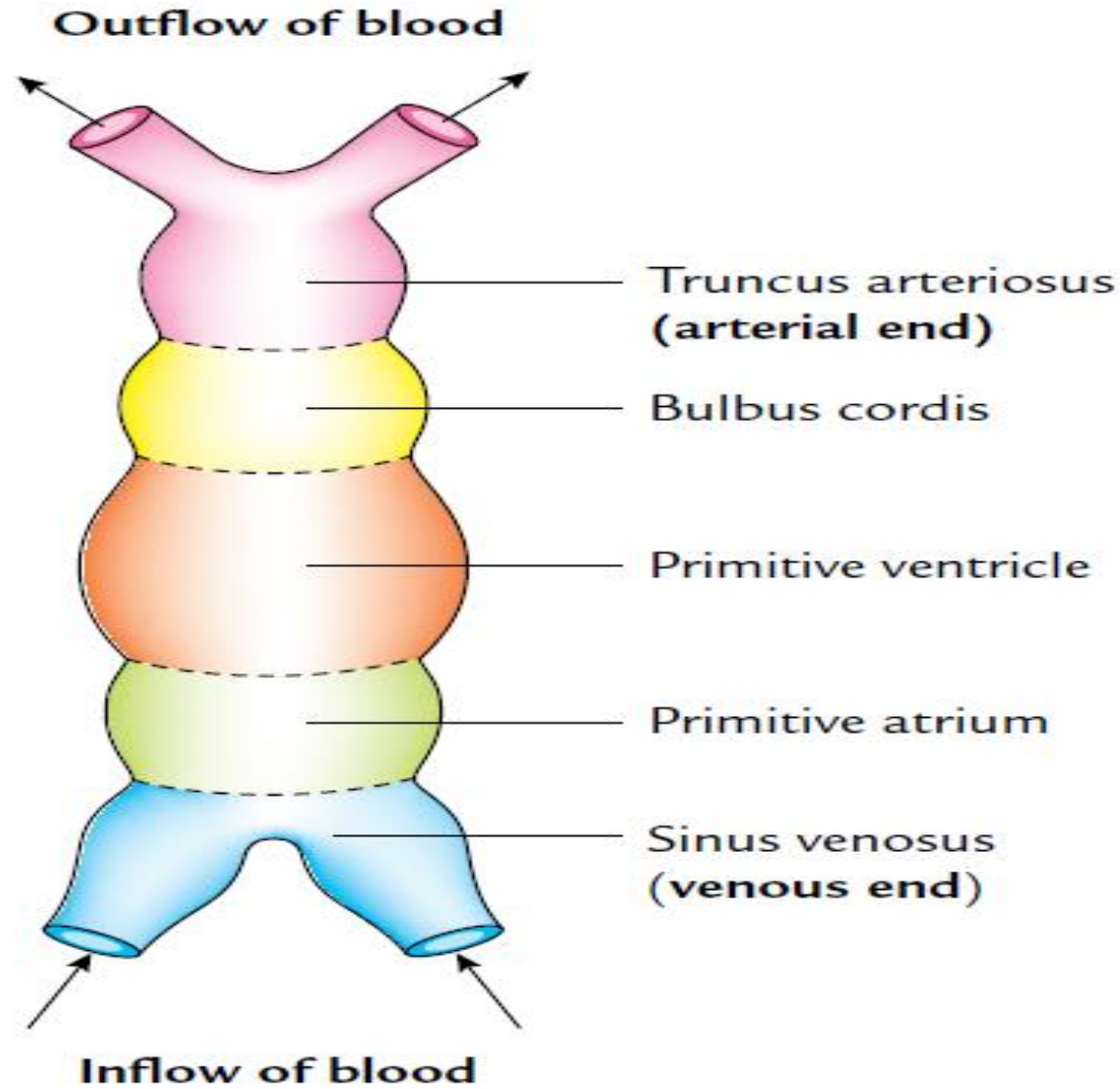
3-primitive ventricle

4-primitive atrium

5-sinus venosus

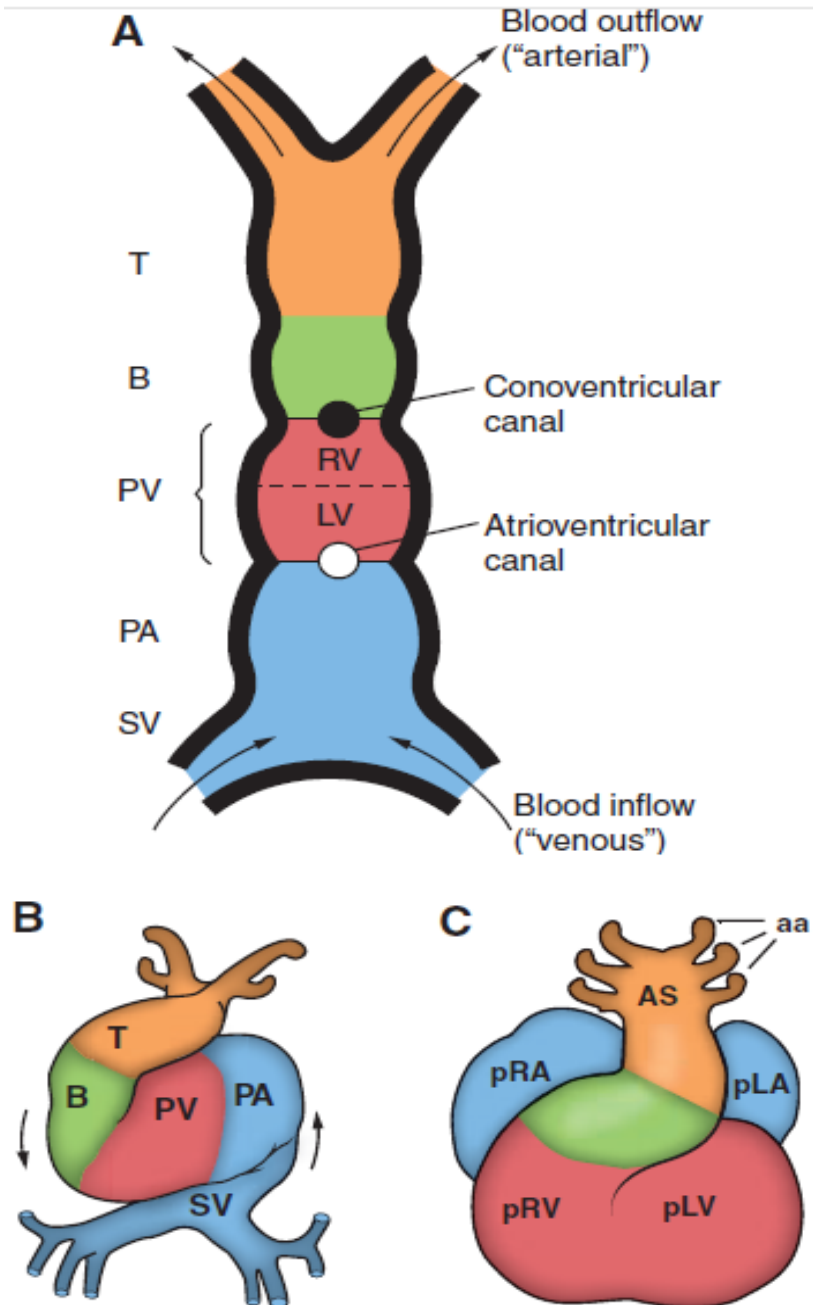


Dilatation of primitive heart tube,,,



❑ Five embryonic dilatations (subdivisions) of the heart tube.

Dilatation of primitive heart tube,,,



Embryonic Dilatation	Adult Structure
Truncus arteriosus (T)	Aorta Pulmonary trunk
Bulbus cordis (B)	Smooth part of right ventricle (conus arteriosus) Smooth part of left ventricle (aortic vestibule)
Primitive ventricle (PV)	Trabeculated part of right ventricle (RV) Trabeculated part of left ventricle (LV)
Primitive atrium (PA)	Trabeculated part of right atrium Trabeculated part of left atrium
Sinus venosus (SV)	Smooth part of right atrium (sinus venarum) ^a Coronary sinus Oblique vein of left atrium

^a The smooth part of the left atrium is formed by incorporation of parts of the **pulmonary veins** into the atrial wall. The junction of the trabeculated and smooth part of the right atrium is called the **crista terminalis**.

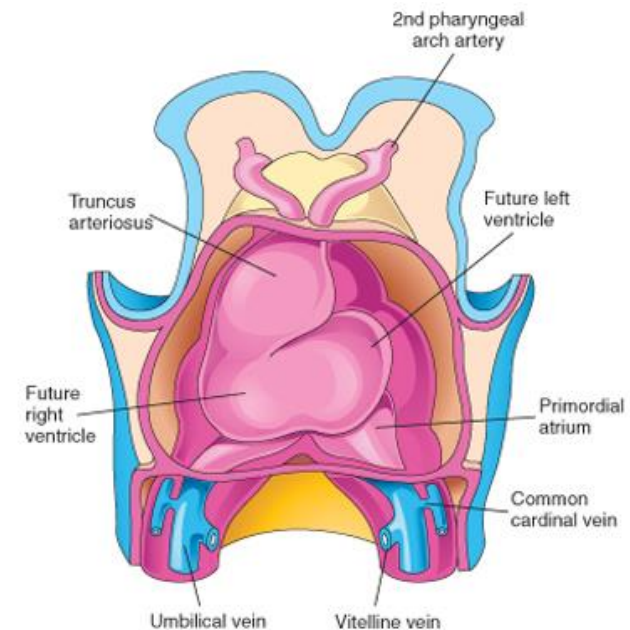
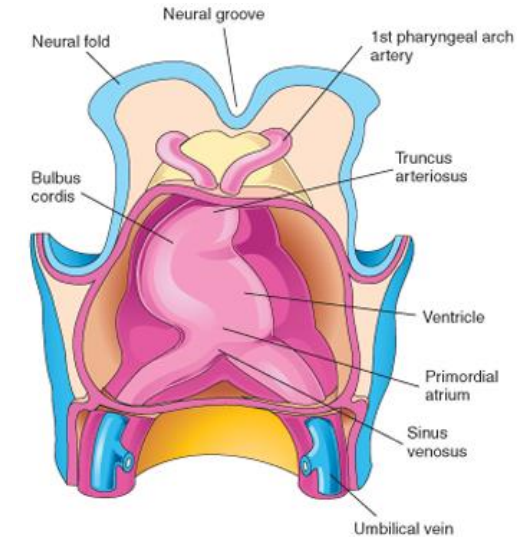
Fate of the embryonic dilatations of the primitive heart tube

Dilatation of primitive heart tube,,,

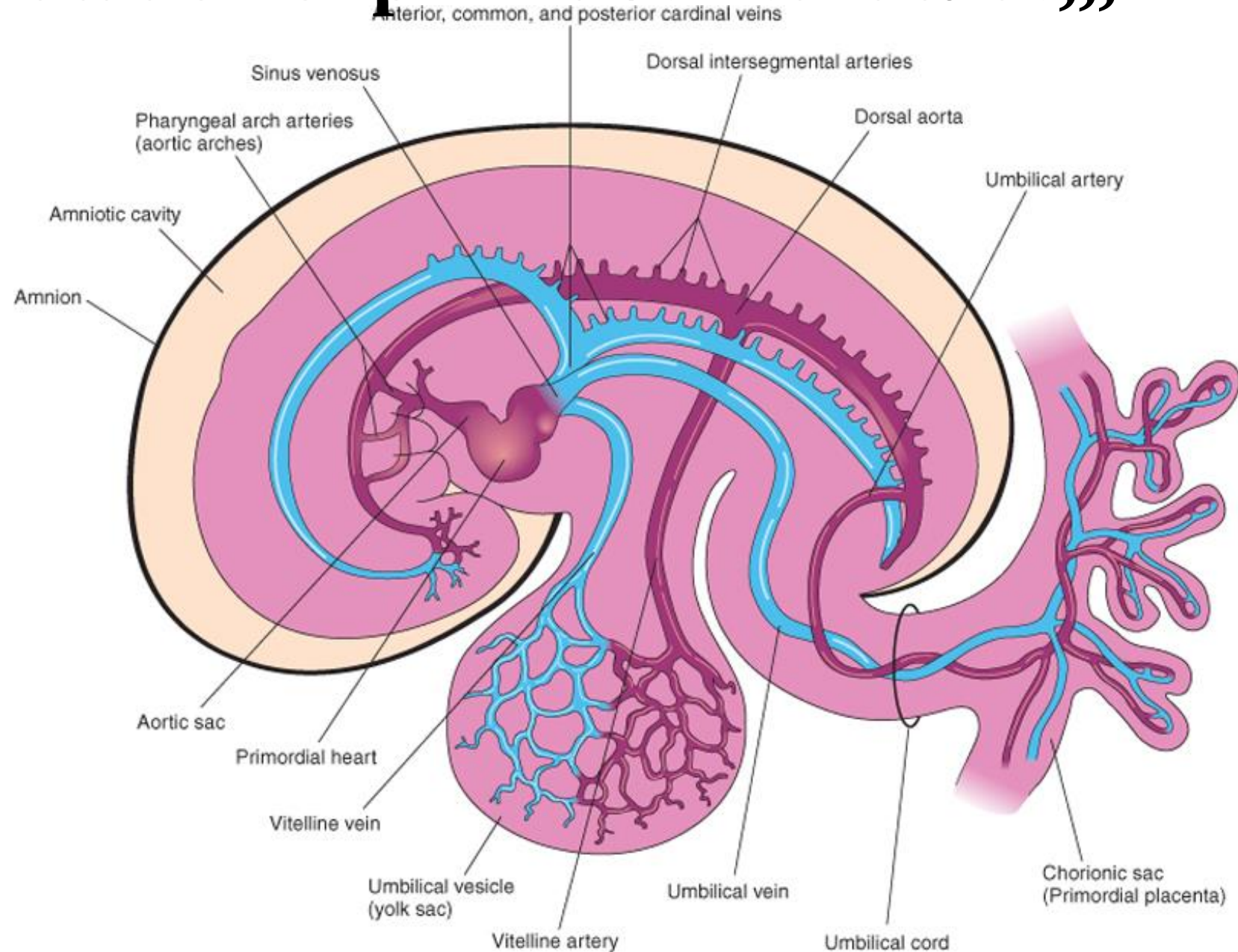
➤ **Truncus arteriosus** is continuous cranially with aortic sac, from which pharyngeal arch arteries arise

➤ **Sinus venosus** has right & left horns

➤ Each horn receives umbilical, vitelline & common cardinal veins from the placenta, umbilical vesicle & embryo, respectively.

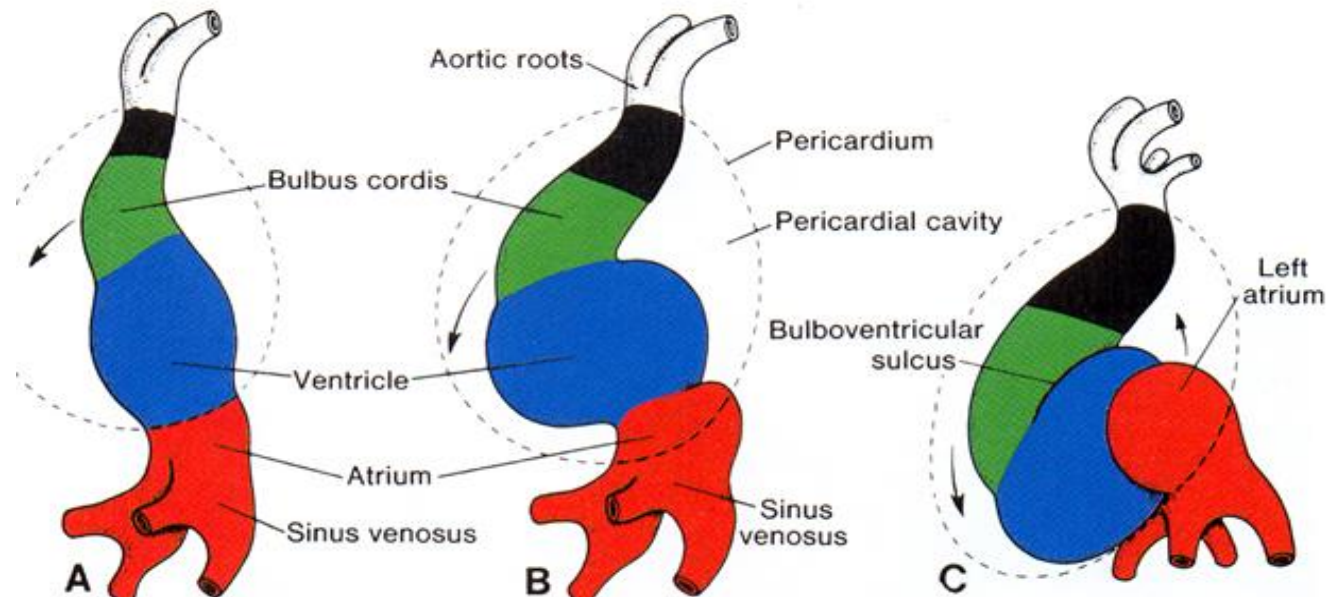


Dilatation of primitive heart tube,,

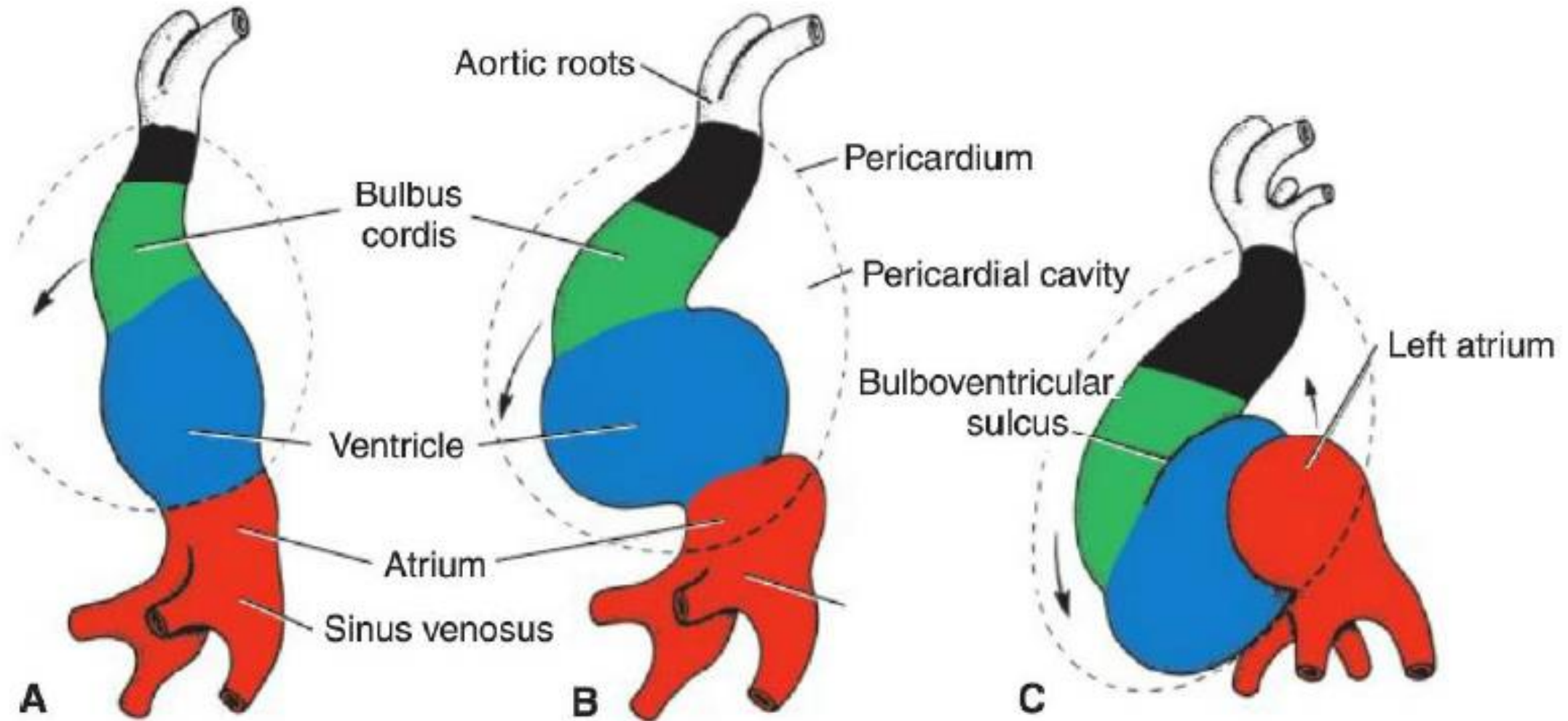


Dilatation of primitive heart tube,,

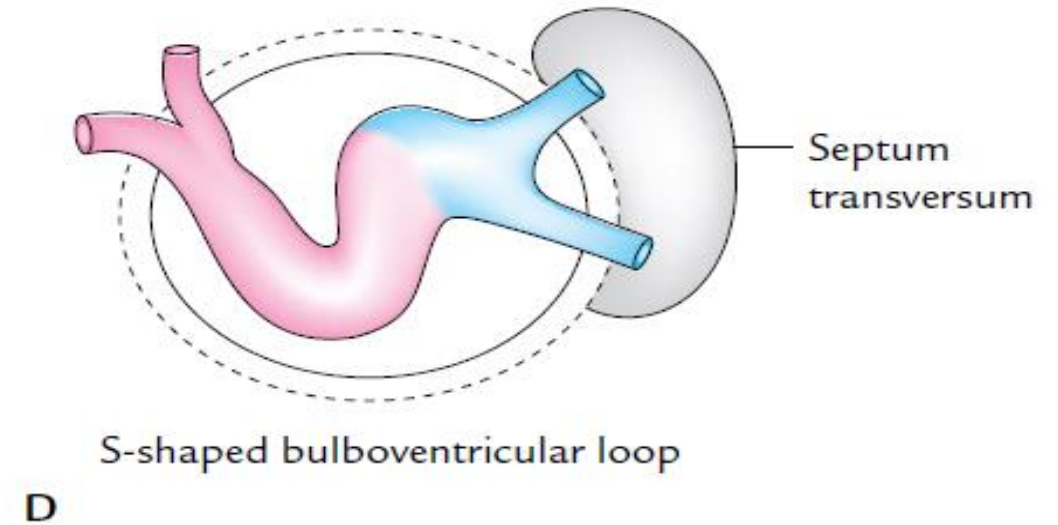
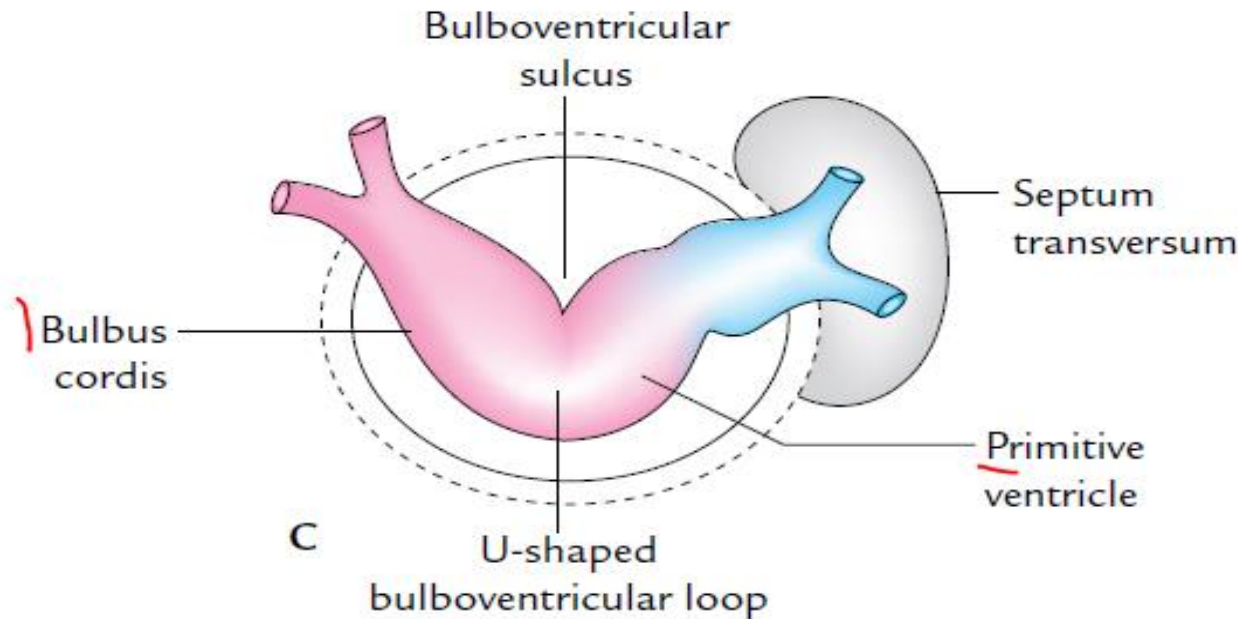
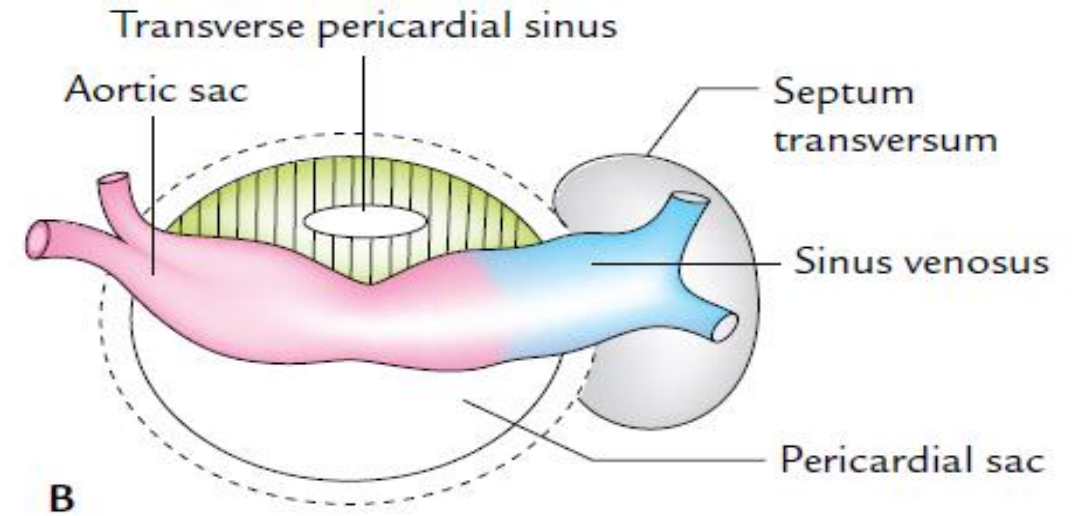
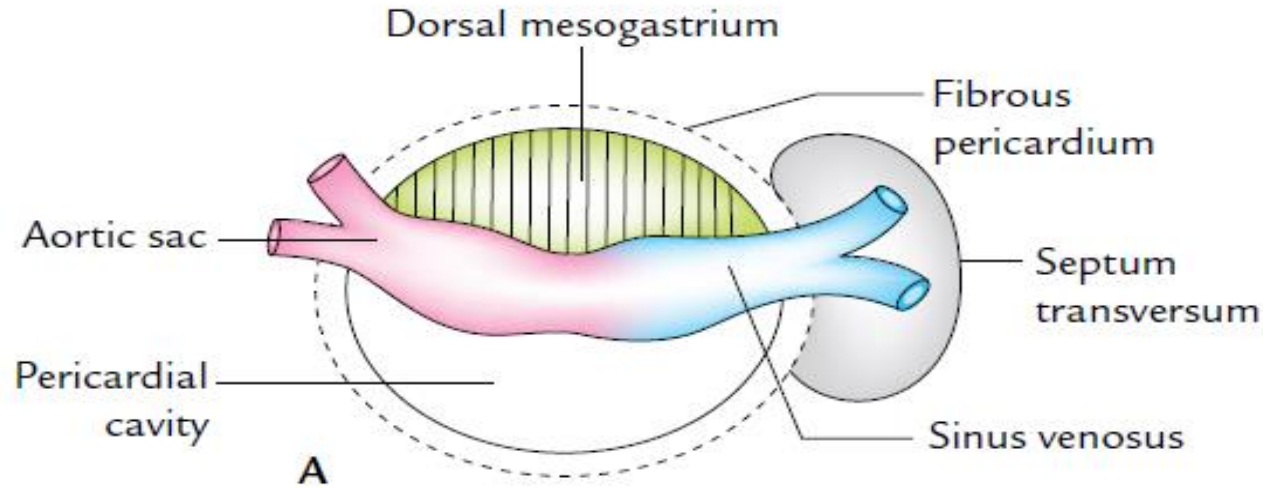
- As the development progresses, the heart tube particularly **bulbus cordis** and **primitive ventricle** grows **rapidly** in a limited space dorsal to the pericardial cavity.
- The heart tube continues to elongate and bends upon itself (on day 23) giving rise an S-shaped heart, then U-shaped.
- Bulbus cordis & ventricle grow faster than other regions, so the heart bends upon itself forming U-shaped **bulboventricular loop** (by the end of 4th week).
- The **atrium** & **sinus venosus** also come to lie dorsal to **truncus arteriosus, bulbus cordis.** & **ventricle.**



Dilatation of primitive heart tube,,,



Dilatation of primitive heart tube,,,



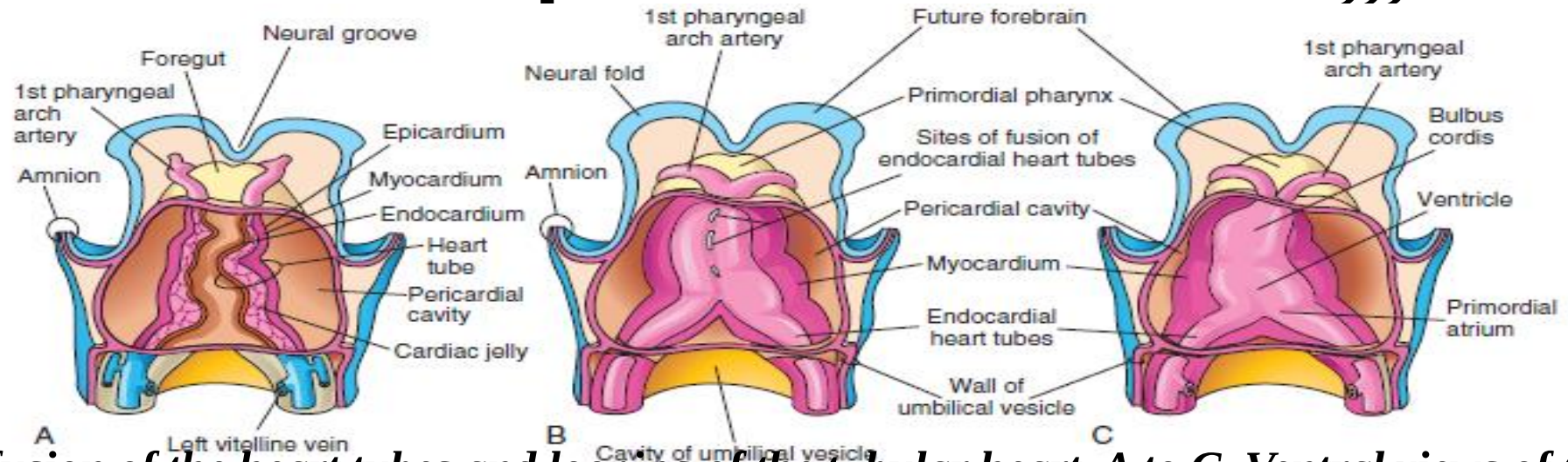
Activate V

Invagination and bonding of the heart tube within pericardial cavity.

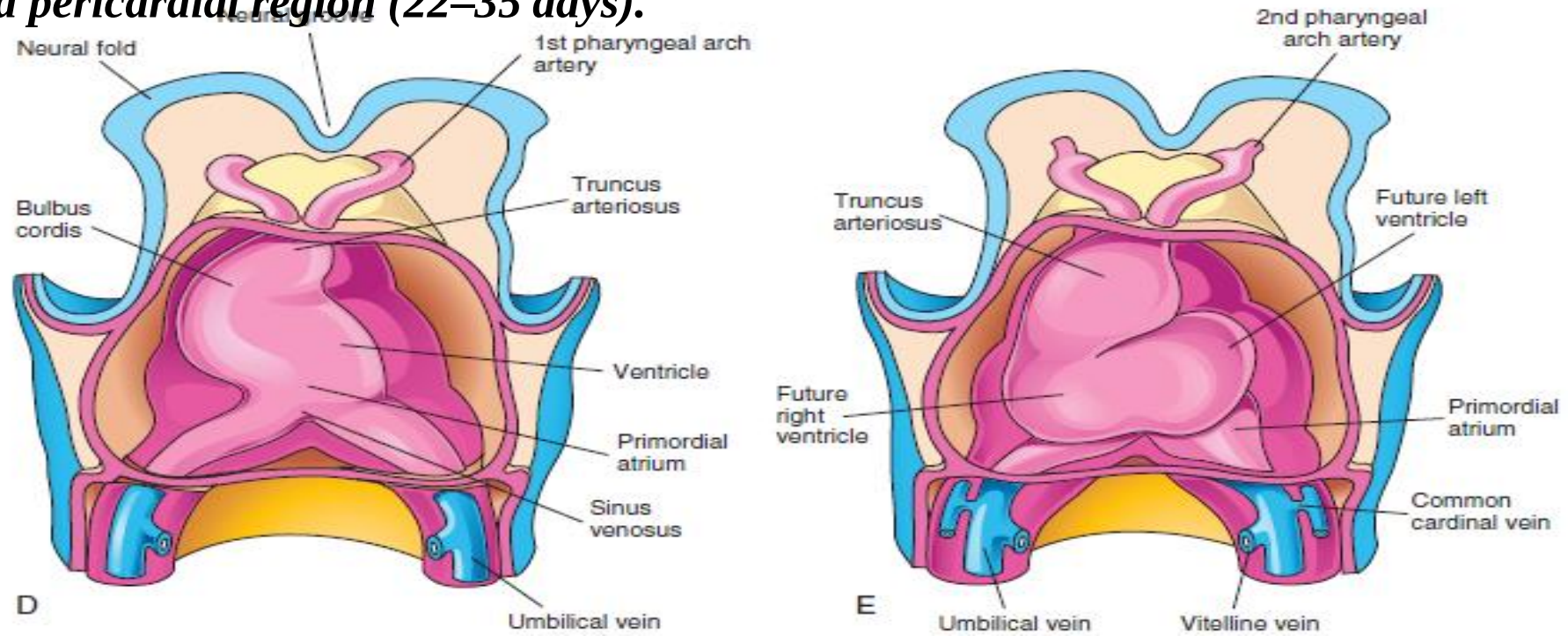
Dilatation of primitive heart tube,,

- Initially the bulboventricular loop is suspended from the dorsal wall of the pericardial sac by a mesentery called **dorsal mesocardium**, but the central part of this mesentery soon degenerates forming a communication between the right and left sides of the pericardial cavity—the **transverse pericardial sinus**.
- As the transverse pericardial sinus forms, the bulboventricular loop lies free in the pericardial cavity, but **the primitive atrium and sinus venosus** lies outside the pericardial cavity embedded in the septum transversum.

Dilatation of primitive heart tube,,



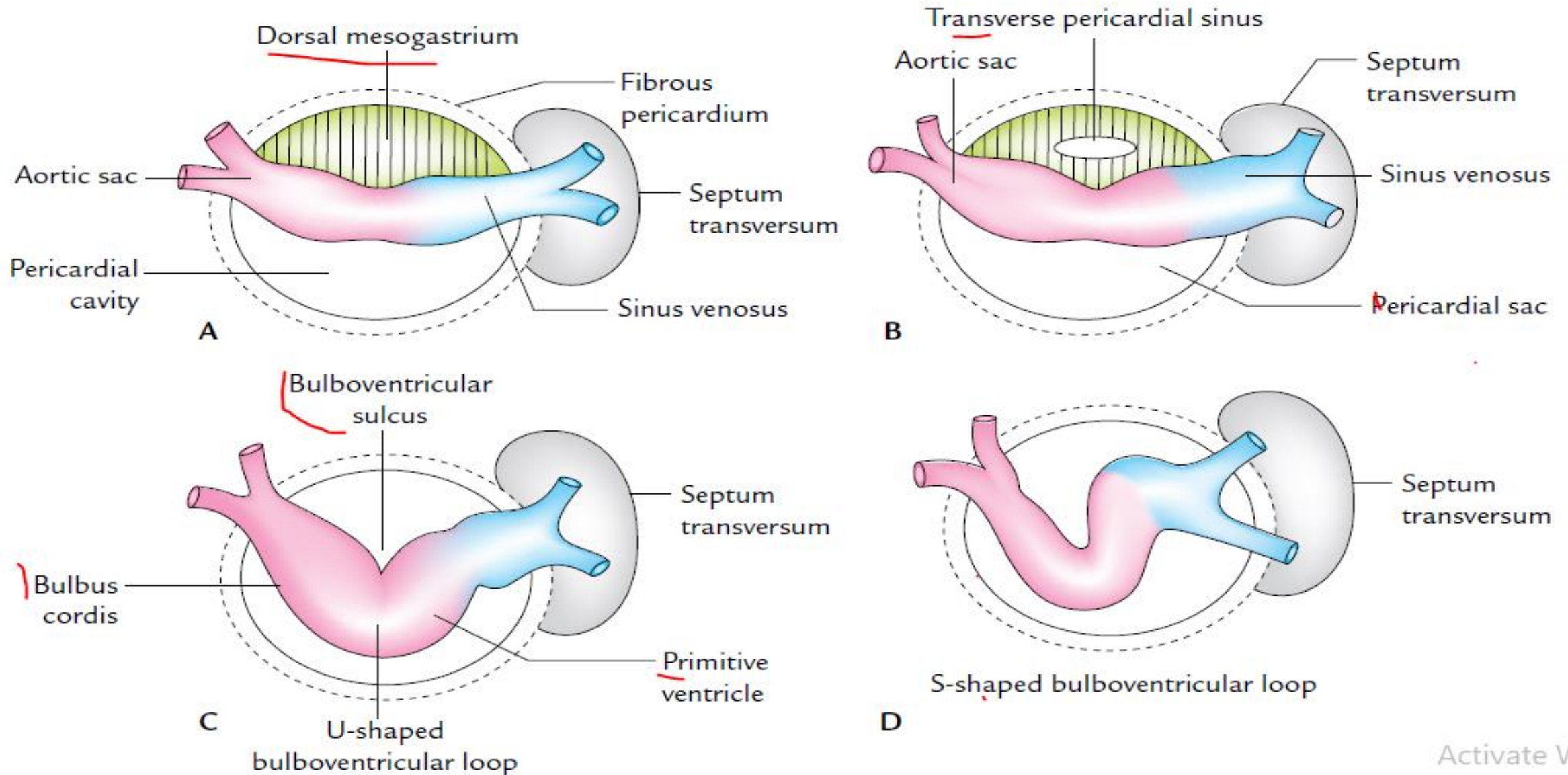
Drawings showing fusion of the heart tubes and looping of the tubular heart. A to C, Ventral views of the developing heart and pericardial region (22–35 days).



Dilatation of primitive heart tube,,

- As the primitive atrium and sinus venosus get freed from the septum transversum, they enter in the pericardial cavity and occupy position posterosuperior to the ventricle.
- The heart tube now becomes S shaped.
- **The bulboventricular sulcus disappears, and bulbus cordis and primitive ventricle together form a common chamber.**
- The atrial chamber that lies behind the above truncus arteriosus expands.
- As it does so parts of it come to project forwards on either side of truncus arteriosus as auricles.

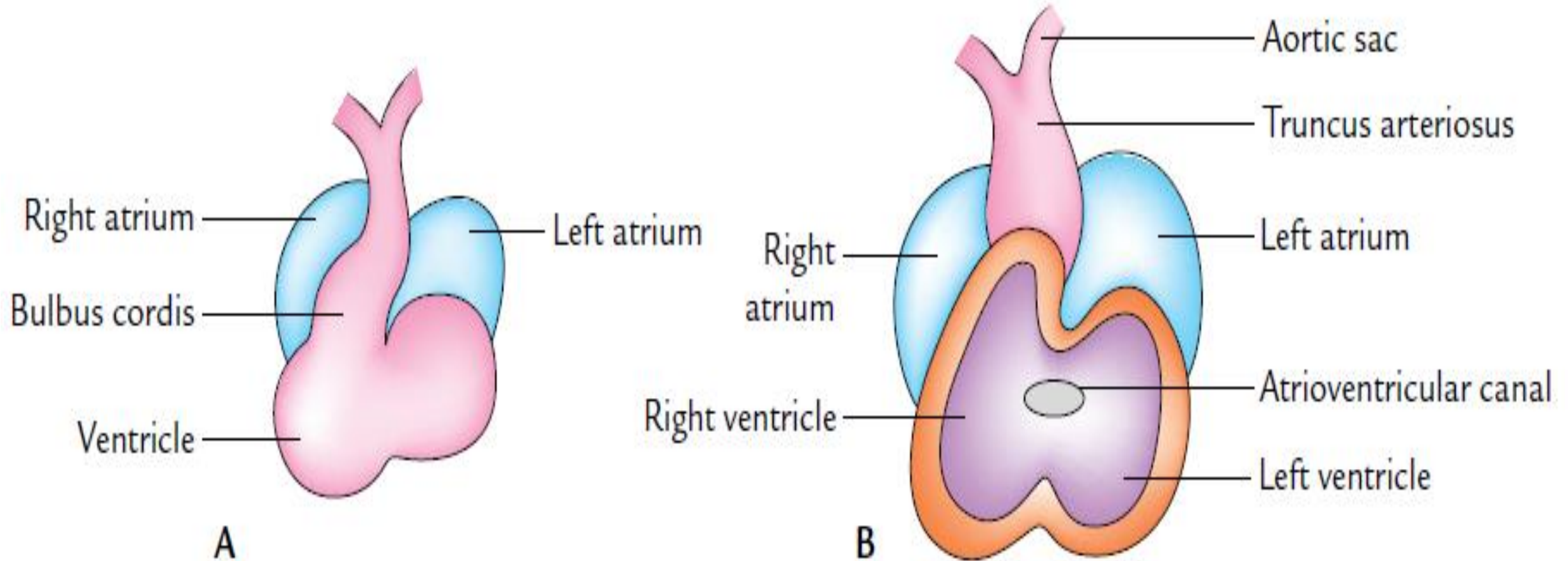
Dilatation of primitive heart tube,,,



Activate V

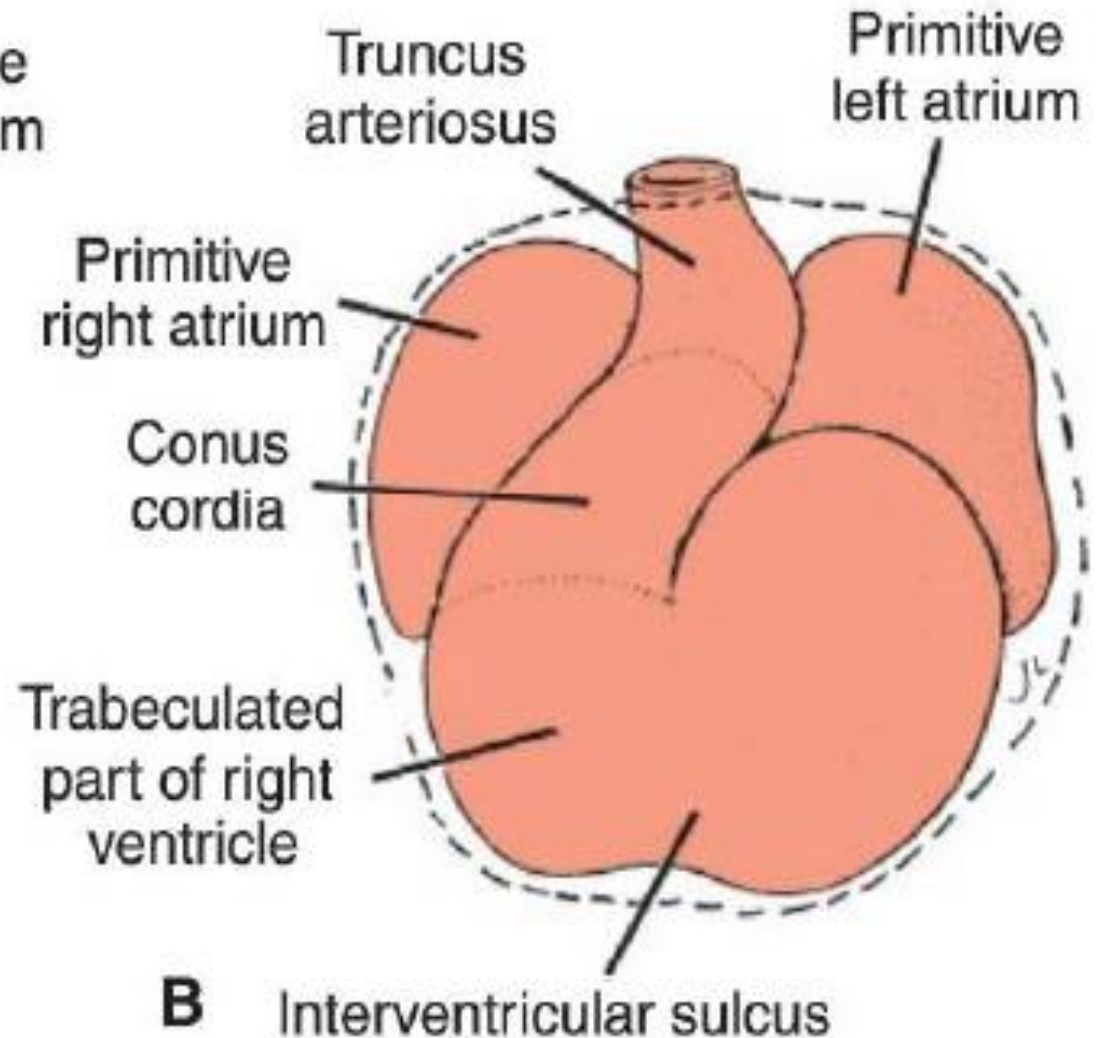
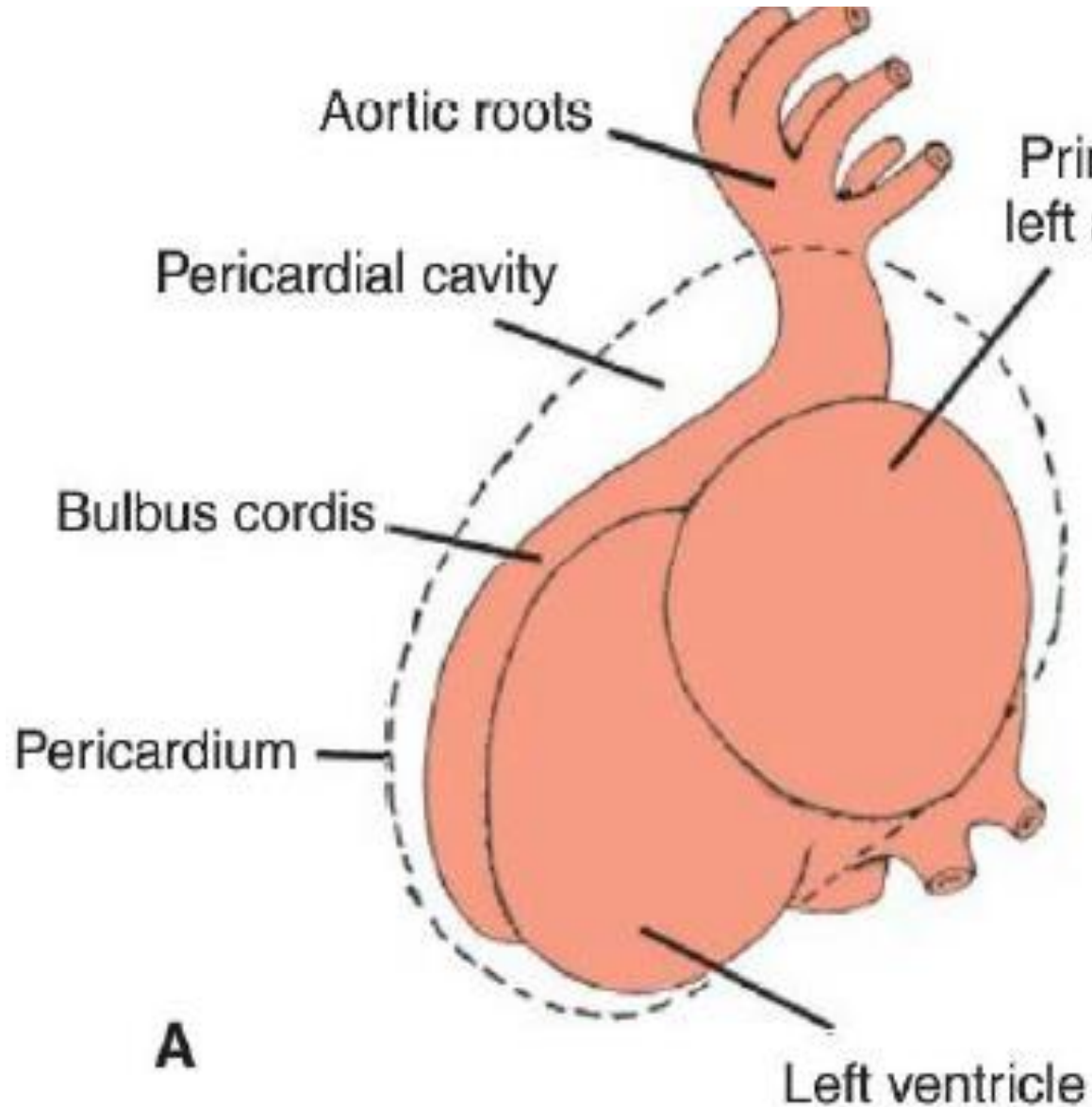
Invagination and bonding of the heart tube within pericardial cavity.

Dilatation of primitive heart tube,,



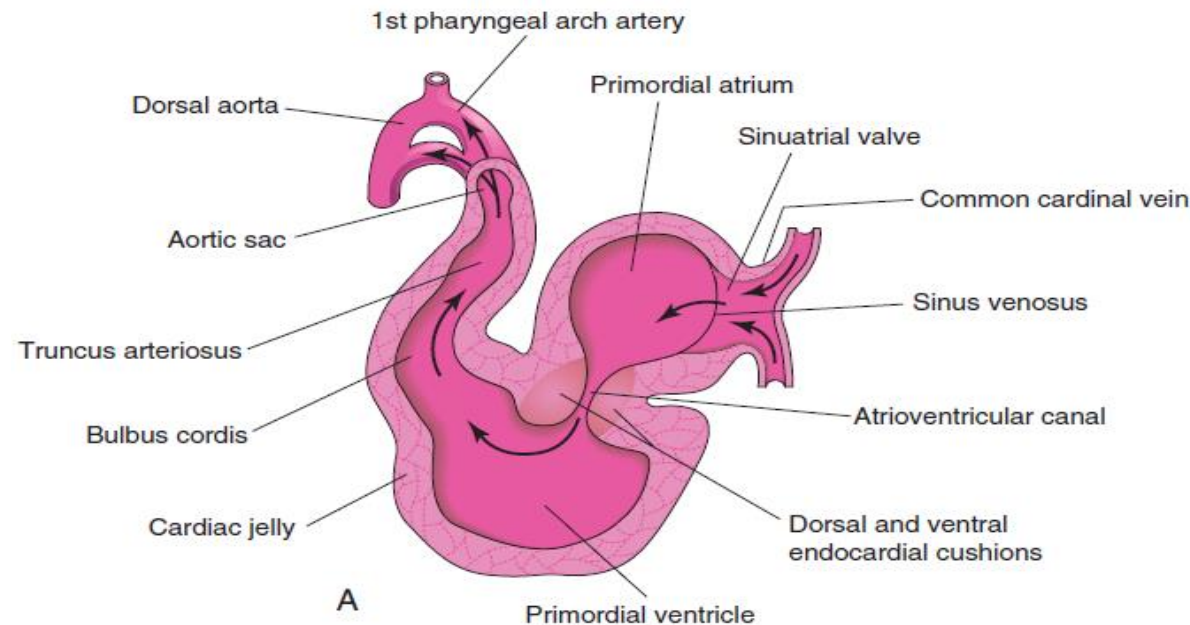
- *Establishment of the external form of the heart (A).*
- *The interiors of developing ventricles are shown on the right (B).*

Dilatation of primitive heart tube,,

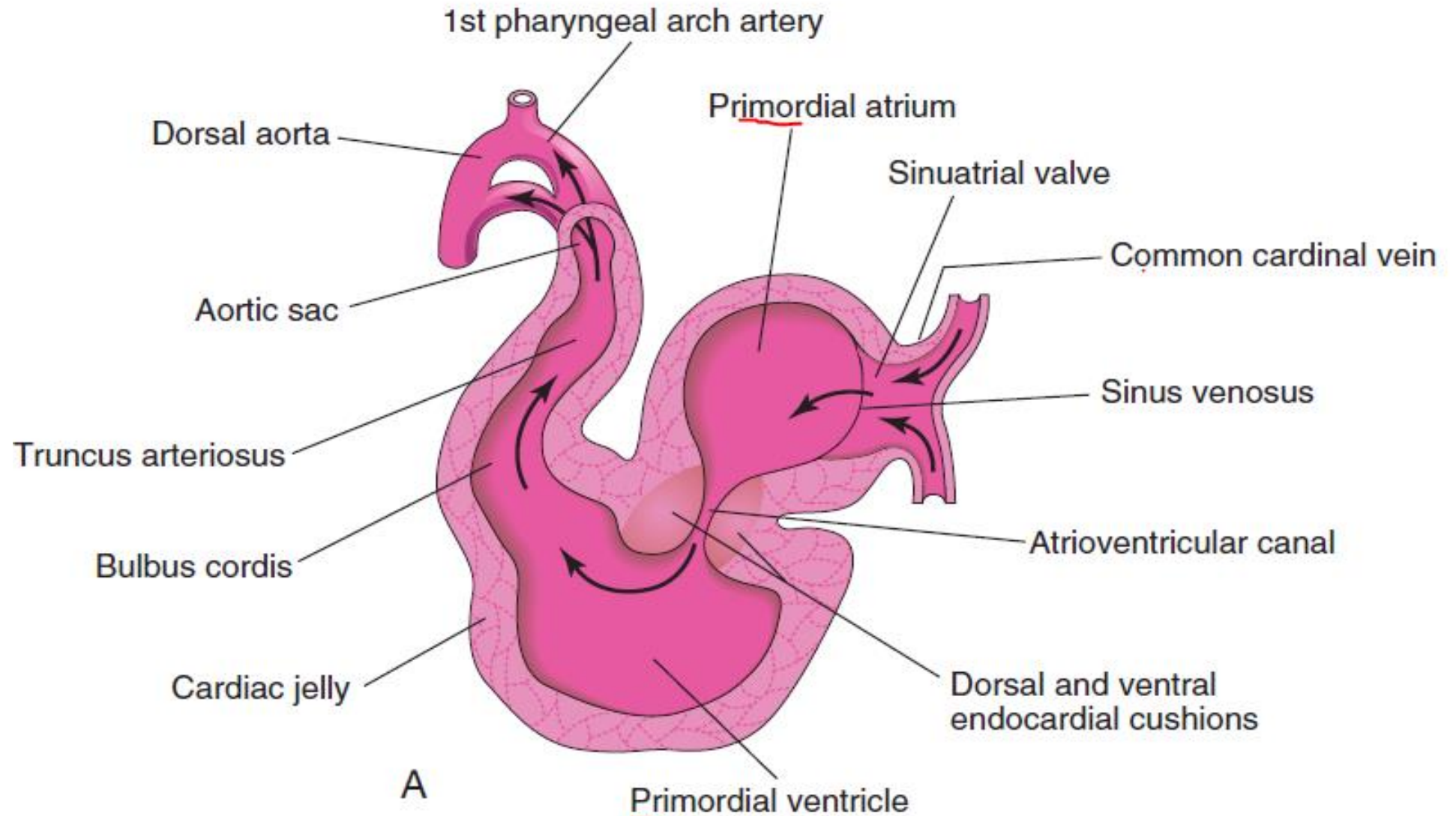


Dilatation of primitive heart tube,,

- *The heart begins to beat at 22 to 23 days*
- By the end of 4th week, unidirectional blood flow begins at sinus venosus by peristalsis- like waves.
- Blood passes through sinuatrial valves into atrium...> Atrioventricular canal ...> ventricle..> Bulbus cordis...> Truncus arteriosus... >aortic sac...> aortic arches (arterial channels) ...> 2 dorsal aortae...> into body of embryo, yolk sac and placenta.

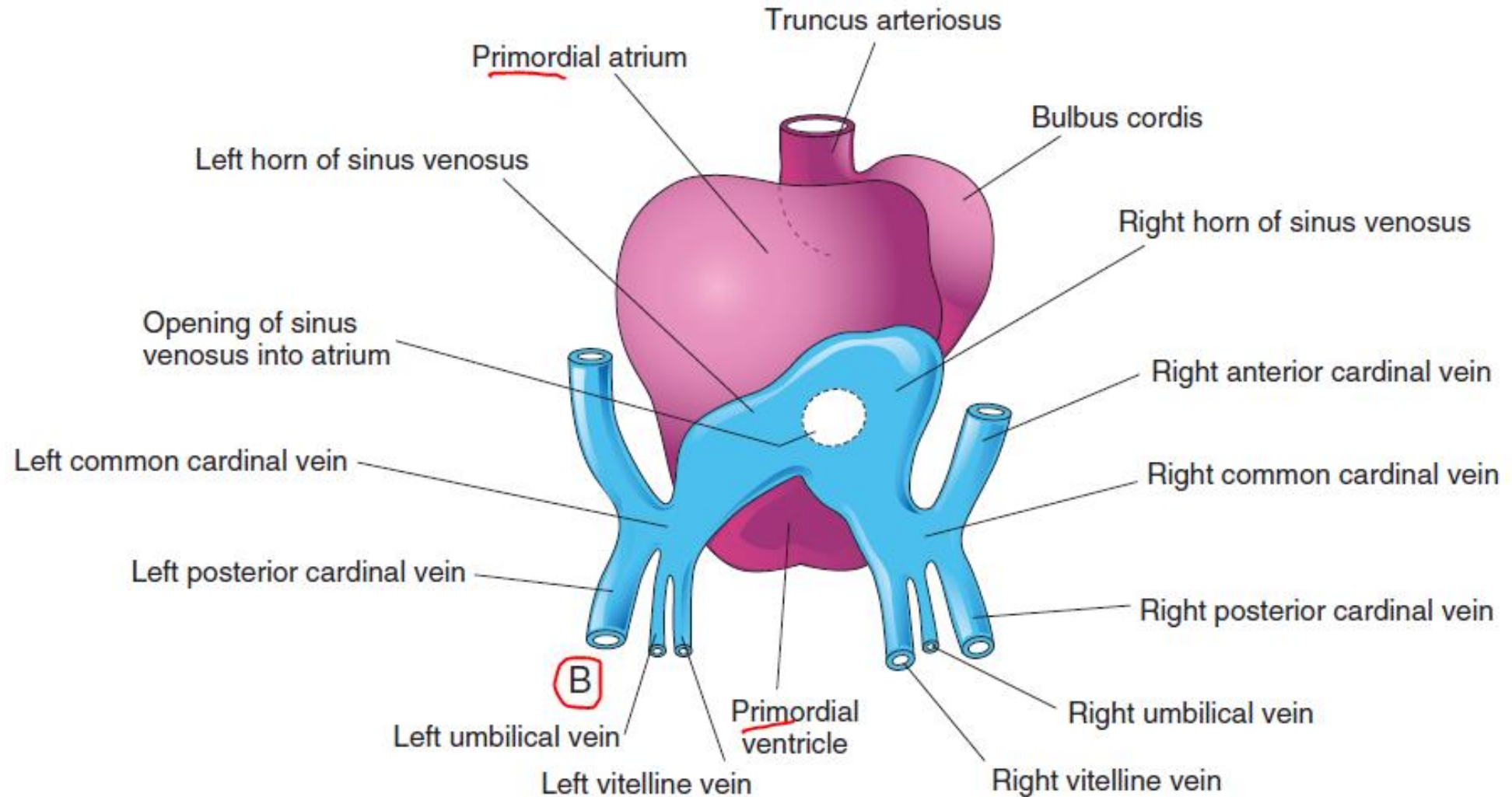


Dilatation of primitive heart tube,,,



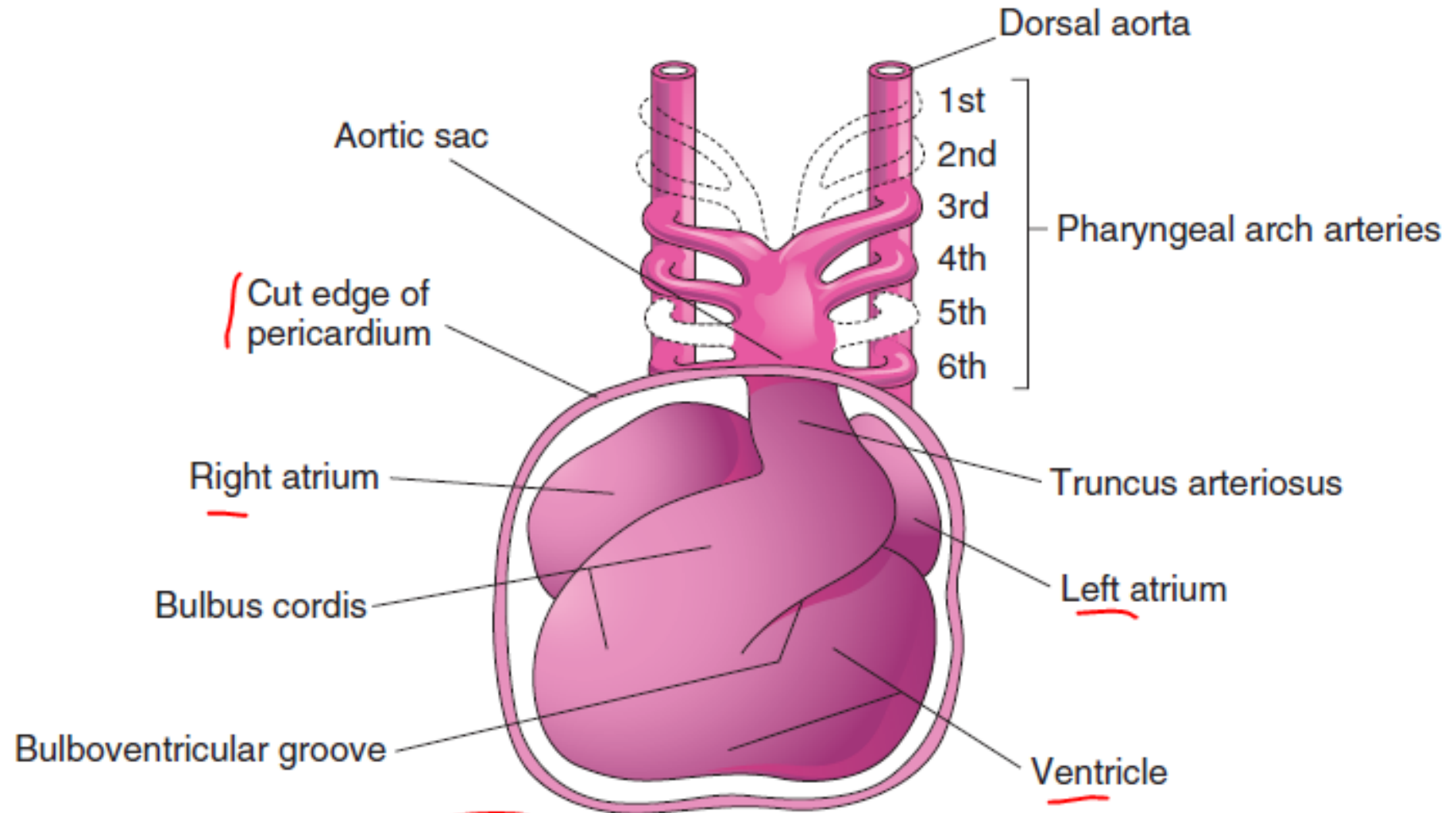
A, Sagittal section of the heart at approximately 24 days, showing blood flow through it (arrows)

Dilatation of primitive heart tube...



B, Dorsal view of the heart at approximately 26 days, showing the horns of the sinus venosus and the dorsal location of the primordial atrium.

Dilatation of primitive heart tube,,,



C, Ventral view of the heart and pharyngeal arch arteries (approximately 35 days). The ventral wall of the pericardial sac has been removed to show the heart in the pericardial cavity.

Partitioning of Primordial Heart and Development of its chambers

Partitioning of the primitive Heart

- The primitive heart tube has a single lumen.
- This lumen is partitioned into four definitive chambers by the formation of four septa.

These septa are:

I. Atrioventricular septum- separate atria from ventricles

II. Interatrial septum- separate the Rt atrium from Lt atrium

III. Interventricular septum- separate the Rt & Lt ventricle

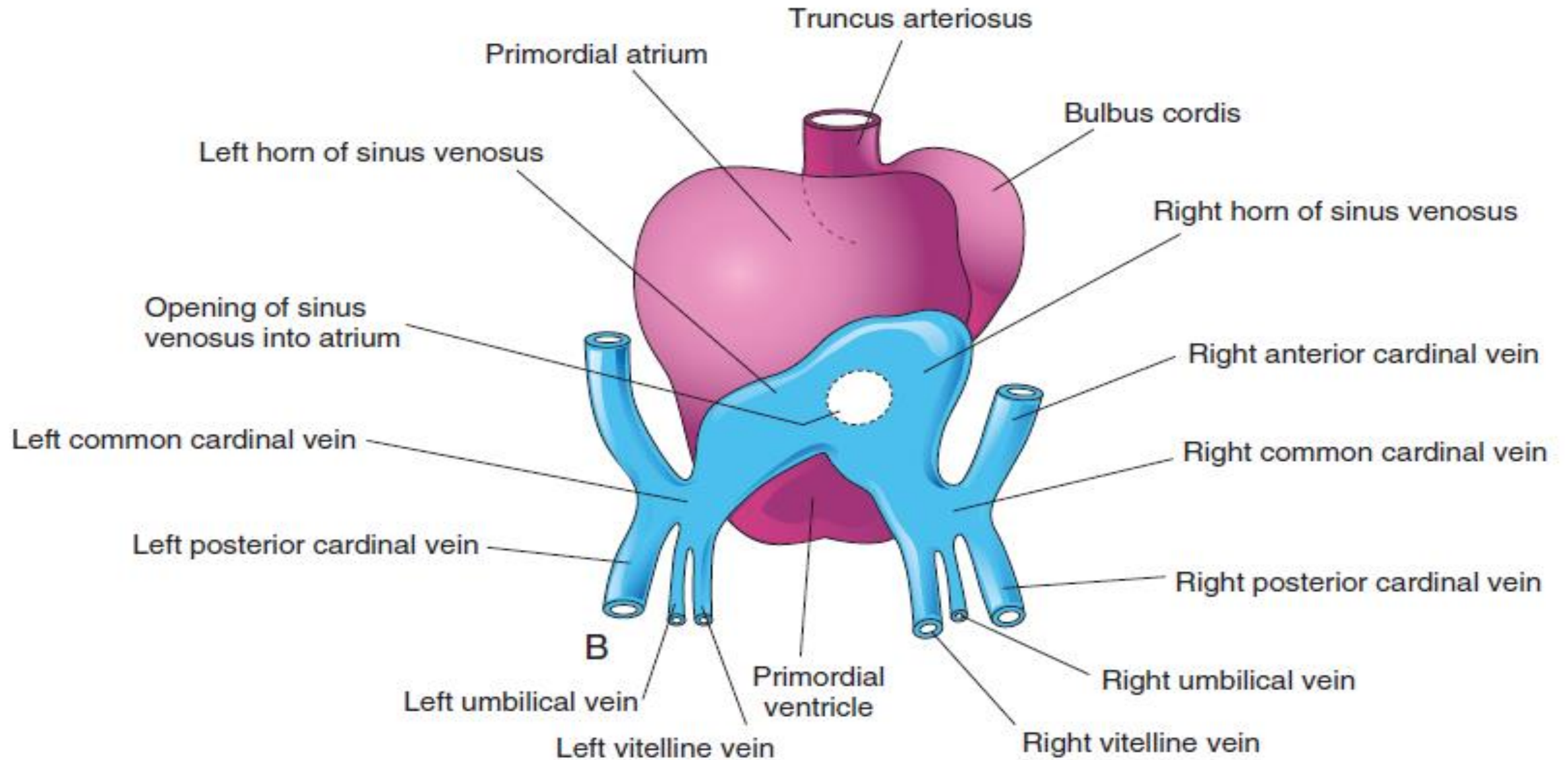
IV. Aorticopulmonary septum- separate the outflow part of Rt ventricle and pulmonary trunk from outflow of Lt ventricle and ascending aorta

- ❑ The major septa of the heart are formed b/n the 27th and 37th days of development, when the embryo grows in length from 5 mm to approximately 16 to 17 mm

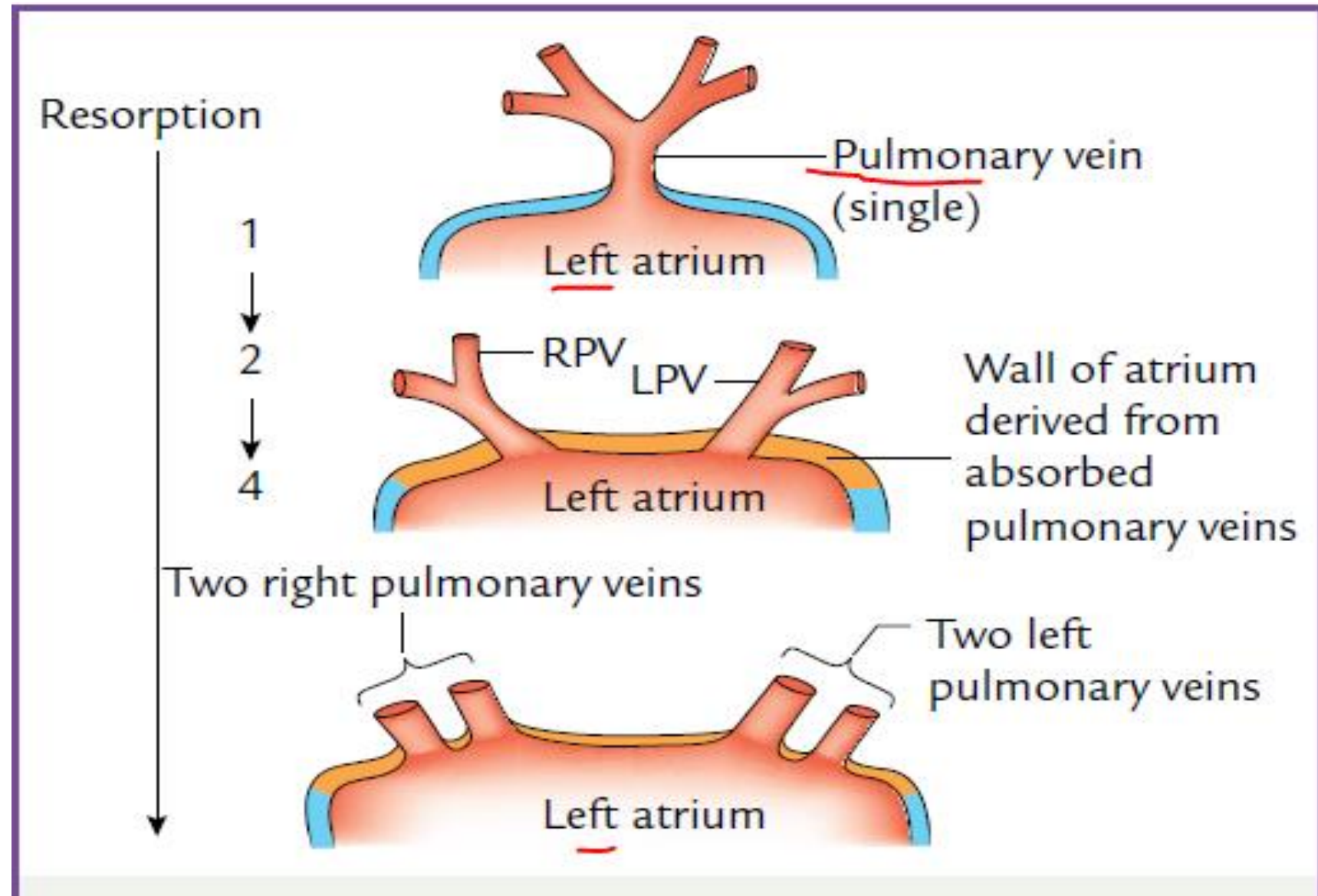
Partitioning of the primitive Heart,,,

- Partitioning of the atrioventricular canal, primordial atrium, ventricle, and outflow tract begins around the middle of the 4th week and is essentially completed by the end of the eighth week.
- In addition to the formation of **septae**, **other embryological** events such as **regression and absorption of sinus venosus into the right atrium** and **absorption of pulmonary veins into the left atrium** also help in the formation of definitive chambers of the heart.

Partitioning of the primitive Heart,,

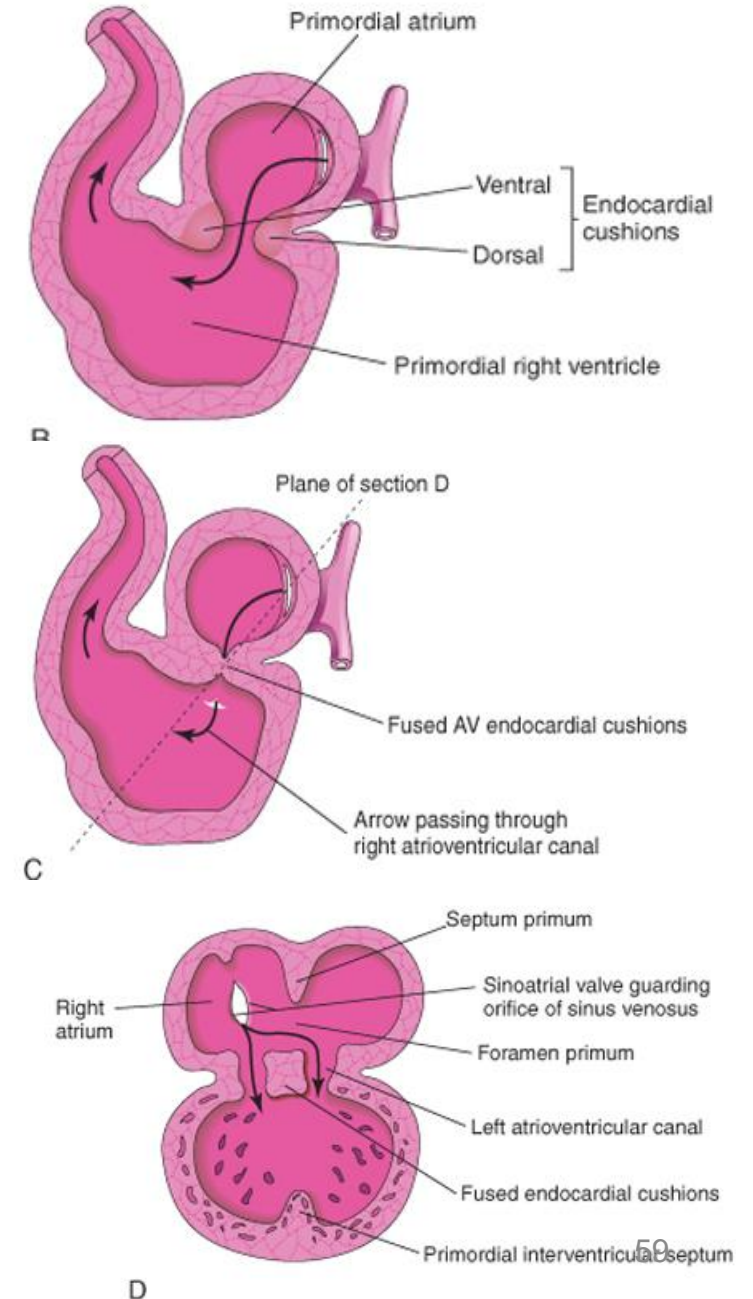


Partitioning of the primitive Heart,,

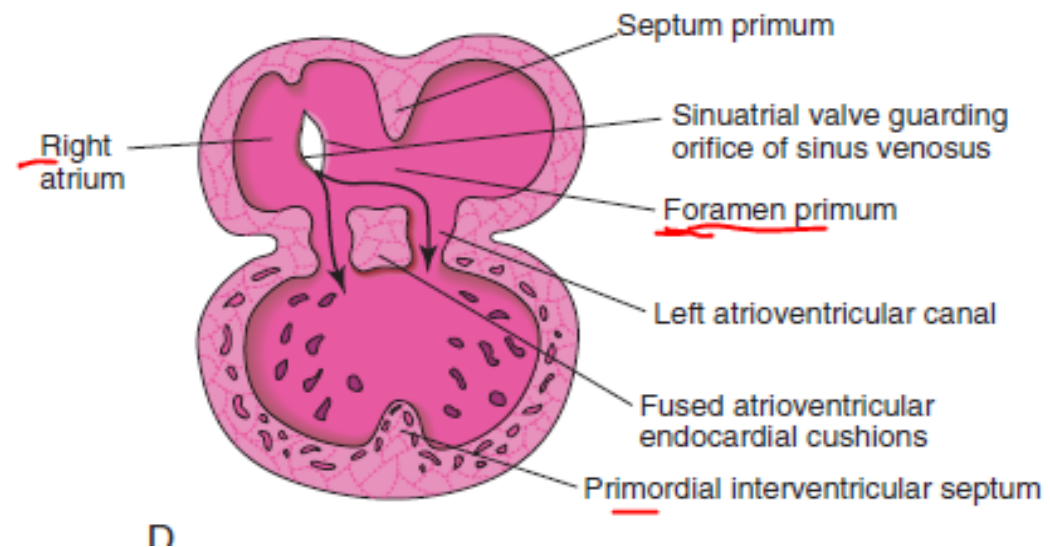
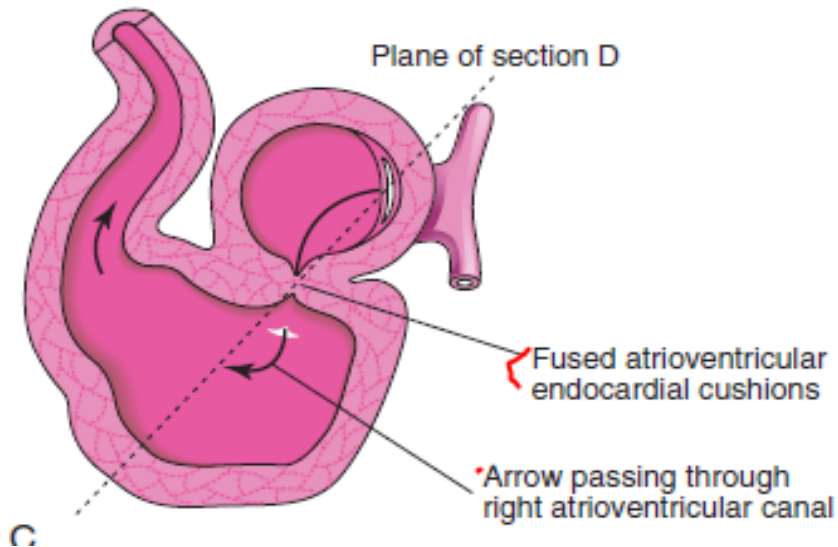
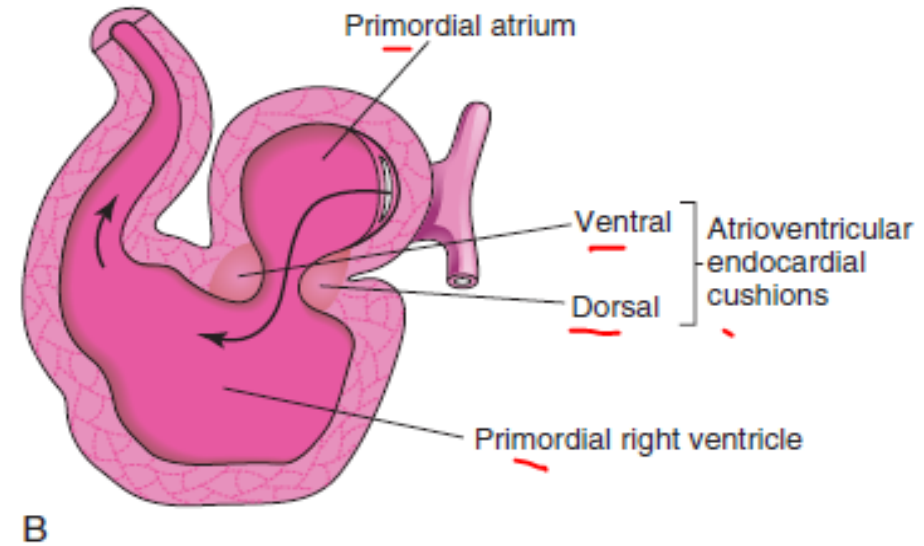
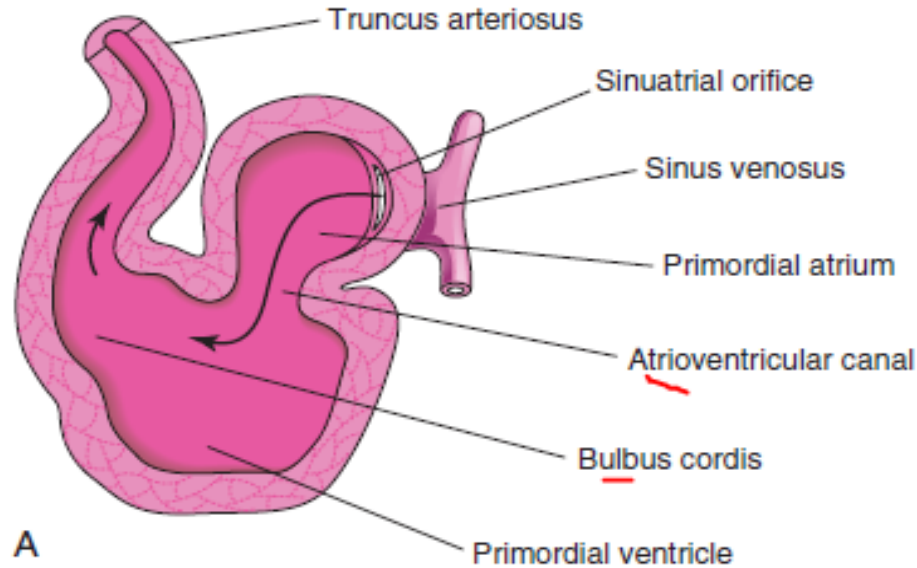


Partitioning of Atrioventricular Canal

- At the end of 4th week, 2 endocardial cushions on **dorsal & ventral** walls of atrioventricular canal, develop from mesenchymal cells of cardiac jelly. (B)
- During 5th week, the AV- endocardial cushions meet and unite in the middle line to form a septum and divide the common A-V canal into right & left A-V canals. (C,D)
- Endocardial cushions also form the AV- valves + membranous septa of interventricular septum.
- Note in D, coronal section, beginning of development of interatrial & interventricular septa.



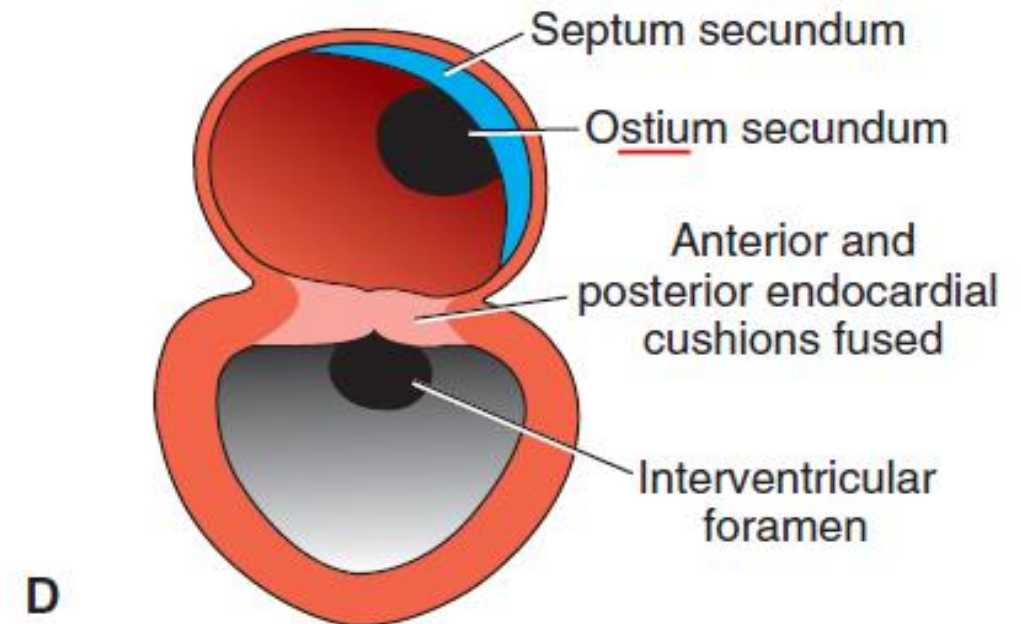
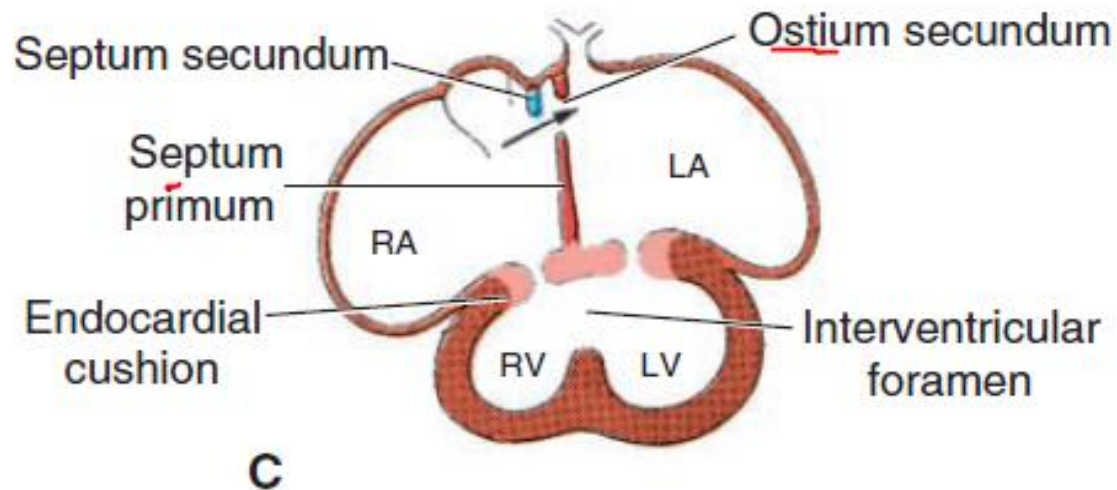
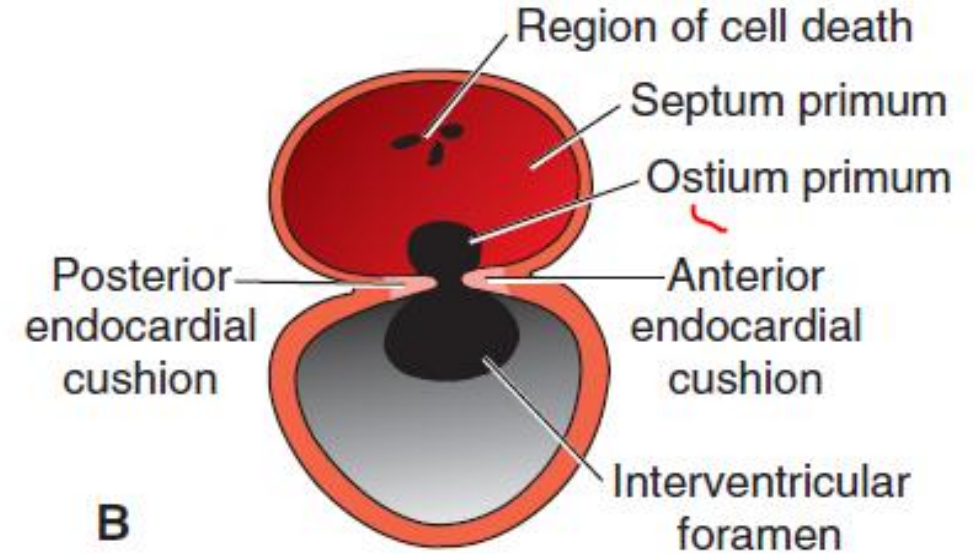
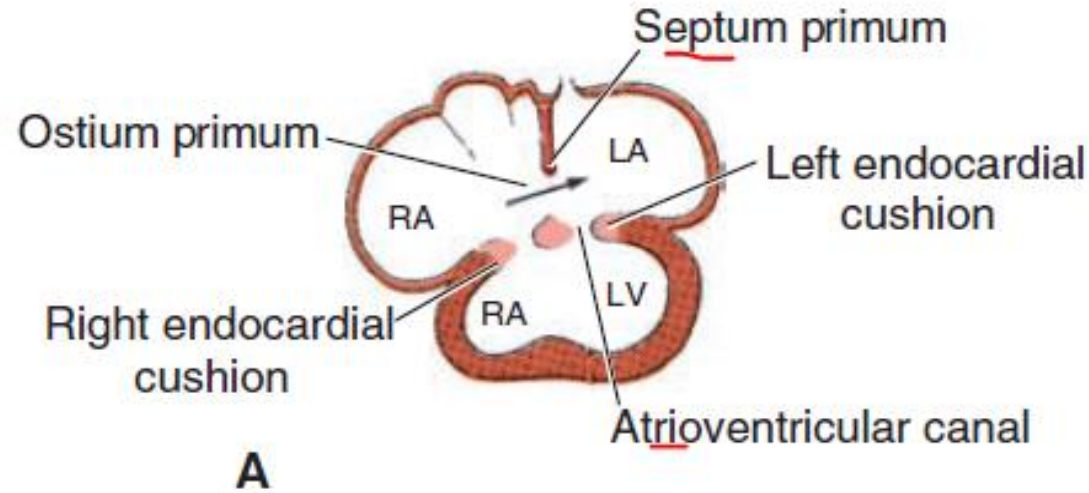
Partitioning of Atrioventricular Canal,,



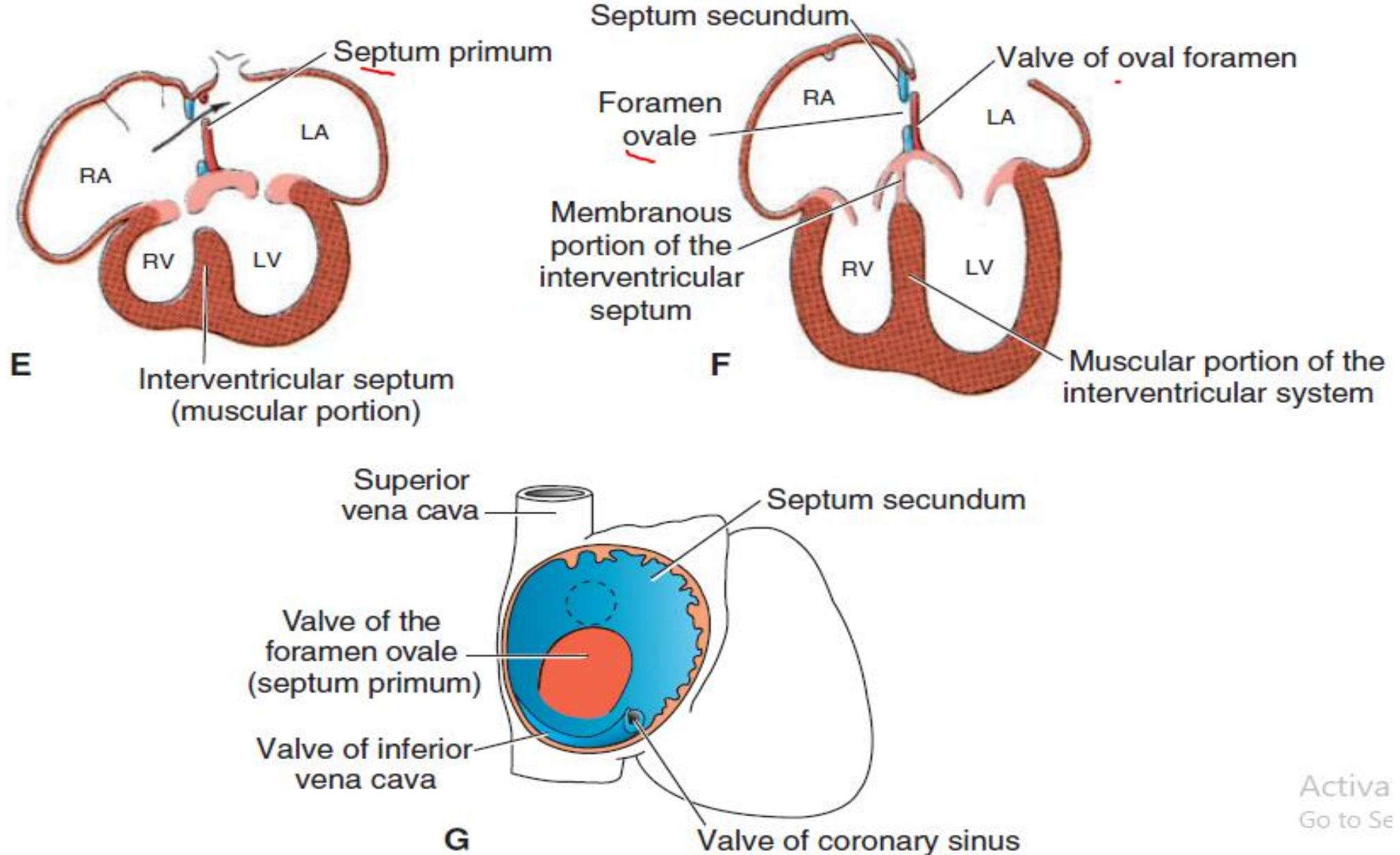
Partitioning of primordial Atrium

- It begins at the end of 4th week by development.
- The interatrial septum divides the primitive atrium into the right and left atria.
- ❑ The interatrial septum is formed by 2 septa *septum primum* & *septum secundum*
 - septum **primum** – it appear 1st
 - septum **secundum**- it appear 2nd
- **1-Septum primum** : a thin crescent-shaped membrane grows from the roof of common atrium into the fusing endocardial cushions dividing common primitive atrium into right & left halves.
- Foramen primum** is formed to pass **blood from R/t atrium to L/t atrium**
- ❑ Before closure of **foramen primum** , perforations appear in central part of septum primum... coalesce to form **Foramen Secundum** (C1-D1).

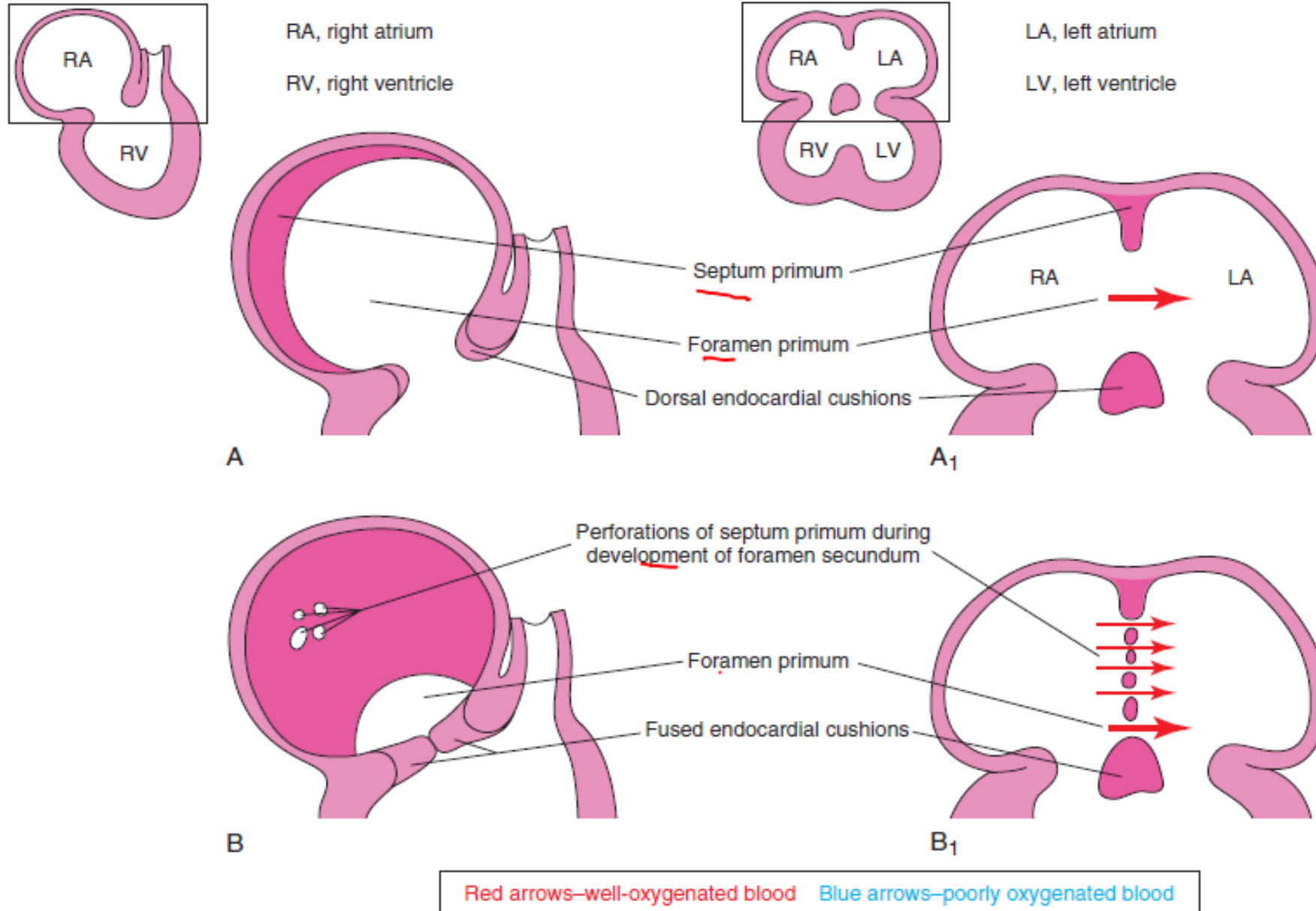
Partitioning of primordial Atrium,,,



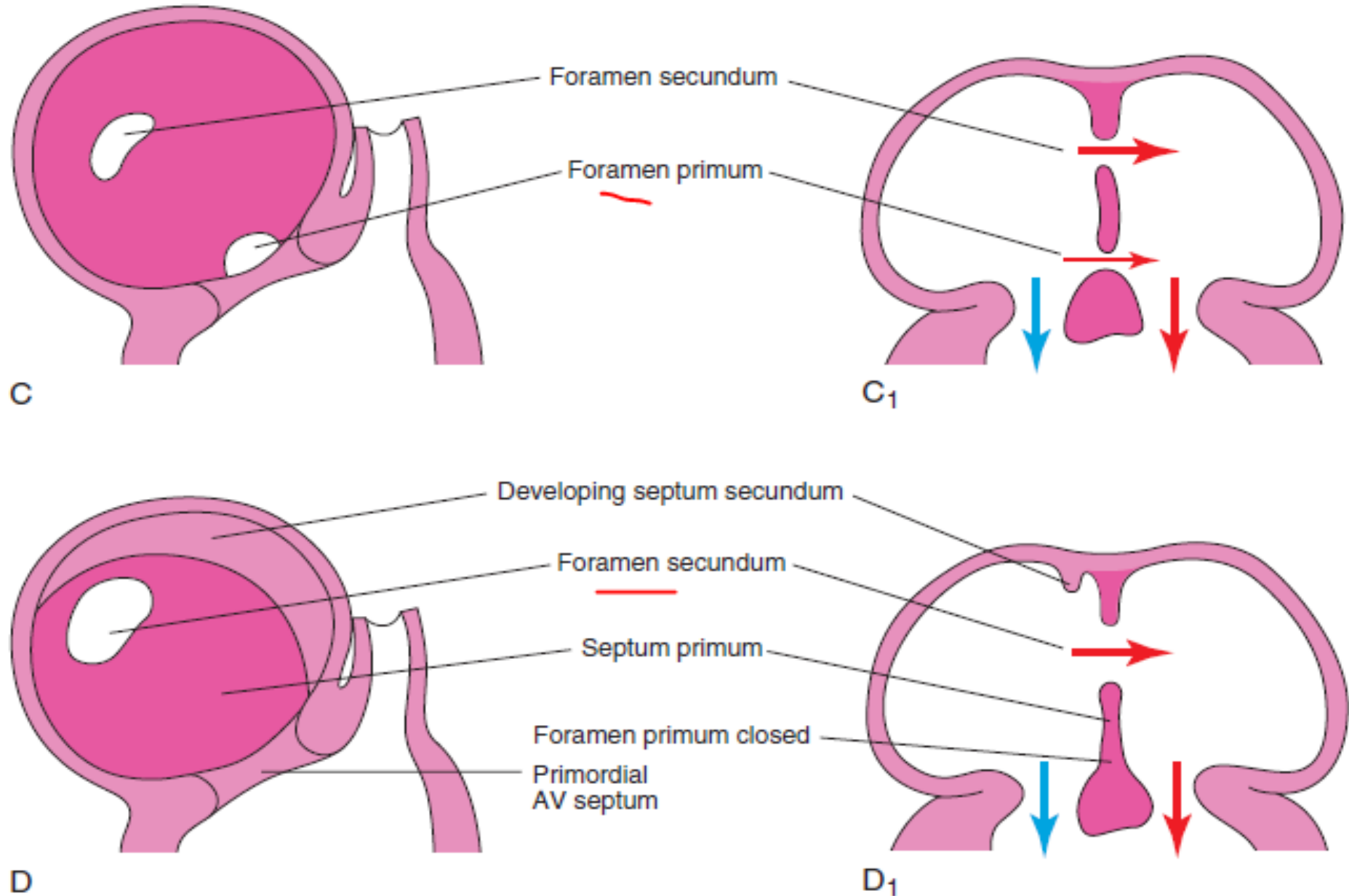
Partitioning of primordial Atrium,,,



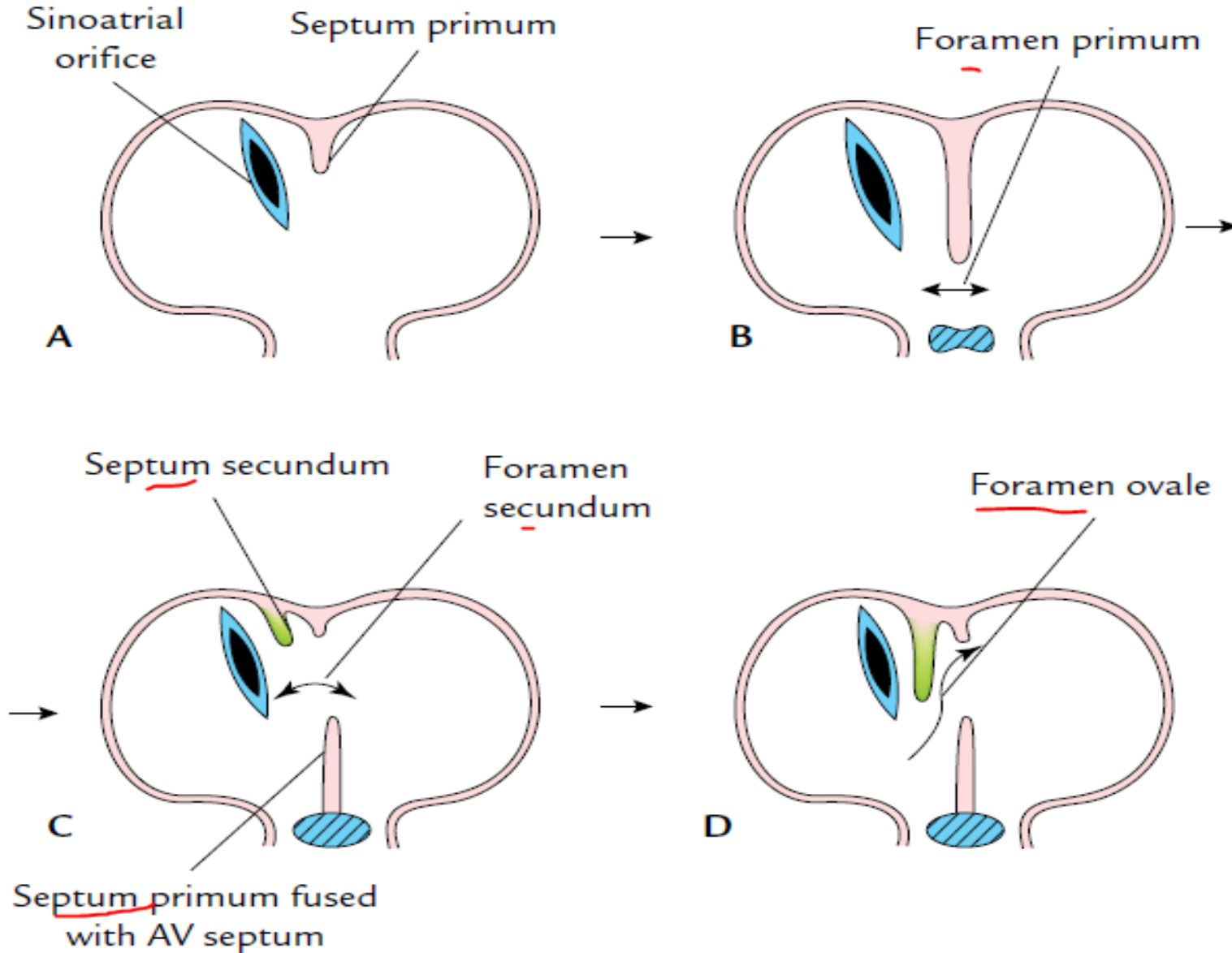
Partitioning of primordial Atrium,,,



Partitioning of primordial Atrium,,,

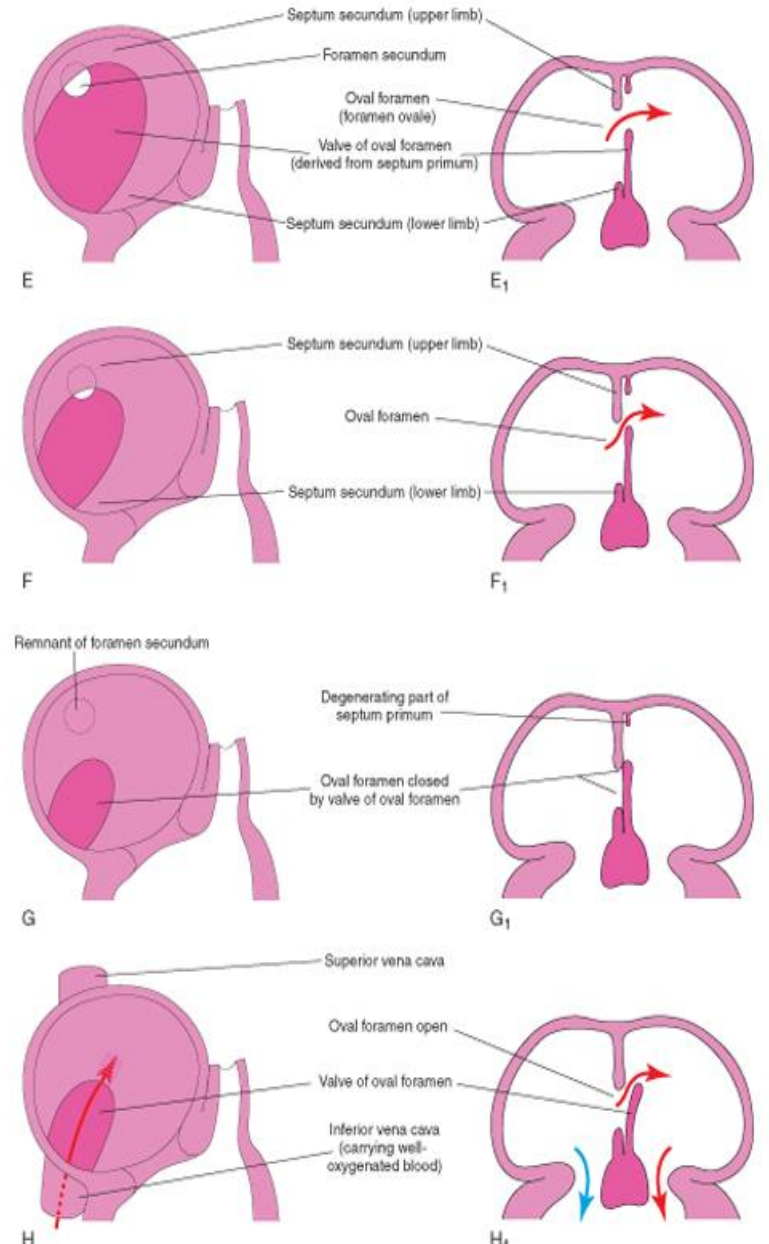


Partitioning of primordial Atrium,,,

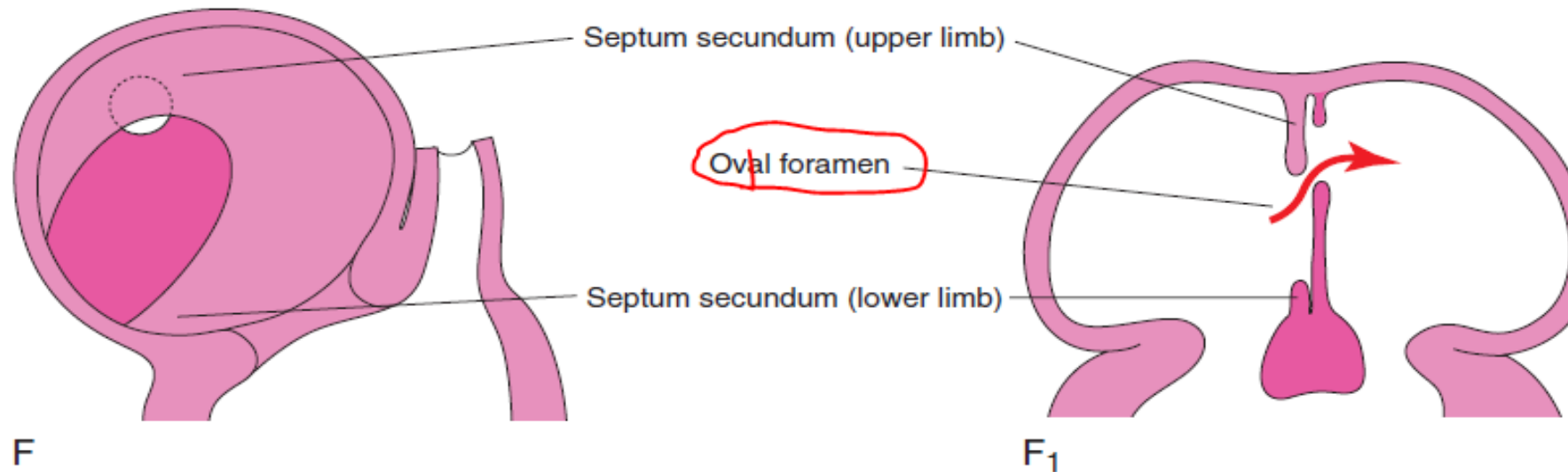
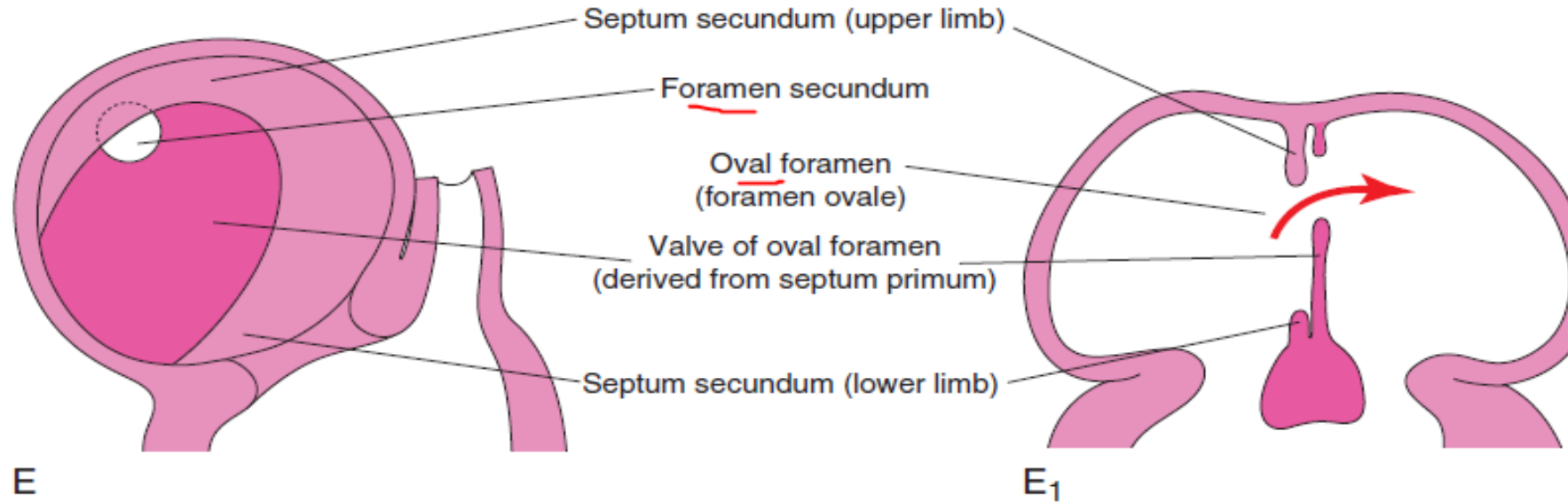


Partitioning of primordial Atrium,,,

- ❑ 2-Septum secundum : a crescentic **muscular membranous** grows and descends from roof of atrium during 5th week.
- ❑ It overlaps **foramen secundum** in septum primum .
- The gap between the lower free border of S.sekundum and the upper edge of S.primum... form “**foramen ovale**”.



Partitioning of primordial Atrium,,,

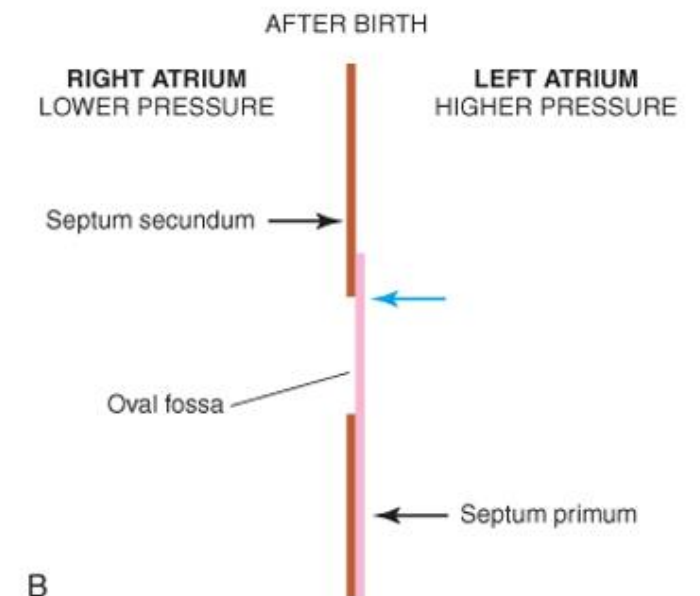
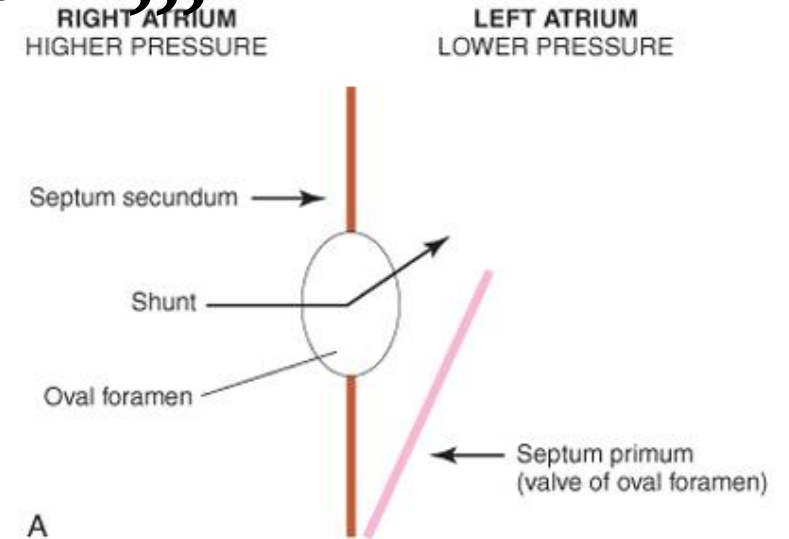


Partitioning of primordial Atrium,,,

➤ In the fetus (**before birth**) ... the pressure is higher in **R/t atrium** than in the **L/t** and highly oxygenated blood flows directly from right atrium to left atrium through **open foramen ovale**.

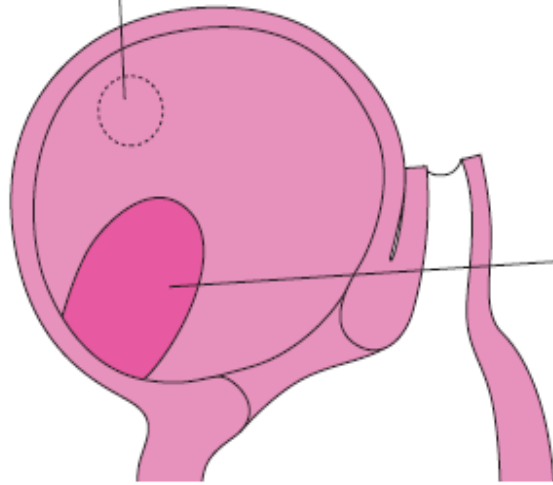
➤ **After birth** ... when the circulation of the lungs begins & the blood pressure in **left atrium rises**, the upper edge of septum primum is pressed against the upper limb of **septum secundum..this** will close the foramen ovale ,forming a complete partition between the 2 atria.

➤ An oval depression in the lower part of interatrial septum of R/t atrium.the **fossa ovalis** is a remnant of the foramen ovale.



Partitioning of primordial Atrium...

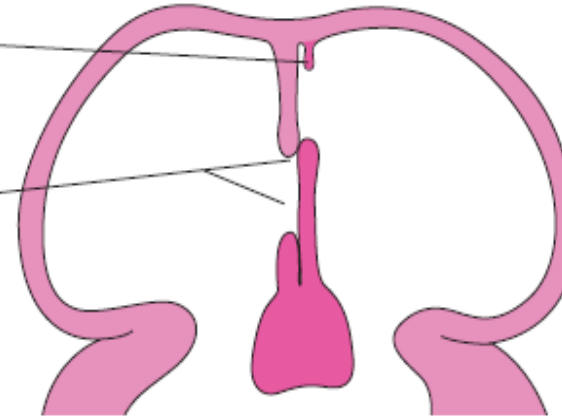
Remnant of foramen secundum



G

Degenerating part of septum primum

Oval foramen closed by valve of oval foramen



G₁

RIGHT ATRIUM
HIGHER PRESSURE

LEFT ATRIUM
LOWER PRESSURE

Septum secundum →

Shunt →

Oval foramen

← Septum primum
(valve of oval foramen)

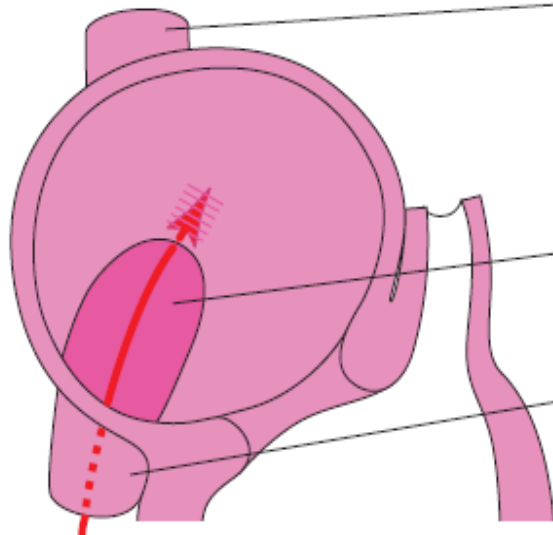
A

Superior vena cava

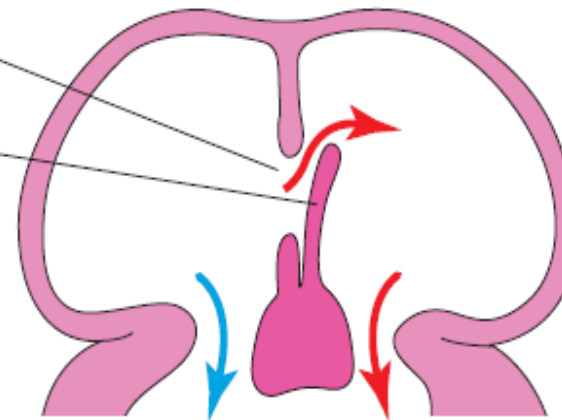
Oval foramen open

Valve of oval foramen

Inferior vena cava
(carrying well-oxygenated blood)



H



H₁

AFTER BIRTH

RIGHT ATRIUM
LOWER PRESSURE

LEFT ATRIUM
HIGHER PRESSURE

Septum secundum →

Oval fossa

← Septum primum

B

Embryological origin of the definitive atrium

Right Atrium

1- Its rough part + auricle from
Right $\frac{1}{2}$ of primitive atrium.

2- right $\frac{1}{2}$ of A-V canal and

3- Its smooth part from
Absorbed right horn of sinus
venosus.

Left Atrium

1- Its rough part + auricle from
left $\frac{1}{2}$ of primitive atrium.

2- left $\frac{1}{2}$ of A-V canal and

3- Its smooth part from
Absorbed part of pulmonary
veins.

Embryological origin of the definitive atrium...

Table

Adult components of right and left atria and their embryonic source of development

Adult component	Embryonic sources of development
-----------------	----------------------------------

Right atrium

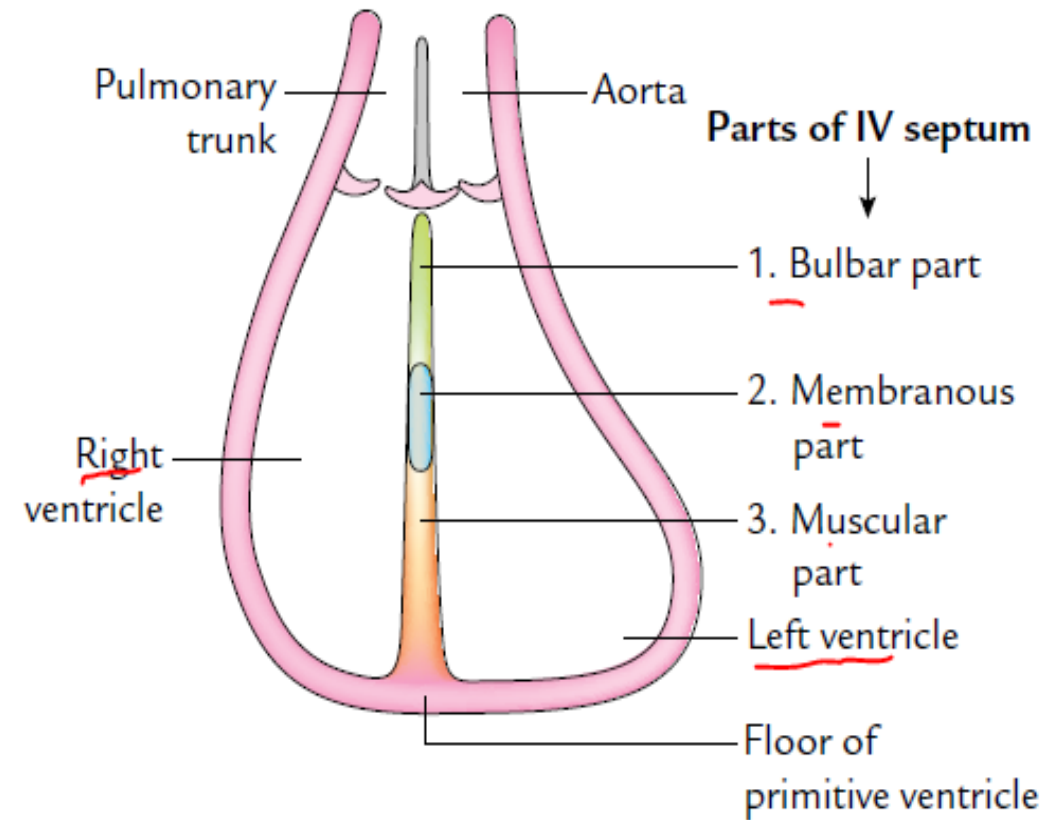
- | | |
|---|-------------------------------|
| <ul style="list-style-type: none"> • Rough trabeculated part (atrium proper) in front of crista terminalis | Primitive atrium (right half) |
| <ul style="list-style-type: none"> • Right auricle | |
| <ul style="list-style-type: none"> • Smooth part behind crista terminalis (sinus venarum) | Sinus venosus |
| <ul style="list-style-type: none"> • Crista terminalis valve of IVC and valve of coronary sinus | From right venous valve |
| <ul style="list-style-type: none"> • Most ventral smooth part | Right half of AV canal |

Left atrium

- | | |
|---|--|
| <ul style="list-style-type: none"> • Posterior smooth part between the openings of pulmonary veins | Absorption of pulmonary veins near the atria |
| <ul style="list-style-type: none"> • Anterior rough part and left auricle | Left half of primitive atrium |
| <ul style="list-style-type: none"> • Most ventral smooth part | Left half of the AV canal |

Partitioning of Primordial Ventricle

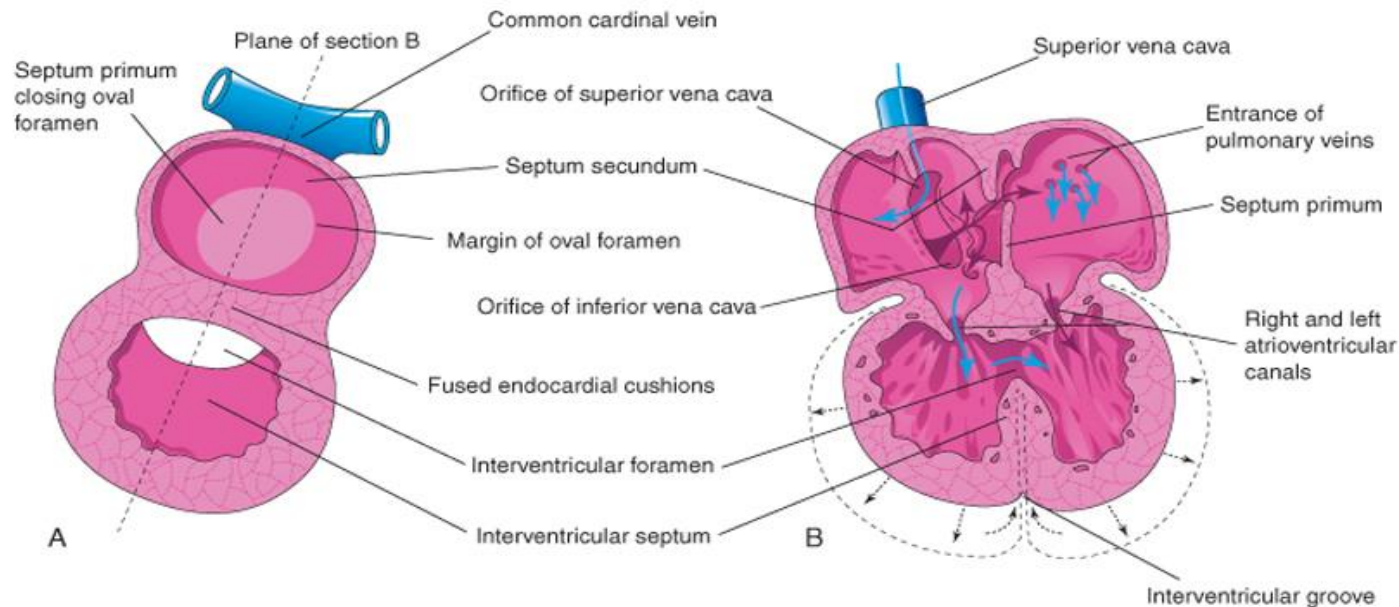
- Division of **primordial ventricle** is first indicated by a median ridge (**muscular IV septum**) in the floor of ventricle near its apex.
- The interventricular (IV) septum consists of 3 parts.
- From below upward these are:
 - muscular part,**
 - bulbar part,** and
 - membranous part.**
- These three parts develop from 3 different sources



- ❑ Division of the primordial ventricle is first indicated by a median ridge—the muscular **IV septum**—in the floor of the ventricle near its apex.
- ❑ **Myocytes from both the left and right primordial ventricles contribute to the formation of the muscular part of the IV septum.**

Partitioning of Primordial Ventricle,,,

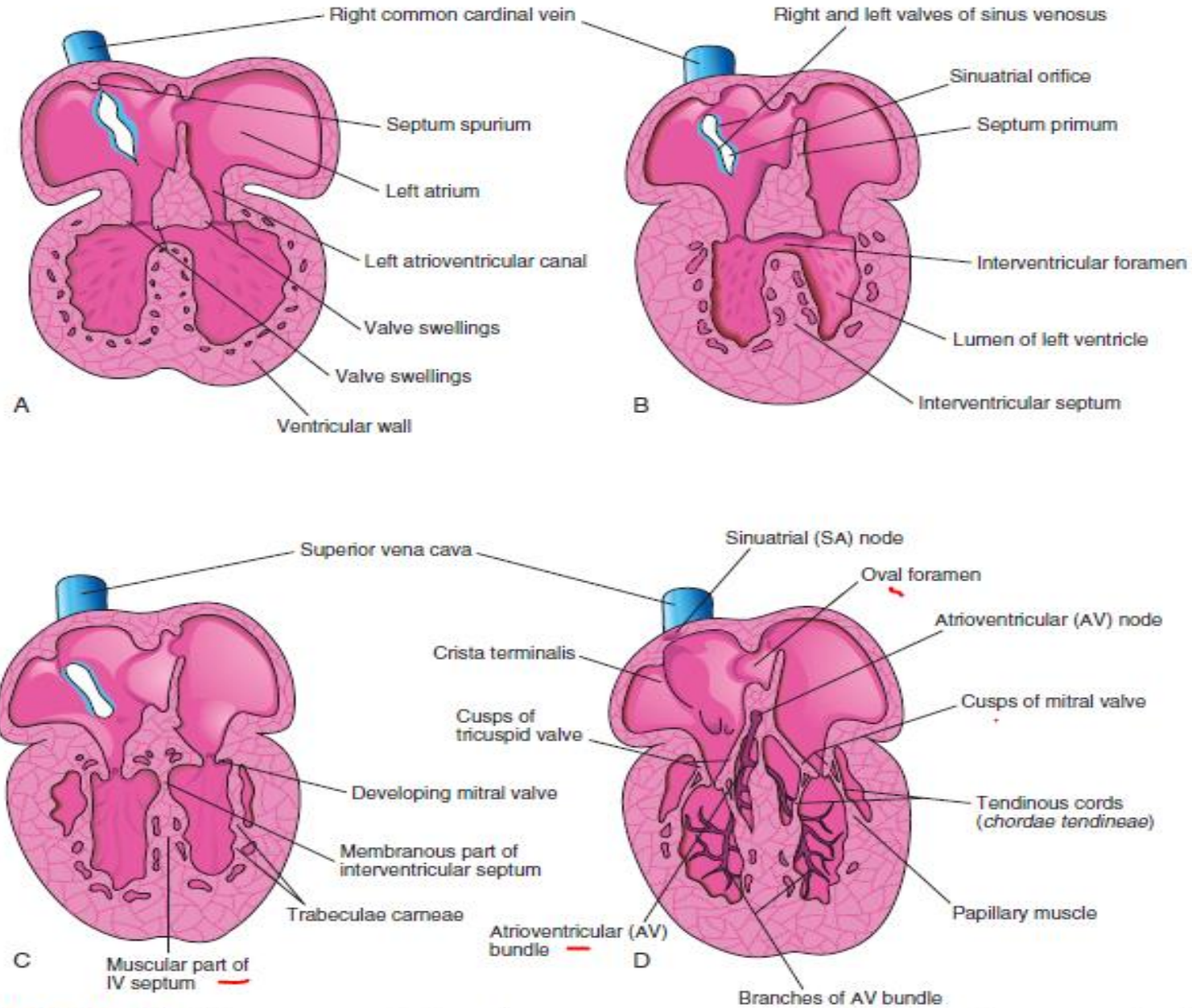
- Primordial **muscular interventricular (IV)septum** arises in the **floor** of ventricle, as thick crescentic fold with concave free edge.
- This septum subdivides the original ventricular cavity **incompletely** into right & left ventricles that communicate together through IV foramen.
- This foramen closes by the end of **7th week** as the **2 bulbar ridges** fuse with the **endocardial cushion**.

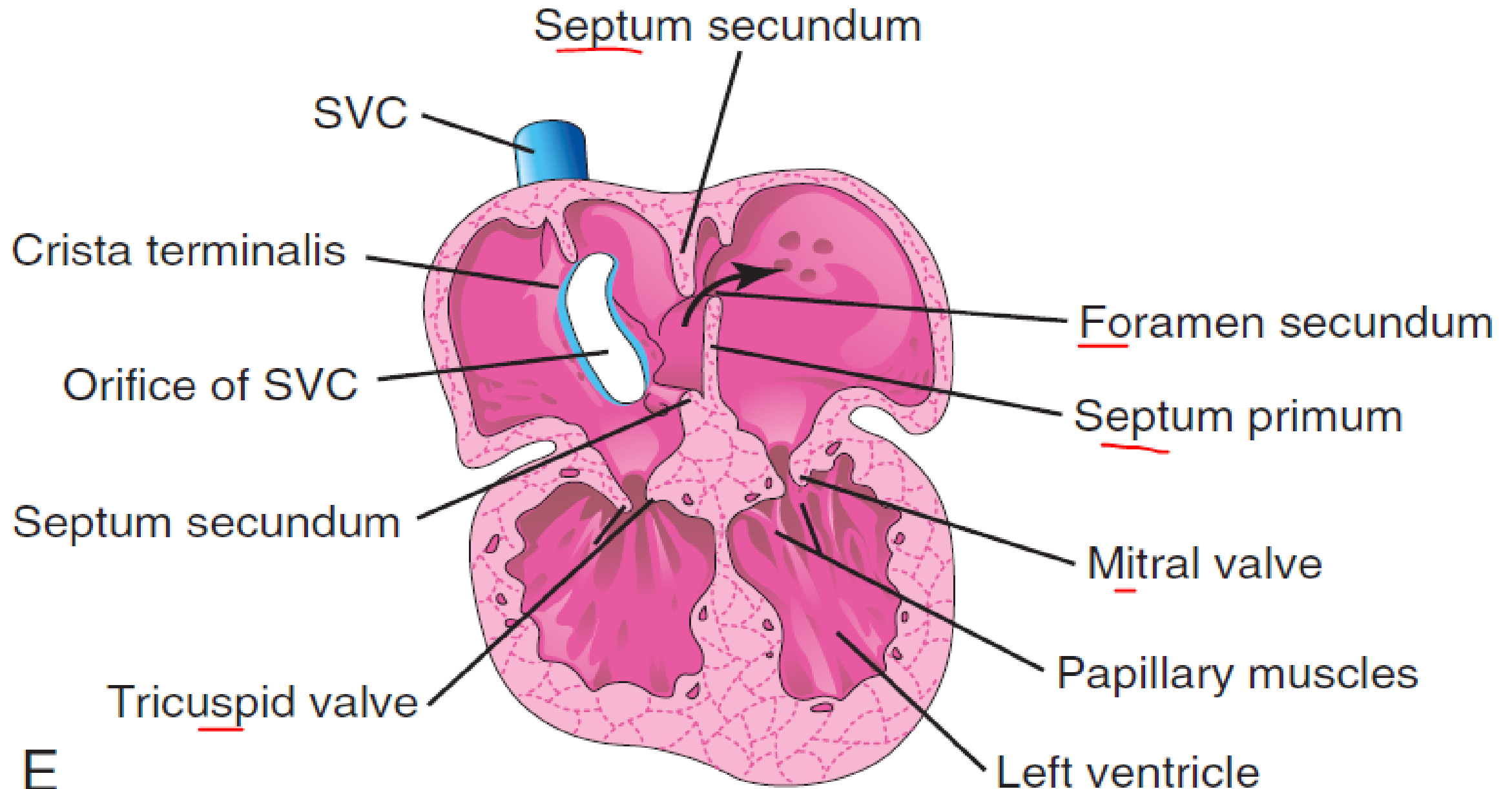


➤ A-sagittal section 5th wk.

➤ Coronal section. 6th wk.

Partitioning of Primordial Ventricle,,

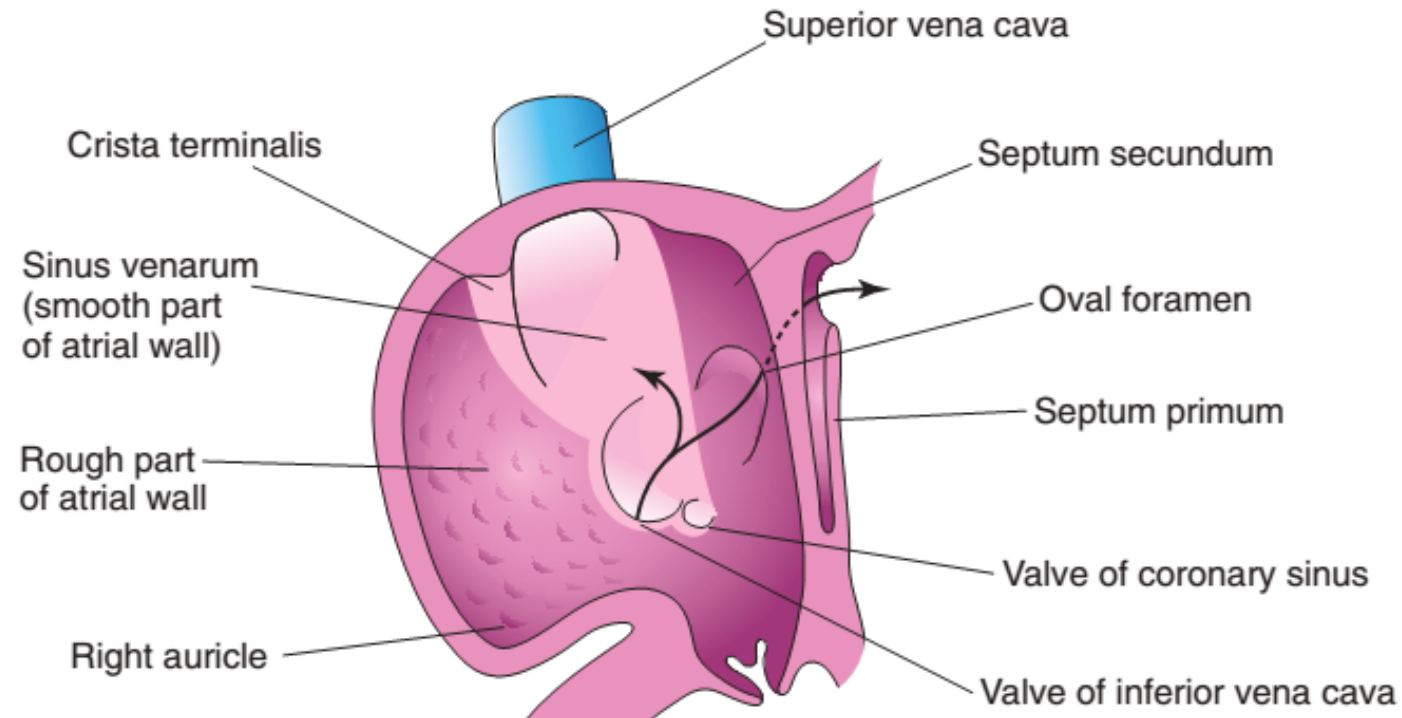




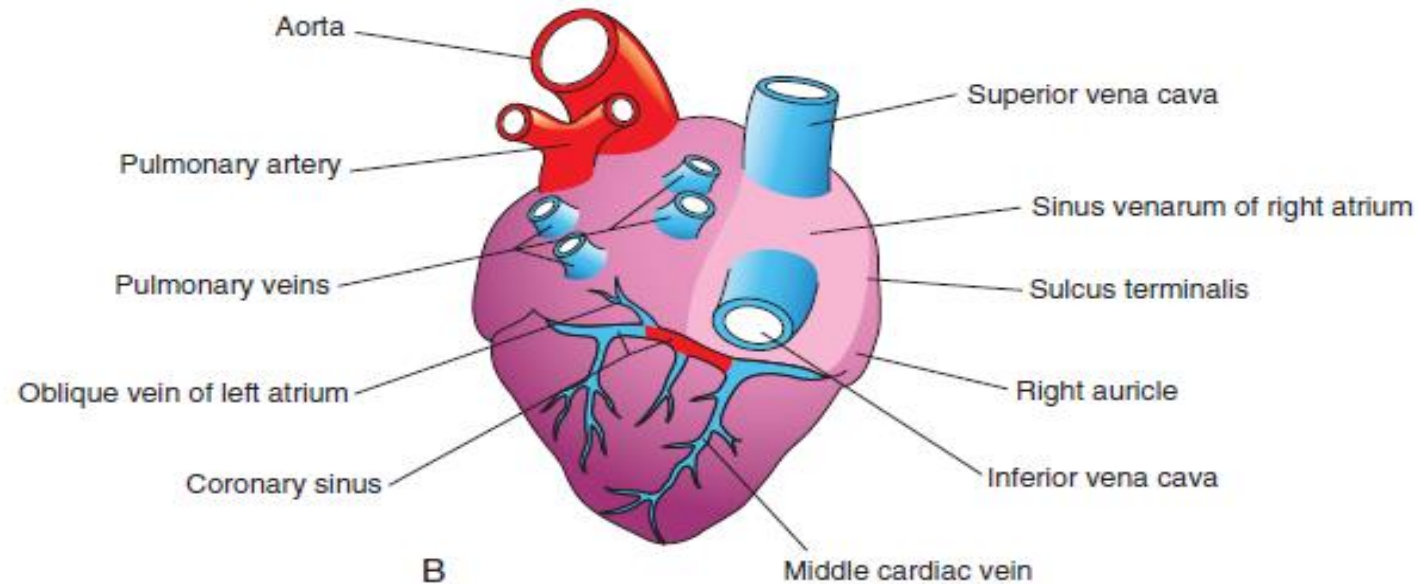
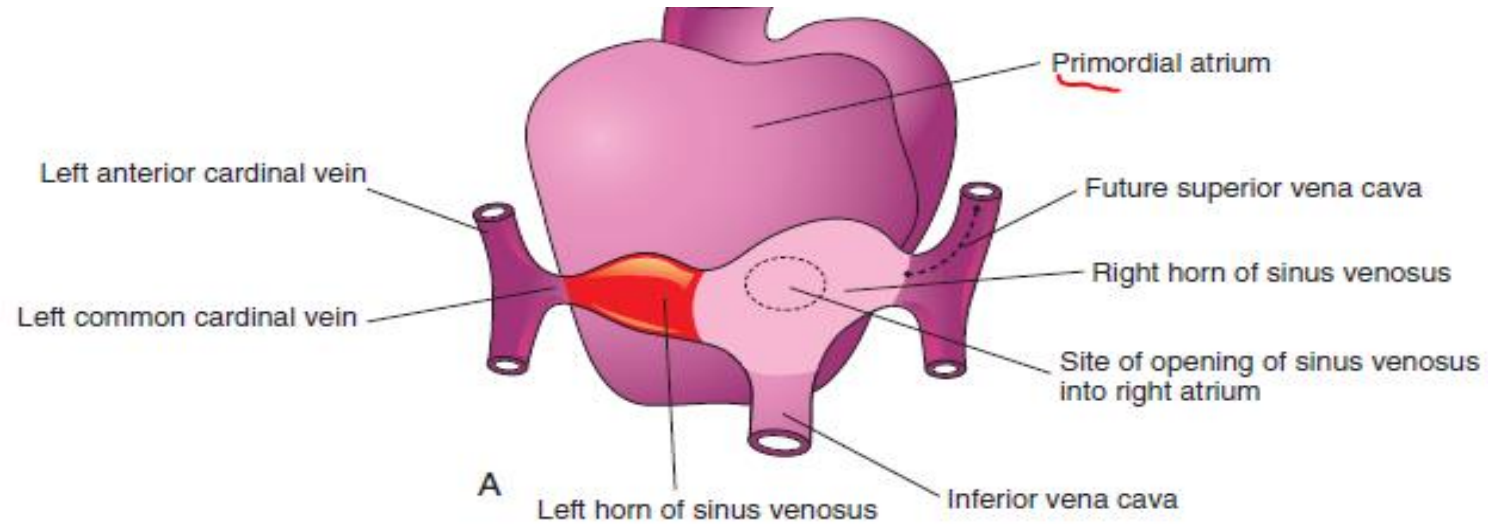
E

Development of Right Atrium

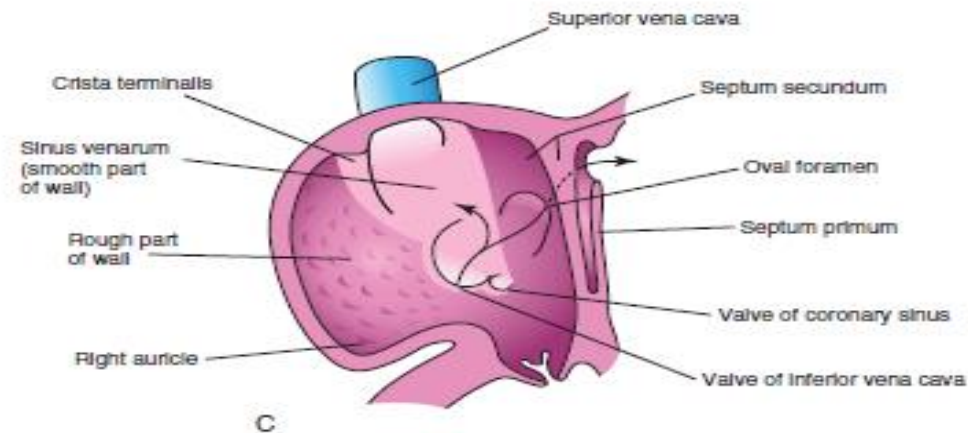
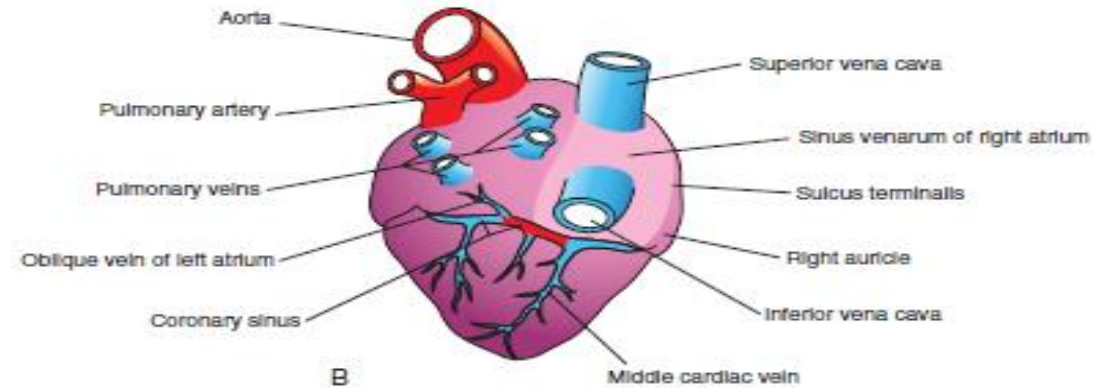
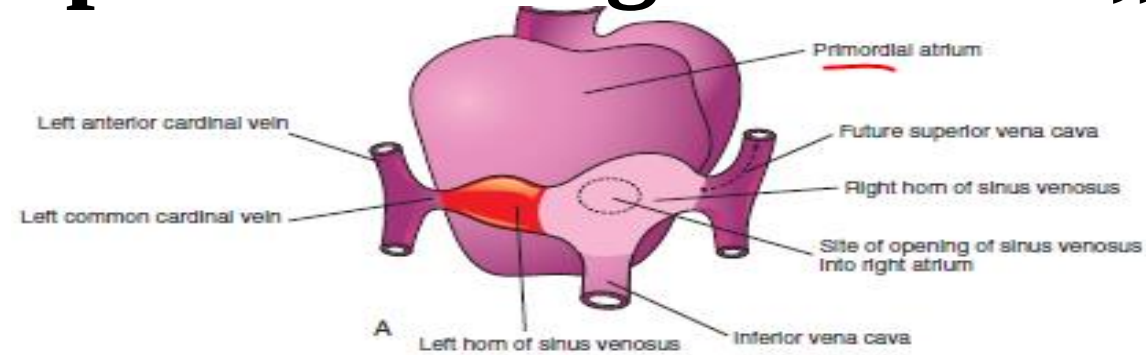
- rough ventral wall and Rt auricle develop from primitive atrium
- Smooth dorsal wall of Rt atrium develop from incorporation of sinus venosus
- **Crista terminalis** divide rough part from smooth part internally



Development of Right Atrium,,



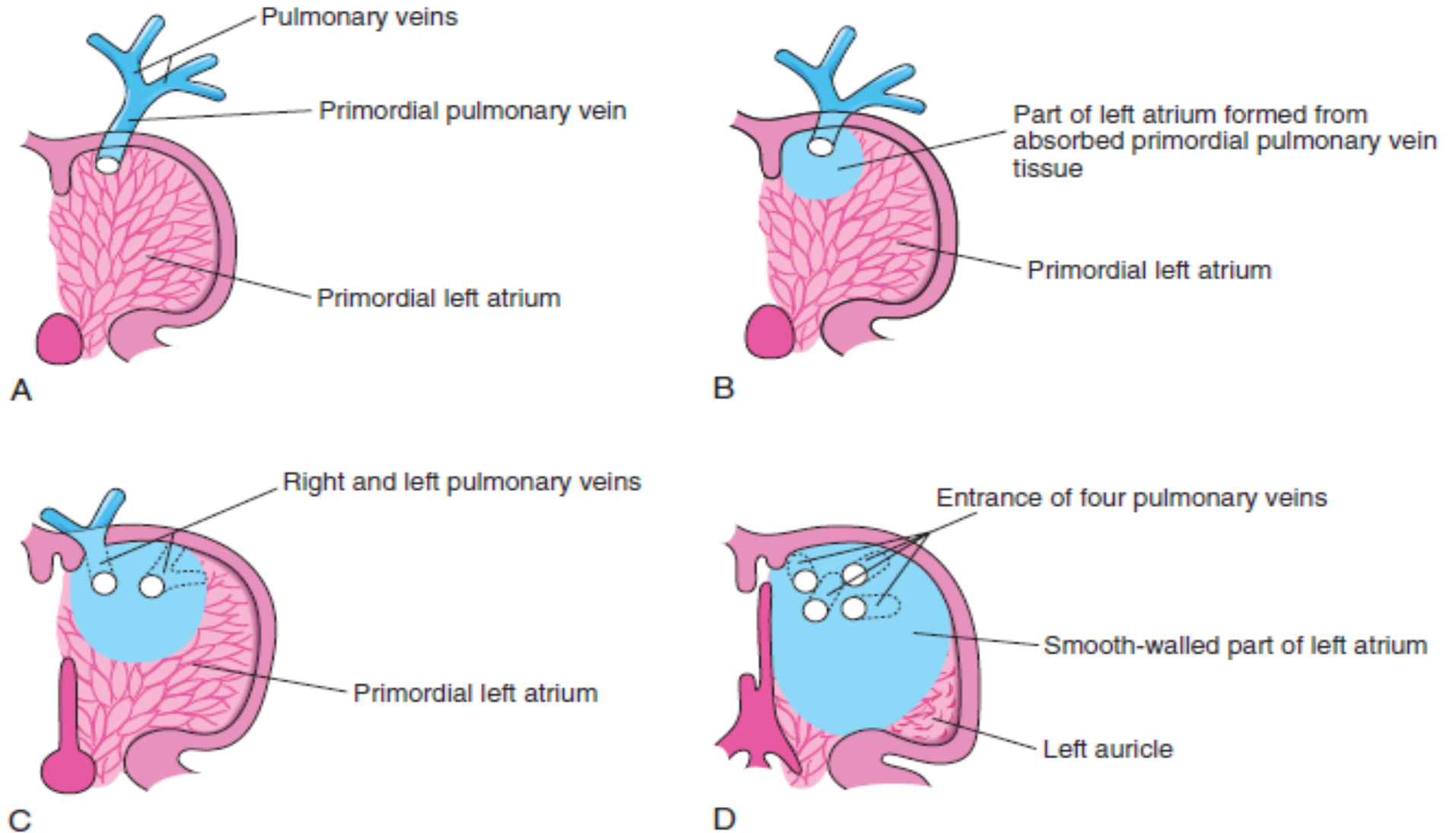
Development of Right Atrium,,



Development of Right Atrium,,

- Most of the wall of left atrium is smooth b/c it is formed by incorporation of primordial pulmonary vein
 - This vein develops as an outgrowth of dorsal atrial wall, just to the left of the septum primum.
- As the atrium expands, primordial pulmonary vein and its main branches are incorporated into the wall of left atrium.
 - As a result, four pulmonary veins are formed
- Rough wall and Lt auricle develop from primitive atrium

Development of Right Atrium,,,



At 5th week(A), late 5th week (B), at 6th week(C) & at 8th week(D)- absorbtion of pulmonary veins into Lt atrium

Incorporation of the proximal part of bulbus cordis into the ventricles

- **Fig. A** sagittal s. at 5th wk, showing the **bulbus cordis** in the primitive heart.
- **Fig. B** coronal s. at 6th wk. after incorporation of the **proximal part of bulbus cordis** into the ventricles to form:
 - In **right ventricle** ... Conus arteriosus (infundibulum), which gives origin of pulmonary trunk.
 - In **left ventricle**.... Aortic vestibule part of ventricular cavity just inferior to aortic valve.

Fig. A sagittal s. at 5th wk

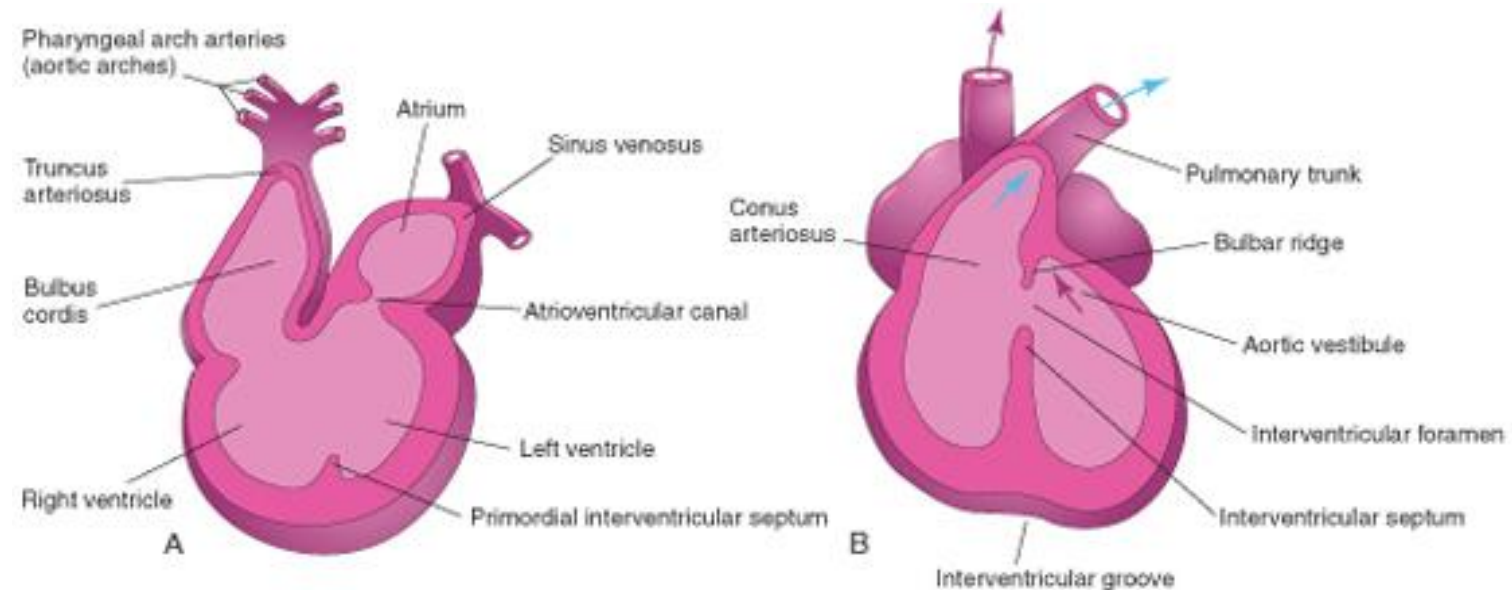
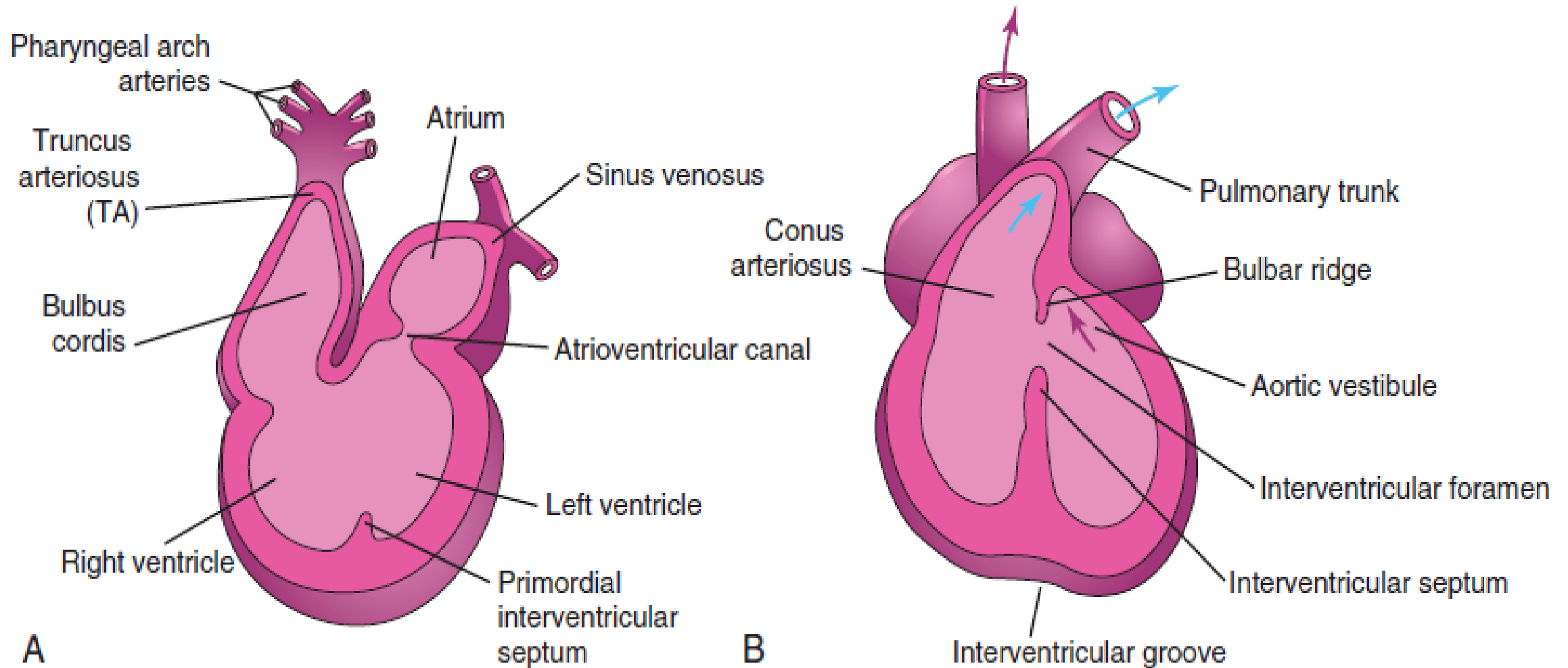


Fig. B coronal s. at 6th wk.

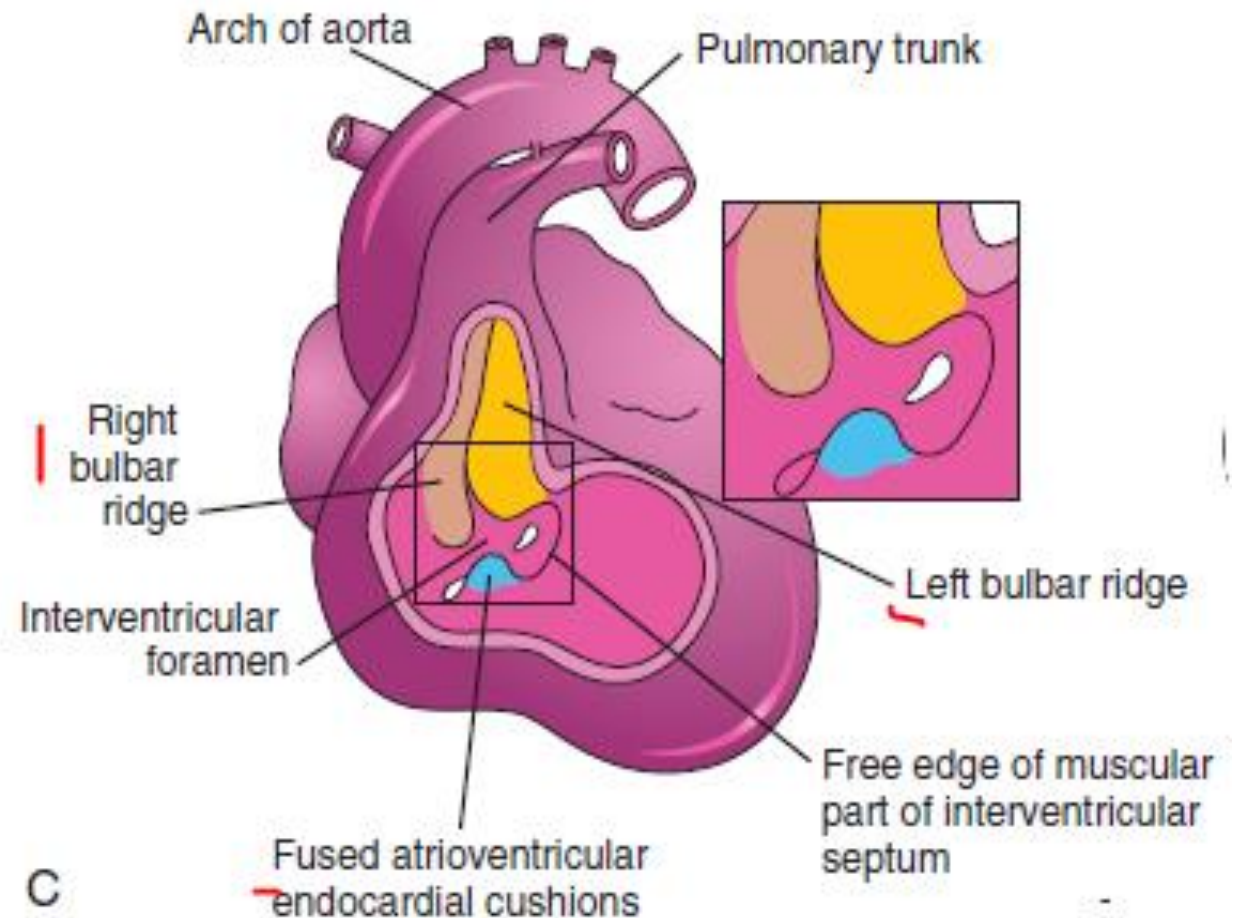
Incorporation of the proximal part of bulbus cordis into the ventricles,,,



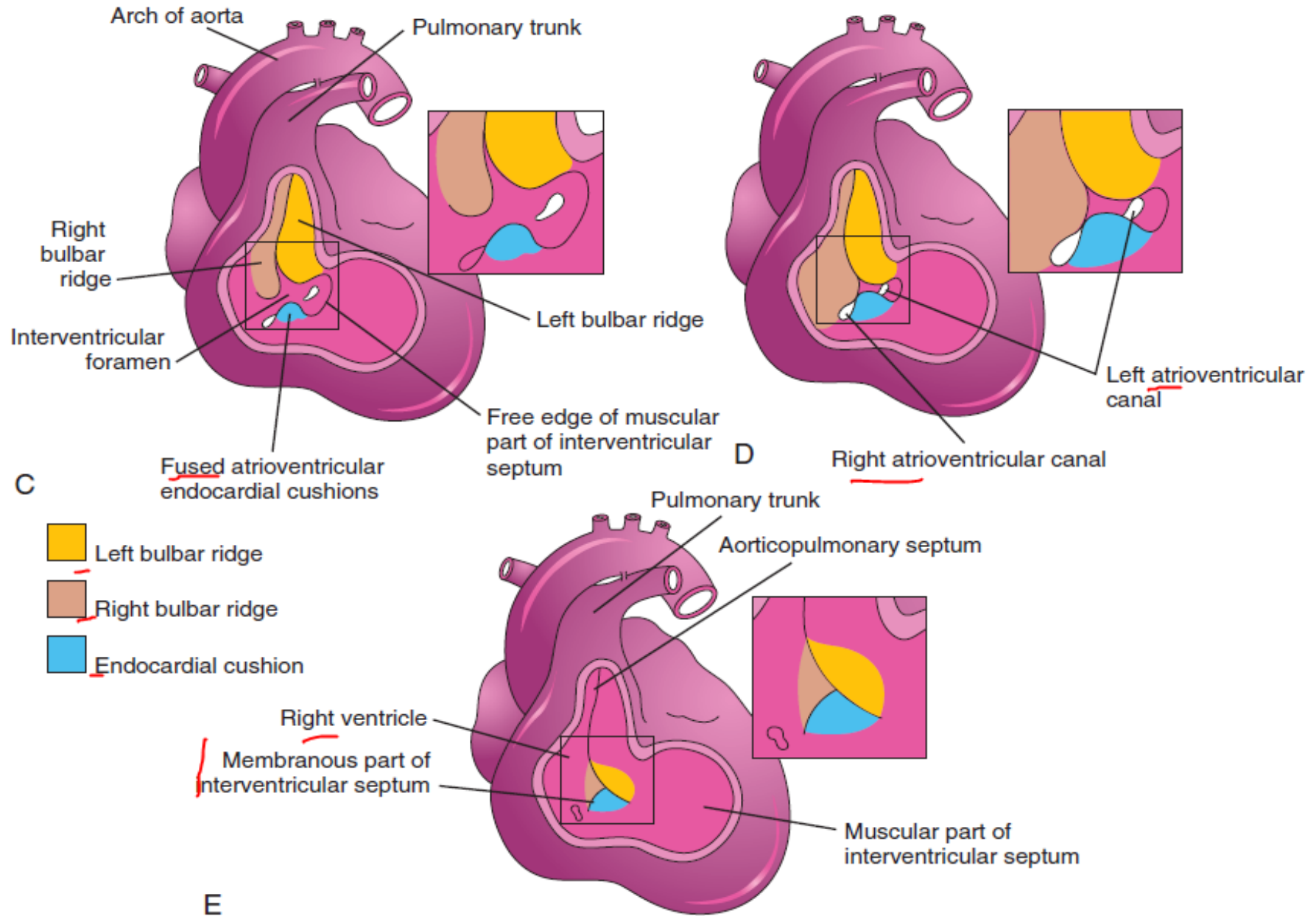
Closure of IV foramen

❖ Closure of IV foramen & formation of membranous part of IV septum result from fusion of the following parts:

1. Right bulbar ridge
2. Left bulbar ridge
3. Fused endocardial cushions



Closure of IV foramen,,

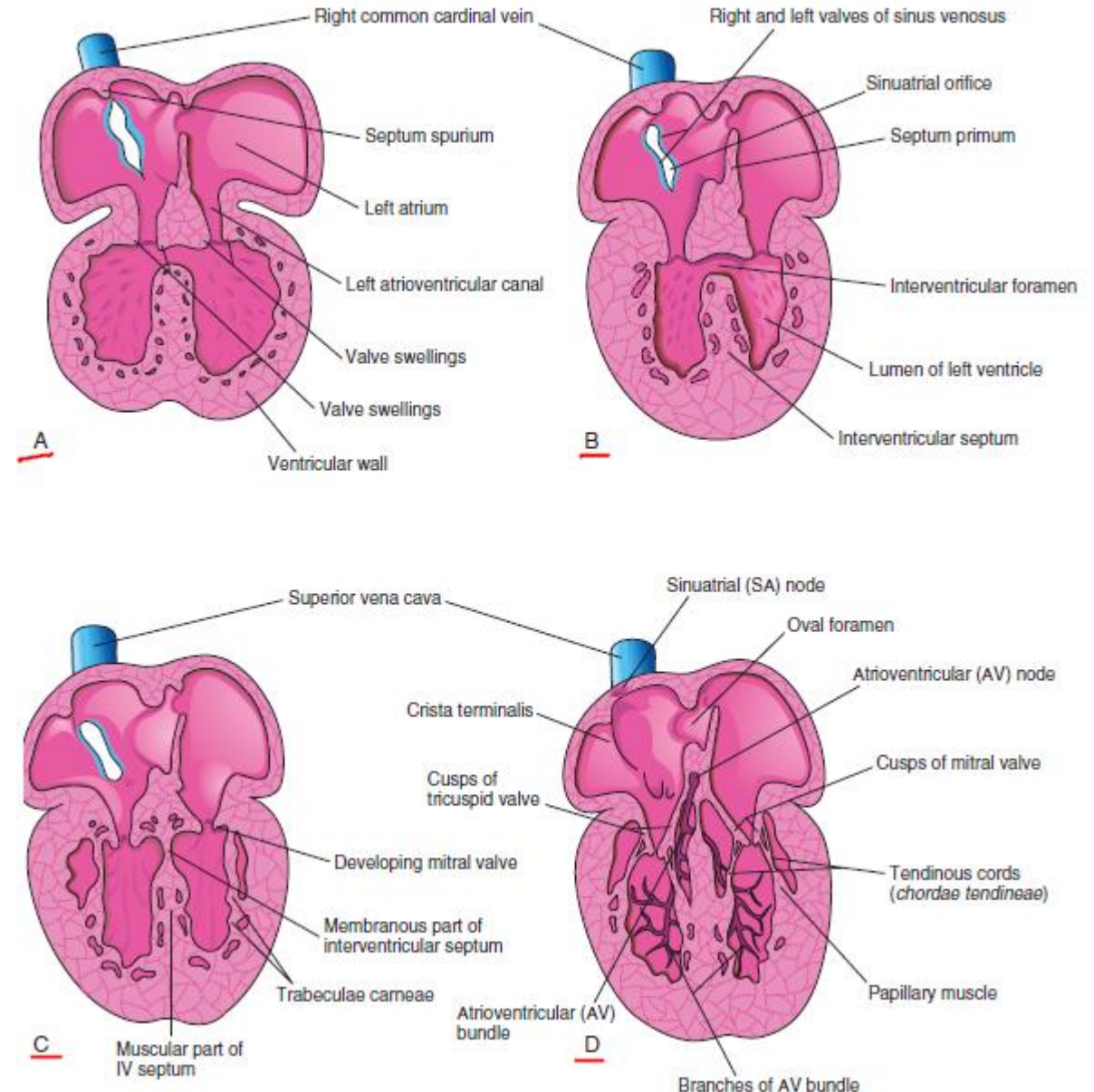


Closure of IV foramen,,,

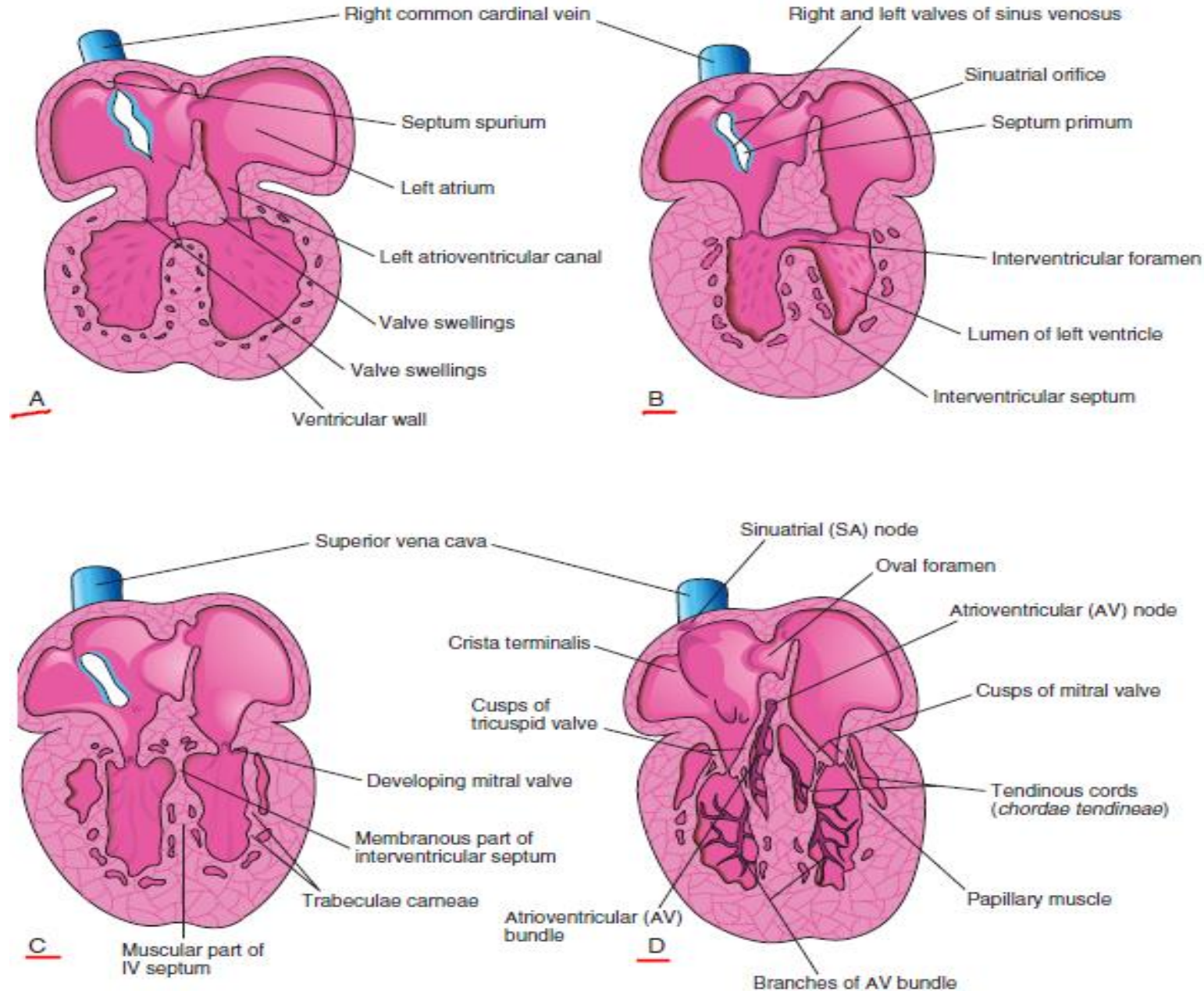
- C, 5th wk showing the bulbar ridges & fused endocardial cushions.
- D, 6th wk, proliferation of endocardial cushions to diminish Interventricular foramen.
- E, 7th wk, fusion of **bulbar ridges** + extensions of **endocardial cushions upward** with **aortico-pulmonary septum** and **down** with **muscular interventricular septum** to close Interventricular foramen, so **membranous Interventricular septum** is formed

Cavitation of Ventricular Walls

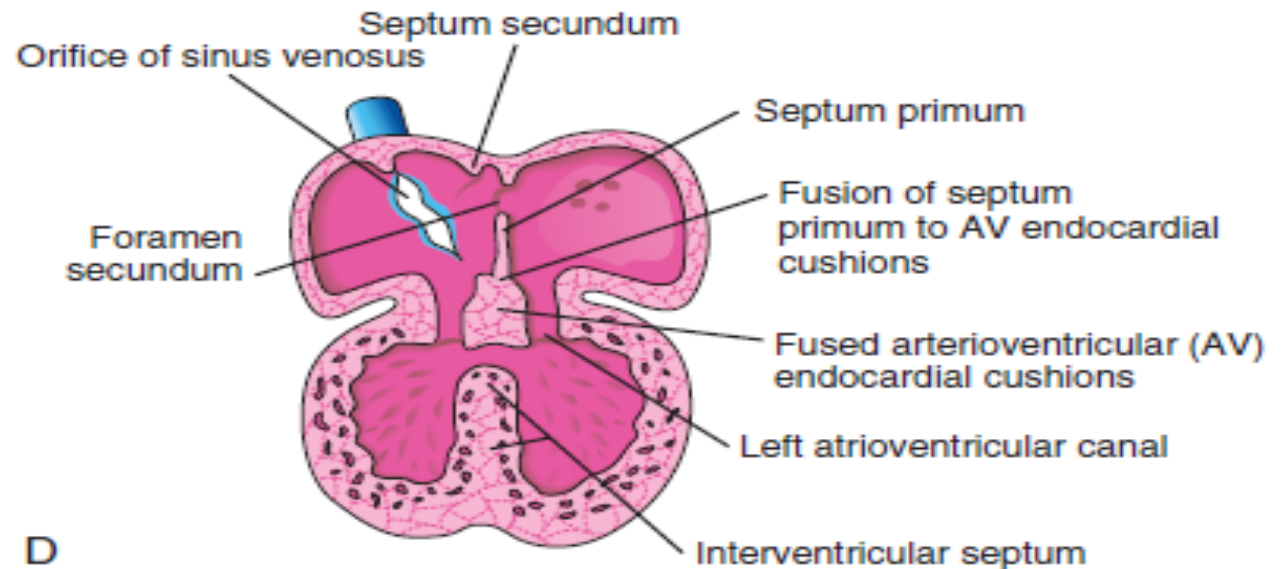
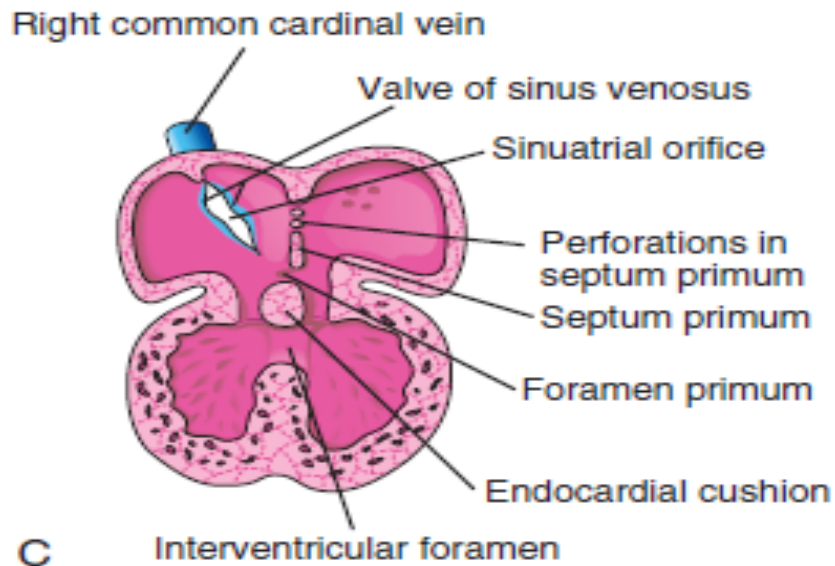
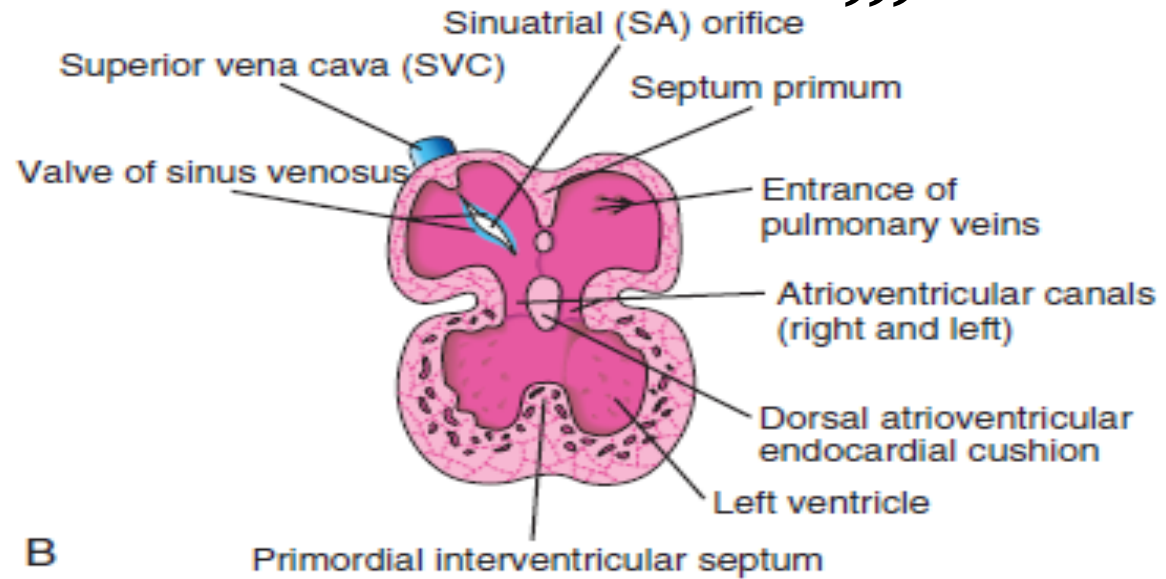
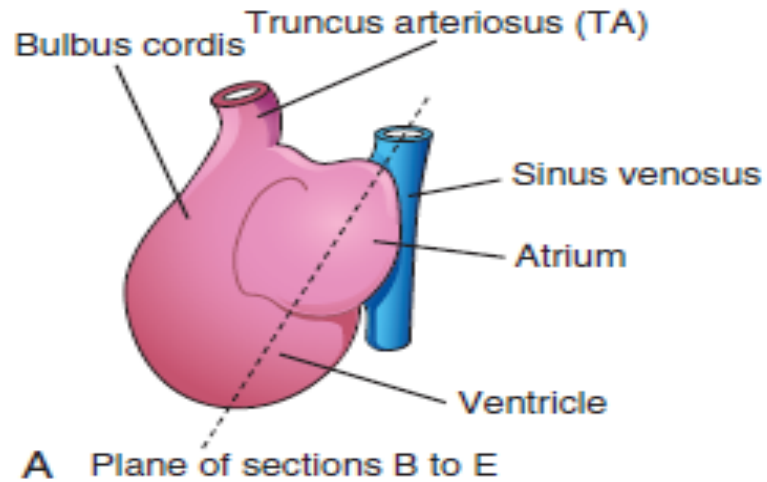
- Leads to formation of **spongy muscular bundles** (trabeculae carneae).
- These bundles become the **papillary muscles** & **tendinous cords** (L. *chordae tendineae*) attached to the cusps of **tricuspid** & **mitral** valves).
- A-5 weeks.
- B-6weeks.
- C-7weeks.
- D-20 weeks.



Cavitation of Ventricular Walls,,

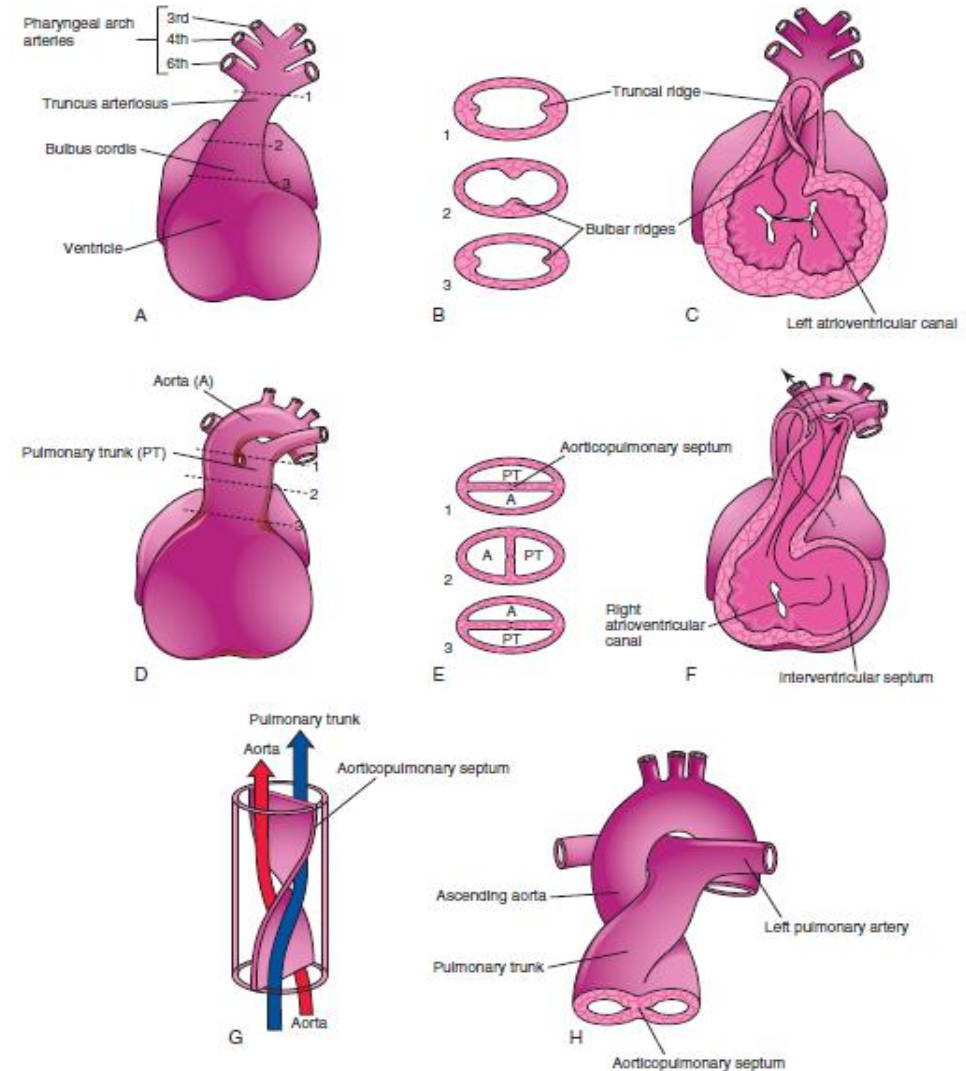


Cavitation of Ventricular Walls,,

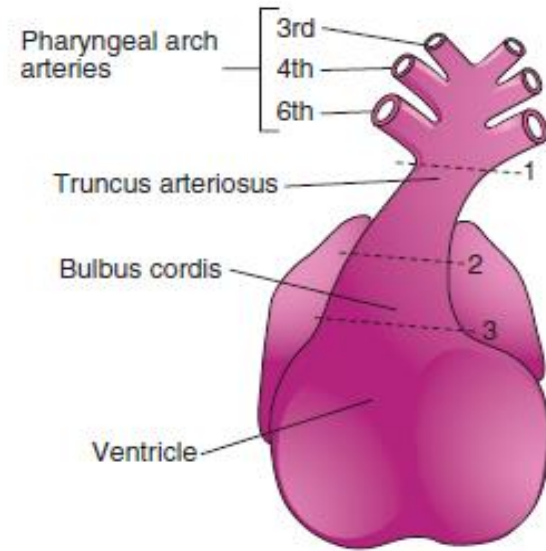


Partitioning of distal part of the Bulbus Cordis & Truncus Arteriosus

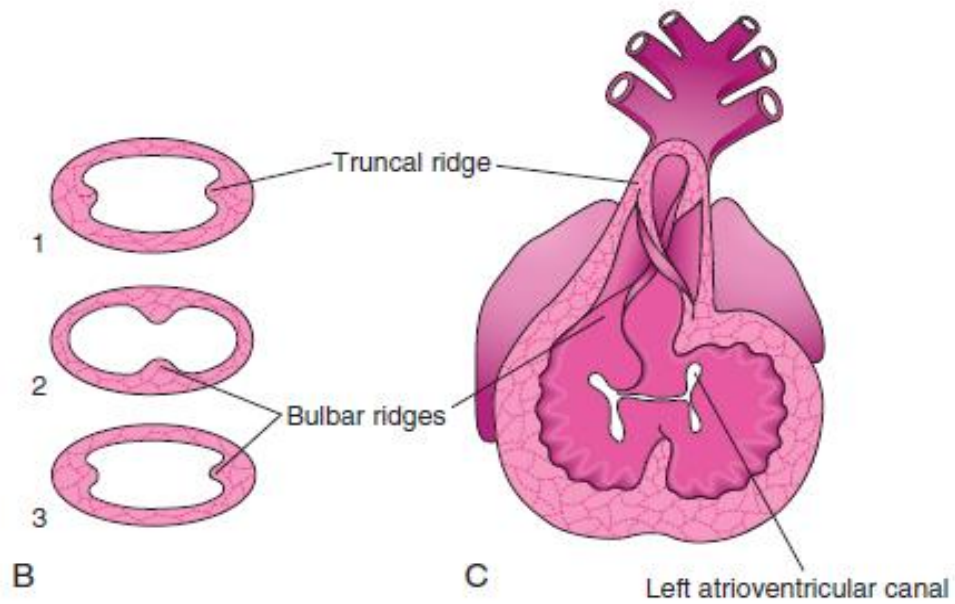
- During 5th w. firstly, a R/t & L/t bulbar ridges are developed in the **lower part**.
- Another **ant.& post. Bulbar ridges** in the **middle part**.
- **Right & left truncal ridges** are developed in the **upper part**.
- **Bulbar & truncal ridges** are developed from **proliferation of mesenchymal cells** of their wall.
- They are also derived from **neural crest** mesenchyme.



Partitioning of distal part of the Bulbus Cordis & Truncus Arteriosus,,,

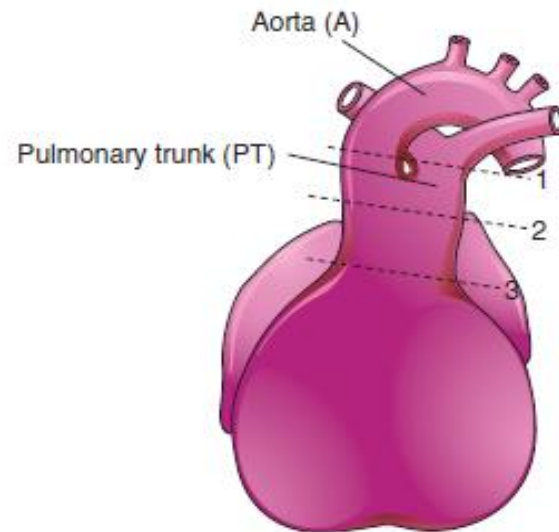


A

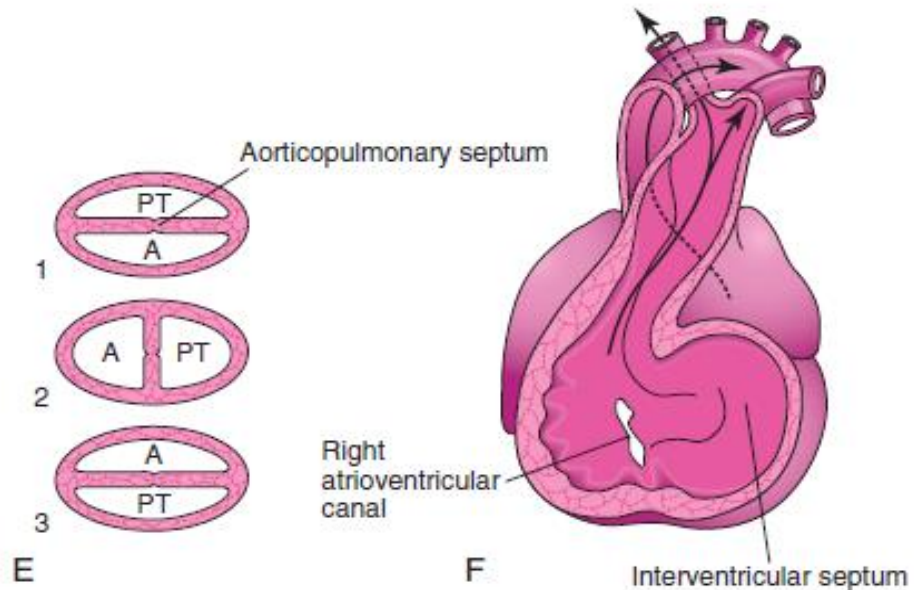


B

C



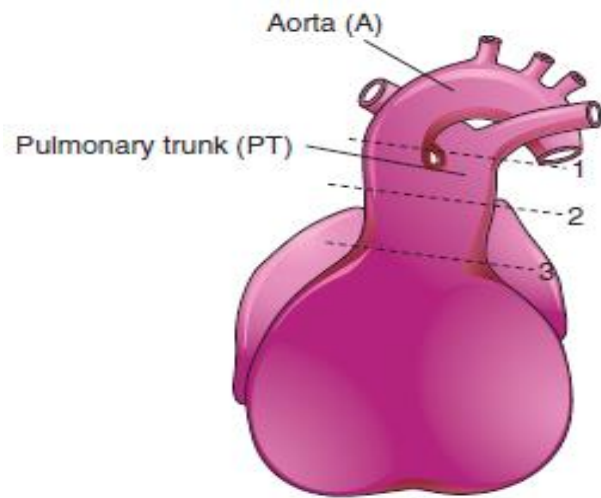
D



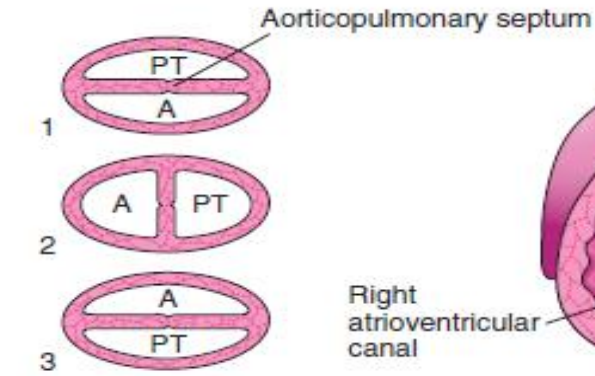
E

F

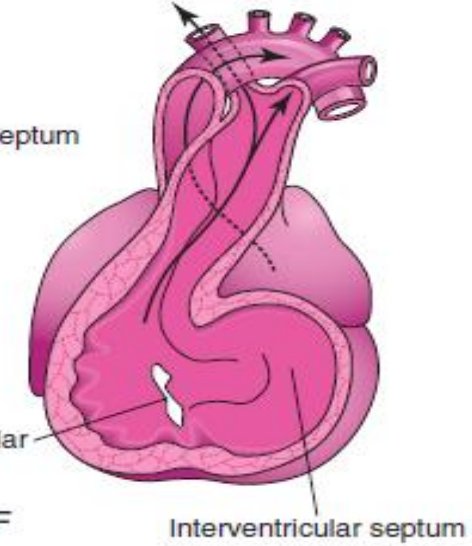
Partitioning of distal part of the Bulbus Cordis & Truncus Arteriosus,,,



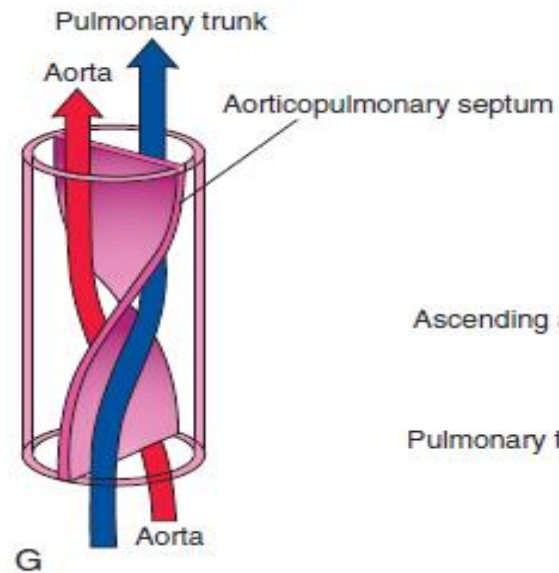
D



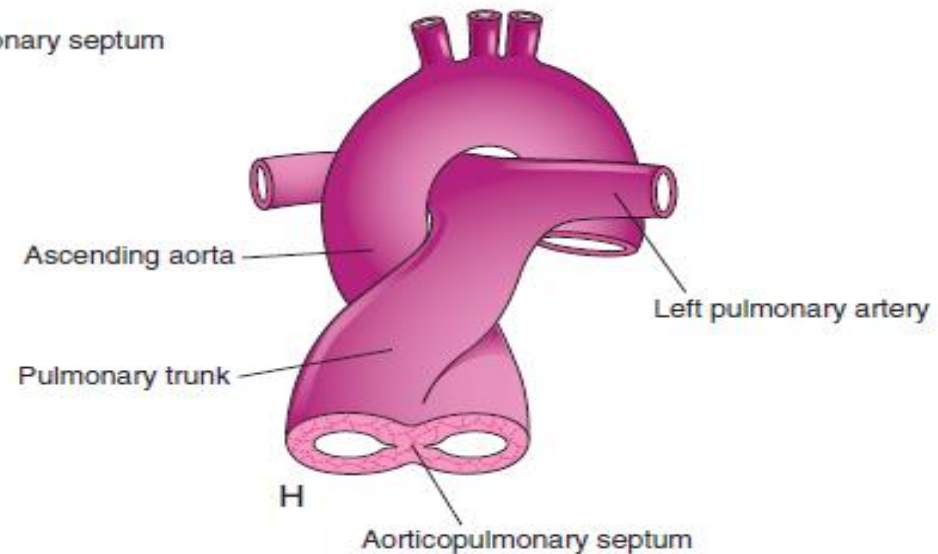
E



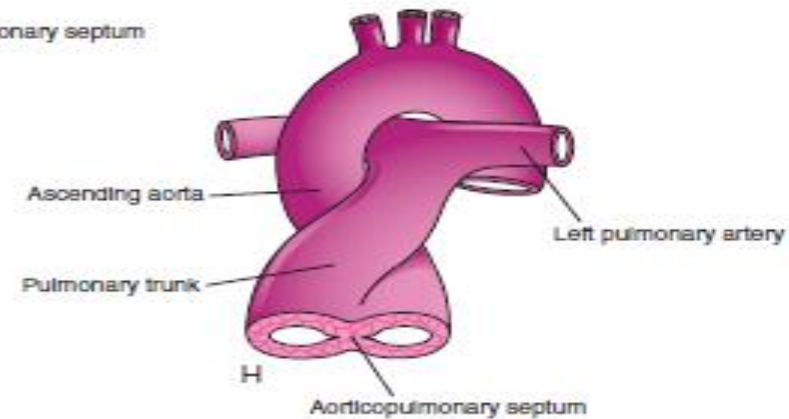
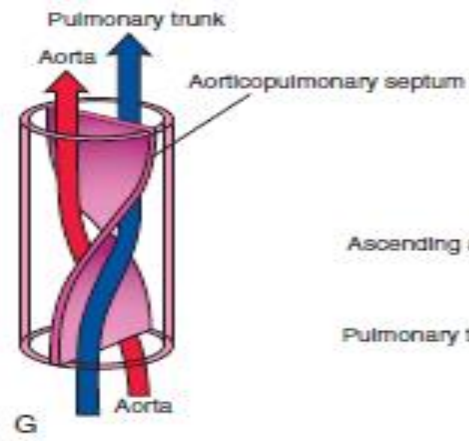
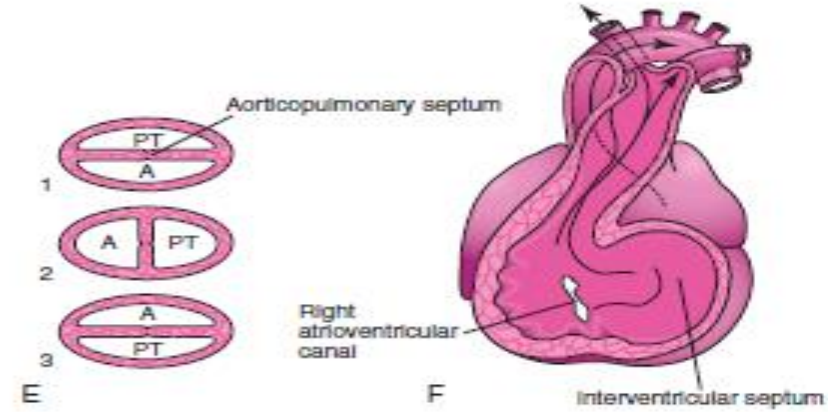
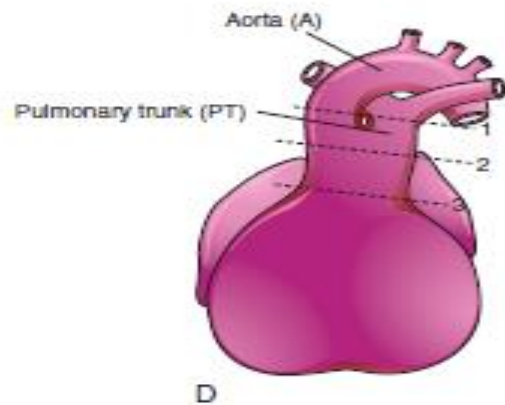
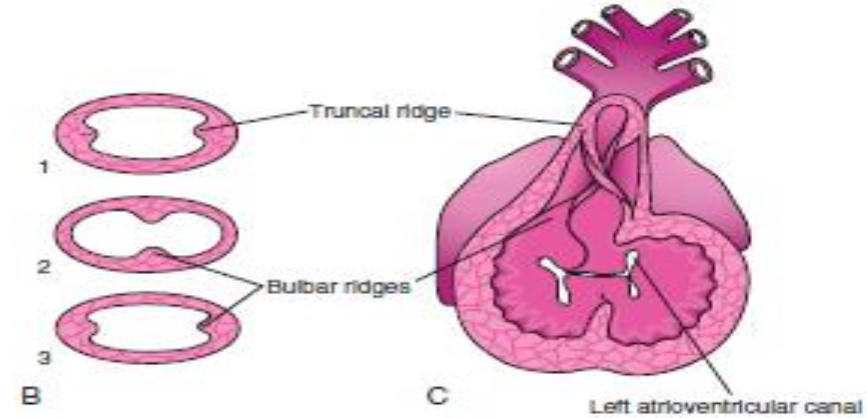
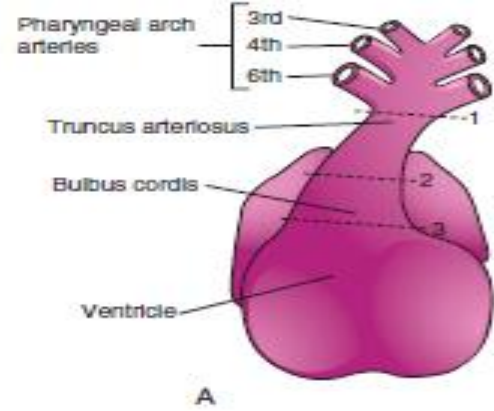
F



G



H



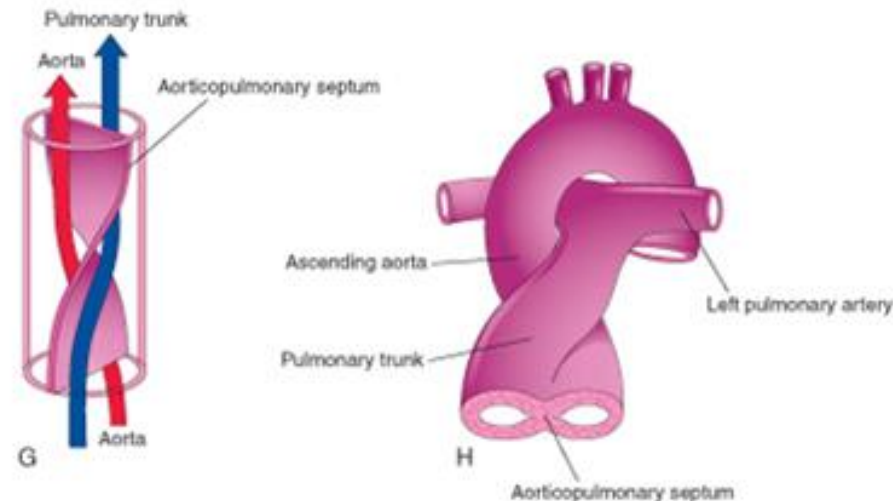
Partitioning of distal part of the Bulbus Cordis & Truncus Arteriosus,,,

➤ The ridges fuse together following a spiral course, forming aortico-pulmonary septum which has a spiral shape at the 6th week, (as in G).

➤ This septum divides **bulbus cordis & truncus arteriosus** into **aorta & pulmonary trunk**.

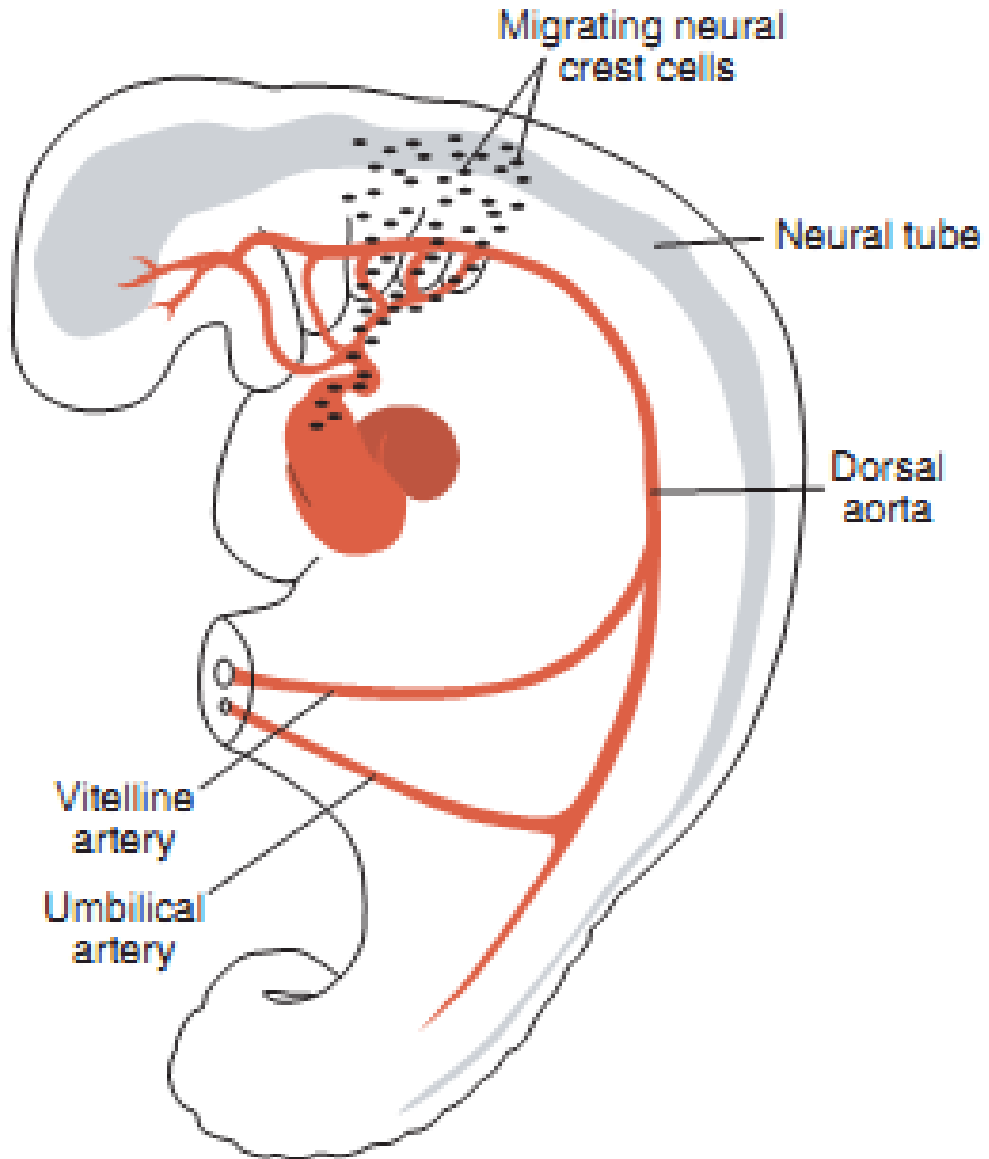
➤ Because of spiraling of aortico-pulmonary septum, **pulm.trunk twists around the aorta**. Firstly **pulm.trunk** lies **ant.& to right** of the aorta near the ventricles, then upward, it lies **post. & to left** of aorta.

➤ H, drawing showing aorta & pulmonary trunk **twisting around each other** as they leave the heart.



Neural crest cells migration to heart tube

- Migrate through **pharyngeal arches 3, 4, and 6** to the outflow tract of the heart.
- In this location they contribute to **endocardial cushion** formation in septation of **truncus arteriosus**.



The end!